

Quantum Statistics and Radiation

Structural Generation of the Bose–Einstein and Fermi–Dirac Distributions

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Abstract

We derive the Bose–Einstein and Fermi–Dirac distributions, their dilute Maxwell–Boltzmann limit, and the radiation laws of Planck, Stefan–Boltzmann, and Wien from the quantum and thermal realizations constructed within Triadic Modular Holography and Dimensional Mechanics. The arithmetic path support, termed *All Possible Paths* (APP), is the discrete torus $(\mathbb{Z}/9\mathbb{Z})^2$, equipped with an additive Cayley graph in the cardinal directions and two evaluations, addition and multiplication. The oriented ternary signature, TRIT, takes values in $\{-1, 0, +1\}$ and distinguishes orientation and regime; the *Topological Phase Kernel* (TPK) composes selection, transport, update, and prolongation while retaining memory. This dynamics produces the regional states and records on which unitary evolution and the calendar action are defined.

Parity determines the symmetric and exterior completions of these realizations: the former admits nonnegative integer occupations; the latter yields algebraic Pauli exclusion. The multiplicities determine the microcanonical counts; Gibbs-state factorization gives the two equilibrium distributions, whose dilute limit is quantified by explicit bounds. For free transverse electromagnetic modes with zero chemical potential, bosonic occupation and mode density produce the Planck spectrum. Convergence of the discrete sums is proved through uniform bounds at low and high frequencies. Their moments determine number, energy, and entropy densities; angular integration gives the Stefan–Boltzmann flux, and extremization of the frequency and wavelength densities gives the two Wien laws.

The calendar and holonomy also produce a capacity renewal $(20, 24, 25, 23)$, a graded carrier with multiplicities $1/380/649$, and a native geometric spectrum of ratio $1/10$. Its tritic refinement retains the residue and gives $1/1000$; the marked $23/26/29$ section evaluates exactly 1.380649×10^{-23} . Explicit isometries transport the energy origin, marker, and provenance. Boundary currents fix the sector offset, while composition with action proves $G_a k_B \Theta_{\text{clk},a} \log 3 = c^4 L_\alpha$ and the effective transducer identity on its positive section.

Configuration entropy, the measure on histories, and the spectrum of a reduced state are related through transport and reduction maps. A relative-divergence identity characterizes the stationary measure of maximal entropy on the areal–volumetric envelope. Thermal transduction connects information, action, and calendar with their dimensional bases. The representation, equilibrium, and convergence conditions specify the domain of each result.

In the finite realization $\Lambda = (\mathbb{Z}/27\mathbb{Z})^2$, the unitary Fourier transform establishes $|\text{supp } \psi| |\text{supp } \hat{\psi}| \geq 729$ and $H_{\text{pos}}(\psi) + H_F(\psi) \geq \log 729$; four subgroup collections saturate both bounds. On the dodecaphasic domain, the F044 readers distinguish ordered correlations not retained by marginal entropy. The prime-vacancy observer R_{vac} composes Sturmian incidence with the logarithmic clock and satisfies a modal identity under symmetric or Fejér summation. Its six cutoffs through 10^8 and the thirty recorded modes are finite witnesses; independent reproduction reaches 10^6 . The critical asymptotic estimate requires modal bounds uniform in k .

Keywords: Bose–Einstein; Fermi–Dirac; Planck law; genealogical states; Fock completion; Floquet dynamics; occupation entropy; modular Hamiltonian; thermodynamic limit; thermal radiation; entropic uncertainty; prime vacancies; ordered memory.

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Introduction

This work derives the Bose–Einstein and Fermi–Dirac distributions from quantum completions of states carrying trajectory and memory, and proves their convergence to the dilute Maxwell–Boltzmann regime. The same bosonic statistics, realized on the transverse modes of the electromagnetic field, determines the Planck spectral law, the Stefan–Boltzmann flux, and the Wien displacement laws. The central problem is to relate state structure, admissible occupations, and thermal observables through explicit operators and measures. Occupation and radiation laws are thus obtained at connected stages of a construction, with the equilibrium and representation conditions appropriate to each.

The foundation is Triadic Modular Holography (HMT), together with its dimensional realization in Dimensional Mechanics. The *All Possible Paths* (APP) construction is organized on the support $(\mathbb{Z}/9\mathbb{Z})^2$. Translations by $\pm(1,0)$ and $\pm(0,1)$ define its additive Cayley graph; the square cells provide a two-dimensional toroidal structure. Addition and multiplication are evaluated on this same support while retaining remainders and quotients. The oriented ternary signature, TRIT, takes values in $\{-1, 0, +1\}$ and distinguishes orientation and regime. Its integer lift retains both the residue and the carry. The *Topological Phase Kernel* (TPK) composes phase selection, transport, update, and state prolongation. Its record retains trajectory and memory: phase return does not reset the state.

Initial conditions and compatible refinement produce the genealogical sections used in the linear realization. Regional generation, the arithmetic of histories, the twelve-phase record, the determination of alpha, and exceptional incidence establish the antecedents of the coefficients and their representations. The state space, unitary evolution, and multiparticle occupation are constructed on this domain. Compatibility with depth restrictions precedes transport between realizations; in the finite instance, the calendar provides an explicit Hamiltonian, propagator, and action functional.

The decisive step toward statistics is the compositional parity character. The even sector is completed through symmetric powers, with nonnegative integer occupations; the odd sector is completed through exterior powers, whose creation operators are nilpotent and whose occupation operators are projections. This yields algebraic Pauli exclusion. The relativistic identification of this parity with spin additionally retains rotation-representation compatibility and locality conditions. The multiplicities determine the microcanonical counts, and factorization of the Gibbs traces yields the Bose–Einstein and Fermi–Dirac distributions. In the thermodynamic regime, variation of the combinatorial entropy under energy and population constraints provides a second characterization of equilibrium. The bosonic sector retains the chemical-potential condition ensuring convergence. Dilute-regime bounds quantify the common Maxwell–Boltzmann limit.

The radiative realization applies bosonic occupation to a three-dimensional periodic cavity with two transverse polarizations, linear dispersion, and zero chemical potential. The constant mode is fixed as background data before the excitation state is formed. Mode density and energy per excitation produce the Planck spectrum. Convergence from the discrete sums is proved by uniformly controlling the neighborhood of zero frequency and the high-frequency tail. Spectral moments give number, energy, and entropy densities; angular integration gives the Stefan–Boltzmann flux. Conversion between frequency and wavelength also transforms the measure through its Jacobian, so the two Wien maxima satisfy different equations. Each displacement constant is characterized by the maximum of the corresponding density.

The relationship between action, calendar, and temperature determines the dimensional normalizations of this realization. Thermal transduction distinguishes decision energy, dimensionless information, and the Boltzmann–temperature pair. Its composition with constitutive speed connects the temporal, energetic, and spatial scales of radiation. Continuity, generator-domain, contractivity, and convergence assumptions remain attached to the operations that require them.

Between transduction and the completions, the carrier-fixing chain consists of tritic renewal, double grading, positive energy memory, Williamson spectrum, threefold refinement, marked section, boundary currents, and centred thermal equivalence. This chain retains the origin that disappears from a normalised density and transports it to area, free energy, gravitation, and radiation. The complete evaluation is therefore presented not as a decimal coincidence, but as the output of typed operators, fibres, and maps.

The entropic interpretation distinguishes occupation configurations, the measure on histories, and the spectrum of a density operator. A calendar's chronological frequency is not the stationary measure on its envelope of histories. The latter is characterized variationally through a relative-divergence identity. In the number-conserving, gauge-invariant fermionic quasifree sector, the one-body correlator $0 < C < I$ and its modular generator K_1 determine each other in finite dimension. Transport, reduction, and conditioning specify which information remains observable. Record recovery and orbital identification of operators are linked through a bilateral balance that preserves state and memory together (Section 17).

The areal unit and entanglement complete this relationship between structure and information. The finite modular identity expresses entropy variations through sectoral areas minus a nonnegative relative-entropy deficit; for $\Delta_{\text{mod}} \neq 0$ and known normalization, area and the modular Hamiltonian recover the occupations of both sectors. The connection between memory, occupation, and radiation thus retains the maps connecting successive objects as well as their scalar values. Appendix A specifies the realization domains and the conditions of these compositions.

The finite realization on $\Lambda = (\mathbb{Z}/27\mathbb{Z})^2$ establishes exact support and entropy bounds, with saturations by subgroups and annihilators. On the dodecaphasic domain, ν_{120} and ν_{270} detect ordered correlations erased by the tritic histogram. On the prime domain, R_{vac} composes the generated Sturmian word, the internal realization of irreducibles, and the $\log p$ clock. Its modal identity is understood through symmetric sums or Fejér means, and its tables constitute finite numerical witnesses.

Relations between the constructions and their results

The following table identifies the contribution of each block to the argument. The exposition preserves the common antecedents and then distinguishes the specific operations: representation, occupation, conditioning, and radiative reading. The graded representation determines occupation; the spatial realization and dispersion determine its radiative application.

Construction and location	Construction and role in the argument
Core and extensions, §§1–9	Admissible states, oriented routes, memory, and refinement; regional outputs, dodecaphase record, and incidence. These antecedents preserve the origin of the coefficients and subsequent representations.
Quantum realization and action, §§10–13	State space, unitary evolution, and calendar action. The realization fixes the operators that subsequently enter the energies and equilibrium states.
Memory and temperature, §14	Readout fibers, reversibility, and the Boltzmann–temperature pair. Dimensionless information, energy scale, and thermal chart are distinguished.
Spectral capacity and marked bridge, §15	Weighted microscopic tree, graded carrier, boundary currents, capacity and Williamson scales, and the variational thermometric transducer.
Parity, Fock completions, and equilibrium distributions, §16 and §§18–23; milestones 16.1, 18.1, 22.1, and 23.1	Parity, Fock space, refinement transport, and the Gibbs state. Multiplicities give the counts; the traces yield the two occupation distributions, whose dilute limit is quantified, §§22–23.
Histories, area, and entanglement	Areal–volumetric incidence determines the history measure; state reduction and the area operator support the modular reading of entropy. See Theorem 24.1 and Section 28.
Fourier uncertainty, ordered memory, and prime vacancies, §29 and §4.16	The finite realization establishes the support and entropy bounds; dodecaphase readers distinguish order from marginals; the prime-vacancy observer resolves the correlation through the symmetric/Fejér modal identity, with a historical witness through 10^8 , independent reproduction through 10^6 , and a critical bound conditional on modal uniformity.
Bosonic occupation and thermal radiation laws, §30; Propositions 30.1, 30.6, and 30.8; limit in Lemma 30.2	Transverse modes, constitutive speed, and bosonic statistics determine the thermal sums. Estimates near the origin and in the tail justify the limit; its moments yield densities, flux, and spectral maxima.

Table 1: Antecedent constructions and their subsequent use in the statistical and radiative argument.

The presence of shared antecedents reflects this actual dependence. Each block preserves the information required by the next operation; a coincident dimension or scalar value is accompanied by the map that allows its reuse. Statistics thus receives a graded representation and a Hamiltonian, while radiation additionally receives a spatial realization and its dispersion relation.

1 Formal kernel of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information needed to prolong them. Numerical realizations are formed only afterwards. A constant recognized at the end of this process does not replace the rule that produced its region or the history that determines its evaluation.

The name *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is Feynman's path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory update, and prolongation. The acronyms are retained only after the objects they represent have been defined.

1.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks on a line graduated from zero to ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must be distinguished from the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (1.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not preserve the integer before reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i , repeated j times, produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 1.1 (Lifted arithmetic evaluation). The APP evaluation that preserves both sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (1.2)$$

Here the term *sheet* denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 1.2 (Reconstruction of the evaluations). *The two pairs in (1.2) determine $i + j$ and ij exactly. The residues alone do not determine these integer evaluations or their quotients.*

Proof. The first statement is (1.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, positions $(5, 5)$ and $(8, 2)$ give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residue projection identifies data that the lifted evaluation distinguishes. $\square$

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9
2	2	4	6	8	1	3	5	7	9
3	3	6	9	3	6	9	3	6	9
4	4	8	3	7	2	6	1	5	9
5	5	1	6	2	7	3	8	4	9
6	6	3	9	6	3	9	6	3	9
7	7	5	3	1	8	6	4	2	9
8	8	7	6	5	4	3	2	1	9
9	9	9	9	9	9	9	9	9	9

One support: 81 positions.

Two evaluations:
 $S(i, j) = i + j$, $P(i, j) = ij$.

Each row is generated by accumulation:
 $P(i, j + 1) - P(i, j) = i$.

The chart represents $0 \pmod{9}$ by 9.

The highlighted block is
 $B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}$,
 with eigenvalues 9 and 5.

Figure 1: APP multiplication table in positive representatives. Non-unit rows and columns are distinguished by gray shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representation edges do not constitute an intrinsic topological boundary of the torus.

1.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southwards and the second eastwards. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement before reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (1.3)$$

Path reversal changes the sign of this vector. Returning to the same residue position therefore does not require the winding to vanish. This is the first example of a distinction that will reappear in phase return: one coordinate may close while another records a nontrivial evolution.

1.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 1.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of both evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S = 810, \quad \sum P = 2025, \quad \sum \Sigma = 405, \quad \sum \Pi = 459, \\ \sum q^+ = 45, \quad \sum q^\times = 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (1.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Removing it would change the arithmetic being transported, even if the residue table remained identical.

1.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readings: its direct reduction modulo three is 0, 0, 0, whereas extracting the factor three and then reducing its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as directly reducing the original number.

1.1.4 Latin realization of the additive evaluation

The difference between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of C^9 . The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in every row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This yields a realization of a Latin square by rank-one projectors, in the terminology of quantum Latin squares [1, 3]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different content: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Thus the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries [2]. This reference provides a precise language for comparing realizations of APP.

1.2 TRIT: orientation and quadratic regimes

Definition 1.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A zero residue does not mean an absence of orientation, phase, or quotient information. Nor does it introduce a physical quantity by itself. Its function is to distinguish algebraically the transport classes that will be realized next.

1.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (1.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 1.5 (Exponentiation of the three types). *In each algebra (1.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (1.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, all terms of degree at least two vanish. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to all three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity through realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction appear jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parametrization; it does not add a fourth algebraic type.

1.2.2 Balanced division and composition of lifts

Ternary reduction expresses an integer evaluation of APP through a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the notation for an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes the compatibility of this rule with sign reversal explicit.

Definition 1.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 and belongs to the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the data needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it is its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined below uses these classes to determine the operating mode. Reading an evaluation and selecting a mode thus share the same ternary reduction, but have different domains and functions.

Proposition 1.7 (Arithmetic of balanced pairs). Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to the pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (1.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (1.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding the quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity ensures that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component suppresses this accounting: $1 + 1 = -1 + 3$, whereas the residue reading retains only -1 . The information removed is determined, in this case, by $\chi(1, 1) = 1$.

The relation with modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images fixes r first, by reduction modulo 3, and then $\bar{c}$. Addition transported by this bijection contains the term $\chi(r, s)$ from (1.7), now reduced modulo 3. Thus the two ternary levels do not compose by independent coordinatewise addition: normalization of the first level modifies the second.

In particular, the classes 3, 6, and 0 of $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving this coordinate prevents the zero residue sector from being artificially collapsed into a single position. Modulus 1000, used later to organize decimal blocks, serves a different positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas coincidence of residues in different bases does not by itself identify their transport states.

1.2.3 Inversion, residue class, and non-visible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum amplitude 1, these are the three possibilities of the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, sequence of modes, or winding. Thus decomposition $\mathcal{B}$ is integrated into the transport state rather than replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reversed order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while retaining the threshold 0. Transport of the remaining trajectory data is specified by the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed below, acts within a type already fixed. This distinction prevents attributing an algebraic equivalence between different regimes to a sign reversal.

1.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (1.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (1.9) does not replace (1.8); it expresses another operation in the declared quadratic codomain.

Definition 1.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 1.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so that $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm does not vanish, the stated formula supplies the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. Here the algebraic norm is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

The three types admit faithful realizations by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. The complex structure, the dual-number structure, and the split real structure are thus represented by a single matrix expression. Each preserves its own law and zero divisors. The form of signature (1, 1) in the last case admits a Lorentzian interpretation in dimension 1 + 1; identification of a physical metric additionally requires a specific space, field, and transport rule.

1.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t + s)J_\tau).$$

Comparing the two components gives the addition laws

$$C_\tau(t + s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (1.10)$$

$$S_\tau(t + s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (1.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution of the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the unit-norm circle. In the hyperbolic type, the eigencoordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

The evolution does not stop because $J_0^2 = 0$. Nilpotency removes terms of order at least two, but preserves the linear term that transports the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with an absence of state or dynamics.

Proposition 1.10 (Composition of affine coordinates). *In the chart where the scalar component does not vanish, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio of components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a well-defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the full algebra; a coordinate singularity does not automatically imply a singularity of the object. This formula will reappear in the elliptic regional reading. Its use requires retaining the denominator and, where applicable, the normalization factor.

1.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus a change of signature organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, the domains, and the effective operator acting at each transition.

In the electronic development of Article I of this series, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution of square I and two nilpotent directions. The joint realization is proved there for that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear within one matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. This relation endows the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its connection to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each retains its characteristic, dimension, and incidence information. Developing the respective operators establishes the connection, not equality of their cardinalities.

Norm-one conservation consequently describes the quadratic realizations of evolutions. Conservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties participate in the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations that give rise to them.

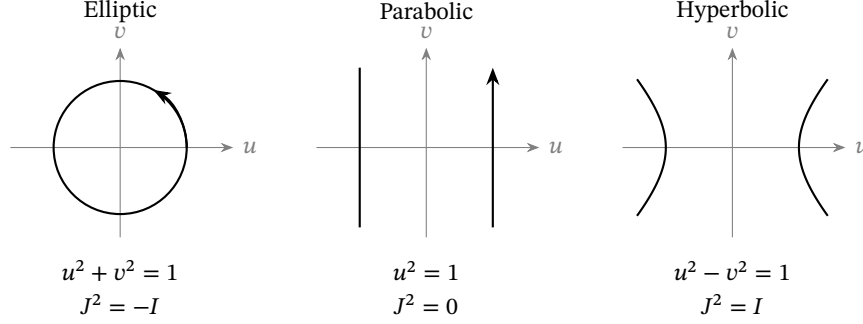


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. Parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

1.3 TPK: Topological Phase Kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation operates. Cardinal transport moves the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore different functions: a ternary sign is not a displacement on the torus.

We shall denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of both cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object by a reduced version that retains only the visible symbol.

Definition 1.11 (TPK update). Let Sel_t be mode selection through M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (1.12)$$

Each application preserves the fields it does not modify and updates the remaining fields by Euclidean division and trajectory concatenation.

The notation (1.12) expresses the order of composition; it does not replace the effective rules. The following section specifies its finite two-cursor realization completely. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

1.3.1 Dynamic state, observables, and operational dependencies

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (1.12) acquires operational content when the coordinates received and modified by each factor are specified. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before reduction. These coordinates are related through the transition and retain different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (1.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary records. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may retain different records.

Operational dependencies have three levels. At the local level, the selector activates a sheet, transport changes its position, and the update computes the new arithmetic coordinates. At the trajectory level, these transitions compose and retain the records of preceding steps. At the refinement level, extension relations determine which continuation preserves the predecessor's prefix, signature, and boundary. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

1.3.2 Tritic selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operating phase and the dodecaphase sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (1.14)$$

The previously defined selector may be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector in the ordered chart of D_9 is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (1.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event forms part of the chronology and its record. Cardinal directions determine the vectors v_d , whereas the phase signature determines which vector is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (1.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records previous crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport data remain reconstructible after the position has been reduced on the torus.

Proposition 1.12 (Conservation of lifted displacement). *For a cardinal trajectory of n steps, with activation indicators a_t ,*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved under concatenation of two compatible trajectories.

Proof. The one-step identity is (1.16). Summing it over the n steps cancels the intermediate coordinates. Splitting the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the previously defined winding number. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its record simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and arithmetic quotient q_t have different definitions; preserving them jointly maintains the geometric and arithmetic provenance of every transition.

1.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After applying (1.15), both directions are reversed at the end of a block of 27 steps. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (1.17)$$

The reading that forms the emission uses the positions before transport, according to the explicit recurrence in Section 3. The observable update and emission therefore compose in a fixed order.

Proposition 1.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. Within a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, the preceding phase is first recovered by subtracting one in Θ_{108} . That phase determines whether directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The following block preserves the activations and reverses the directions, so the integer displacements cancel over 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (1.13). Projecting history onto the calendar makes this finite check possible. The trajectory record, its evaluations, and the new prefixes remain in the dynamic state on which $\mathcal{U}_t$ acts; observable equality does not reinitialize them.

1.3.4 Affine memory transport and composition of states

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , we write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q , and S_q denote, respectively, the sets of residue–quotient data, the abelian transport-memory group, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified component by component in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue–quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart supplies local integer reconstruction. The precision towers additionally incorporate the truncation maps and cylinders developed in 3 and 4.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed through maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{1.18}$$

Here k_{tr} denotes the transport cocycle; the rational coordinate κ of record K retains its notation in the dodecaphase-closure chapter.

Transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{1.19}$$

The second equation is an affine action: the linear term transports previous memory and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$ and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (1.16). This concrete instance connects the abstract law with the cardinal operations already defined.

Proposition 1.14 (Compatibility of the update with concatenation). *Updating by (1.19) along γ_1 , followed by updating along γ_2 , coincides with updating along $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (1.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The statement for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 1.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves arithmetic transport and boundary prolongability simultaneously. This is the structure in which returns with memory and successive refinements take place.

1.3.5 Dodecaphase record, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The record preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window's subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence record. We adopt the chart developed in 7.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal records. Composition of segments and their increments is governed by (1.19). By contrast, the

group C_{12} describes translation of the window index; Θ_{108} preserves the step calendar. This distinction specifies what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancing 9 steps preserves r and increases b by one. Addition of phases introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar preserves $[b]_{12}$; a record of turns preserves $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 3 proves the nonsplit exact sequence (3.5). Its function is to describe the relation between these coordinates before they are used to construct the signed state.

The signed record receives phase and memory data through the composition developed in Section 7:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (1.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between this signed state and K , together with the rational reading of K , is proved in the corresponding chapter. Record operations precede these transformations. Reconstructing a record from its cyclic differences verifies its recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) retained by the record, the event reader counts codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different record coordinates. Chapter 7 develops this reader, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data this chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

1.3.6 Phase incidence and internal channel construction

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases and $O_{\text{ph}} = D_9 \setminus \{9\}$ the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (1.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 1.15 (Cardinalities and oriented phase incidence). *The sets in (1.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Multiplying by 6, 8, 9 gives 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve the incidence. $\square$

Here the cardinality 271 has a phase-incidence realization. The cubic completion in Section 3 has its own construction of the corona 1000 – 729. The correspondence between the two realizations must preserve the maps and marks identifying them, in addition to their cardinality. Likewise, the selector ε_9 , the incidence (90, 120), and the extractor $\mathcal{E}_{108}^{90,120}$ retain the domains and operations just distinguished.

1.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The reversed trajectory starts at the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and integer displacement changes sign. These identities follow by reversing the order of steps and replacing each cardinal vector with its opposite. For a loop, it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On records, reversal also requires transporting the order of components and the orientation of events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete record. In particular, the regional involution interchanging closure and propagation components preserves the self-scaling axis; its formula is given in Section 4.13. This regional operation acts on a different domain from cardinal trajectory reversal.

Extensions over a route $r : q \rightarrow q'$ are expressed by relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , we require

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

1.3.8 Nonadic holonomy and continuity of refinement

A phase turn composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (1.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the full domain we retain $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed later. The cyclic base has nine phases, while the fiber preserves trajectory, quotients, and boundary. Monodromy is the action of one turn of this base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization through the effective TPK transitions.

If g observes phase and M the number of retained turns,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure and the second identifies the new temporal depth. Coinductive prolongation also preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its antecedents without replacing them by the latest observation. This arithmetic of histories and its unity are developed in 3.8–3.10; the regions and their readers are constructed in Section 4.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the additive and multiplicative signature alphabets, with cardinalities 18 and 26. Their product identifies the catalog $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 3. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for all other phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. History transport remains given by (1.22), with its memory and boundary.

The resulting hierarchy establishes the function of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphase record determines the data entering K and the closure of α . Numerical and incidence realizations receive this preceding structure. This order allows every consequence to be presented in its own chapter while preserving, from the formal kernel onwards, the objects that make it possible.

1.4 Accumulated memory and transport resolvents

Calendar return does not remove increments produced during the journey. To express this distinction in operator form, an accumulated trajectory-event record is constructed first, followed by its generating series. Reduction of memory variables will be performed on this transport, also preserving the source and output reader.

1.4.1 Oriented events and record update

Let (d_t) be the sequence of integer direction codes retained by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification of their alphabets is assumed. We number the windows starting from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphase record. Its orientation is

$$\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1).$$

For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign retained in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (1.23)$$

Each summand is incorporated when its transition occurs. This definition counts trace events; it does not select directions through a value that is to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$ and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated record in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at a different origin, z_0 retains the preceding record. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 1.16 (Update and return with memory). *The record in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (1.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (1.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (1.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1} \mathbf{e}_0$, since $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (1.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, which proves (1.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^T D_3 + D_4^T D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^T B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the record; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 1.17 (Event-induced balance). *Update (1.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (1.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$; hence $R^T B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^T B y_m + \frac{1}{2} a_m^2 B_{00}$. The coordinate sum is invariant under permutation R , giving the second equality. $\square$

1.4.2 Generating series and uniform memory control

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 1.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)) \quad (1.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (1.29)$$

The derivative series converges uniformly on every disk $|u| \leq r < 1$.

Proof. Multiplying (1.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves its uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with another reader's variable requires preserving that clock and the map relating the two readings.

1.4.3 Memory elimination: operator, source, and reader

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 1.19 (Schur reduction with source and reader). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (1.30)$$

Its complete reading satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (1.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting it into the first gives (1.30); applying ℓ_0 and ℓ_1 to the two components gives (1.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes departure into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: the source and reader must also be transformed.

The distinction is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (1.32)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example concerns the twelve-window record, not the entire TPK state.

1.4.4 Segment composition and coherence of reduction

Let three composable affine transports be $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete journey is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (1.33)$$

The proof is successive substitution of the three affine formulas. Grouping the first two segments first, or the last two, produces the same expression; the factors transport each increment from its point of origin. In (1.24), $R_i = R$ and b_i is the event of its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (1.34)$$

Proposition 1.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (1.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting it into the first two produces (1.34). Eliminating the second component then gives the left-hand side of (1.35). Solving the two memory equations together gives the right-hand side, by Proposition 1.19. The formal solution is unique because $I - uT$ has constant term I . Both procedures also transform the source and reader according to (1.30)–(1.31). $\square$

Balance (1.27) describes record increments. The preceding identities allow them to be transported when memory is reused in dodecaphase closure; they do not by themselves determine additional analytic coefficients.

2 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

2.1 Phase selection, transport and enriched state

2.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (2.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t-1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (2.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b-1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

2.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i+1, j+1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (2.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

2.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

2.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (2.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (2.6)$$

Proposition 2.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (2.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

2.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (2.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

2.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (2.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

2.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 2.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

2.4 Six windows and decimal block emission

2.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 2.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (2.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

2.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (2.9)$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \quad (2.10)$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 2.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (2.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

2.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (2.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 2.5 (Positional recovery of the emissions). *The encoding (2.11) determines the complete word by*

$$U_m = \left[\left[\frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right] \right]_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

2.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 2: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

Table 3: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2),$$

$$\mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

2.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (2.12)$$

where $G^{(6)}$ applies (2.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

2.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 4: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 5: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 2.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 2.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 2.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (2.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 2.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

2.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (2.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$$\begin{array}{c|cccccc} |\Psi^{-1}(w)| & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \text{number of words } w & 132 & 66 & 18 & 3 & 6 & 18. \end{array} \quad (2.14)$$

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 2.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (2.13).

2.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (2.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (2.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 2.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (2.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 2.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 2.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 6; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 6: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

2.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (2.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (2.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (2.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 6.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (2.12); its scope is the same finite domain.

2.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (2.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

3 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions for the dynamics. An initial condition must also specify an orientation, and the additive and multiplicative sheets have independent cursors. This distinction allows exhaustive reconstruction of the emission without selecting a numerical region in advance.

3.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.1)$$

The independent annex enumerates all these conditions; the body presents the law transforming them. The rows of an emission table are fibers of this initial-condition domain.

The finite realization uses six windows of nine transitions. At each instant t , the phase signature is $M_{\text{ph}}(1 + ((t - 1) \bmod 9))$. If its value is $+1$, the positive residue $\Sigma(i + 1, j + 1) = \rho_9(i + j + 2)$ of the first cursor is read and that cursor is then moved; if its value is -1 , the positive residue $\Pi(i + 1, j + 1) = \rho_9((i + 1)(j + 1))$ of the second cursor is read and the second cursor is then moved. Zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, cardinal orientations are reversed. Integer evaluations and their quotients are retained in the record, but the emitter uses the residue readings specified here.

In the unit sector, the multiplicative reader uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For finite emission it is extended by zero to the non-unit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory record.

Let S_m be the sum of the three positive additive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the record. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (3.2)$$

The integer coefficients and calendar of this recurrence are explicit parts of the declared emission law. Values of π, φ, e or α are not received. Distinguishing an emission rule from a constant recognized from its output prevents the genealogy from being hidden in a decimal table.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule is written

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (3.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalog adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transport of subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with TRIT is performed after (3.4).

Proposition 3.1 (Injective factorization of emission). *The word $U = (U_1, \dots, U_6)$ determines both signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient set of 729 words.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence of (3.3). Hence coupling is injective on the signature product.

Enumerating the 324 conditions for one cursor through the preceding recurrence produces eighteen additive signatures, each with eighteen preimages, and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. The annex preserves these signatures and their initial conditions, allowing the finite enumeration to be repeated without a reference numerical catalog. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\begin{array}{ccc} \underbrace{432}_{18 \cdot 24} \text{ fibers of } 144, & 18 \text{ fibers of } 432, & 18 \text{ fibers of } 1944. \end{array}$$

The sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (3.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. Domain (3.1) is complete for the declared support and emitter; it does not contain every prefix of every prolongation in advance. Depth belongs to the system of compatible extensions constructed over those conditions.

3.2 Calendar closure and memory advance

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (3.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 3.2 (Nonsplitting of the calendar). *Sequence (3.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{lcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring a prolongation prefix. Nonadic holonomy Γ_9 denotes transport through one turn on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (3.6)$$

Phase equality and memory increment are compatible and constitute two observations of the same transport.

3.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order allows its transitions to be expressed as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. The state space, temporal order, and transported information are thereby specified.

The signature τ selects a quadratic regime and admits a temporal reading relative to this parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present, and future. The correspondence concerns the operational position of a transition within a history: retained antecedent, update, and admissible prolongation.

This interpretation acquires content through three distinct operations. Restriction of a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport, and recording in the observed transition. Prolongation considers continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deleting the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unlimited continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions each depth imposes on preceding depths. Existence of a compatible collection is established in the corresponding prolongation domain; merely specifying the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses order and changes the sign of each component. The trajectory is also subject to the transformations of position, sheet, and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its other coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical record is nonzero.

The relation with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_{\tau})$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to that local representation. Phase return of Γ_9 , by contrast, acts on the trajectory with memory, according to (3.6). A trajectory can recover its phase while retaining a new antecedent: historical data have changed although the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, whereas the TPK determines their selection and effective concatenation.

The aperiodic suspension in Section 4 will make this distinction explicit through a finite phase factor, an irrational rotation, and a memory degree. The link between TRIT and temporality is thus expressed by operations on trajectories and their quadratic representations. Its content exceeds a classification of signs: it preserves the regime, the direction of transport, and the relation between antecedent, transition, and continuation.

The monodromic representation of this return is developed in Section 4.10. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 4.12 defines the aperiodic time-crystal model, and Section 4.14 composes this temporality with the three Euler realizations.

3.3 Observable compression and orthogonal conservation

An isometric realization provides a quantitative expression of what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of transport through one turn, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (3.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 3.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r} D^* D T^r + T^{*n} T^n. \quad (3.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r , then summing, produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n} T^n$. $\square$

The final summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of conservation of unity under compression: the norm is distributed among the visible component, memory, and terminal, rather than being replaced by a single residue.

3.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonals modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unity of stochastic normalization and conservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This yields a local realization in $SO^+(1, 1)$; no prior introduction of four-dimensional spacetime is required to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, and $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of a single operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and let Δ_x, Δ_y be forward differences. Link variables are differences of potentials, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (3.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixture changes the sign of plaquette curvature. Summing over a rectangle of sides a, b cancels interior edges, and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

3.5 Cubic completion and two notions of boundary

Expanding the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (3.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The expansion corona contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (3.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent connection to Steiner hexads and octads requires the corresponding combinatorial maps; cube cardinalities do not replace them. Here this completion supplies the relation between bases 729 and 1000; its electronic application is treated in Article I of the series. Exceptional prolongation is developed here in Section 9, after regional generation and the dodecaphase record.

3.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description making its operation explicit. Let E_9 be the set of nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, extending E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without changing the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 3.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 3.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (3.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (3.12)$$

The punctured expansion contains 270 elements in addition to the initial cube; the complete expansion contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning one of the nine classes of E_9 to each remaining coordinate. Classification by the number of additional coordinates is exhaustive and its classes are disjoint. The sole element of S_3 is a_{**} . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH, here denotes this articulation of the initial body, expansion strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure through restoration of the apex. The added unit therefore has an explicit incidence location.

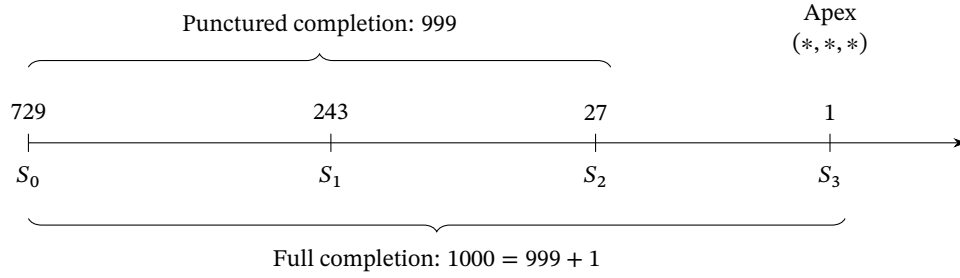


Figure 3: Strata of the cubic completion. The arrangement represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

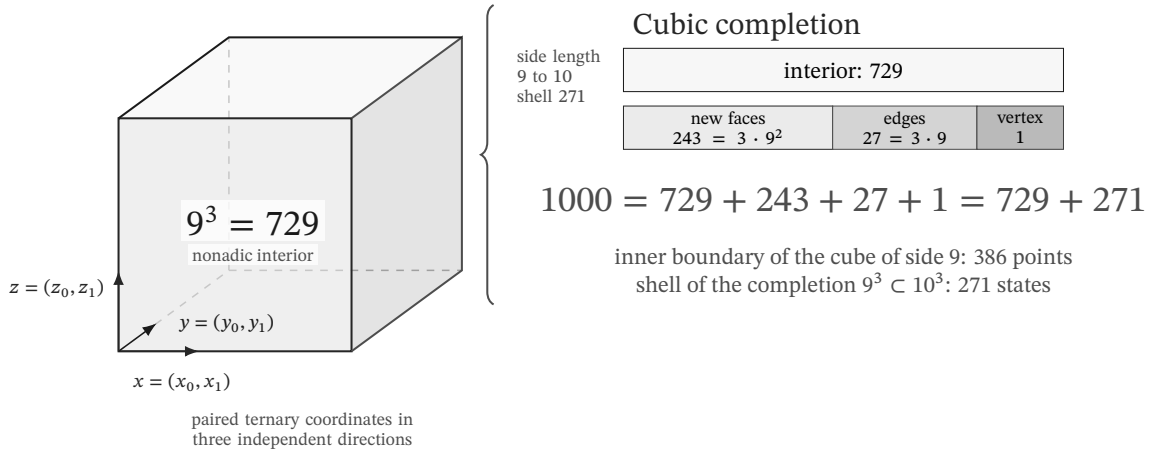


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

3.5.2 Conservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ deletes the last coordinate.

Proposition 3.6 (Projective compatibility of normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r}/10^d$, and the masses of all strata sum to one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (3.11); the binomial $(9 + 1)^d$ gives total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define a different measure: the distinction is operationally relevant. In particular, conservation of probability mass, conservation of transport records, and thermodynamic dissipation concern different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamic and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its expansion. Cubic completion, base-thousand positional expansion, and exceptional-incidence reading share support data; their respective maps determine which information from these data each representation uses.

3.6 Cubic incidence and Lorentzian realization

Cubic completion supplies a relation between interior and boundary. Its geometry can be developed through incidence of the six faces with the eight vertices. Label faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and vertices by $v \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),v} = \mathbf{1}_{\{v_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^T \in \mathbb{R}^6$, let R_F denote face opposition, and let $P_u = uu^T/6$, $P_- = (I - R_F)/2$.

Proposition 3.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^T = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_{\square} = \mathbb{R}^6 / \ker(MM^T) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct non-opposite faces share two vertices. The matrix $12P_u + 4P_-$ has exactly these entries. The uniform subspace has dimension one; differences between opposite faces generate a three-dimensional space orthogonal to it. Its complement has dimension two and belongs to the kernel. $\square$

The decomposition $1+3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_{\square}([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. The domain, decomposition, and sign convention producing the Minkowski realization are thus explicit. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_{\square} = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_{\square}e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_{\square}(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Direction reversal and face opposition thus acquire a concrete geometric realization.

3.6.1 Soldering, reversal, and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the endpoint fibers of an oriented dual edge a , a face label $\lambda_m(a)$, and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibers at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve label and orientation when transported to that common fiber, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (3.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale, and transport, rather than limiting geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four, the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, the eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic application of this domain belongs to Article III of the series and uses the electronic construction of Article I together with its stated coupling.

3.7 Compatibility of modular and positional reductions

The block-decimal base enters through Euclidean division preserving residue and quotient simultaneously:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Reduction of a block therefore requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and 3. This is an operational reason for preserving memory through a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer remainder. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adding six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (3.14)$$

Both identities follow by substitution of the definitions. Comparing endpoints is thus reduced to integer operations retaining each prolongation's contribution. This compatibility makes it possible to refine a region before evaluating its real coordinate.

3.8 Number as a compatible section and value as a reading

Distinguishing state from observation specifies what object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP–TRIT–TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. These are not independent coordinates: earlier arithmetic evaluations and transports impose their relations. A prolongation preserves these relations in each antecedent, even while adding new steps and memory data. The following definition formalizes this arithmetic of histories through its restriction maps.

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined in it.

Definition 3.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (3.15)$$

A reader is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the reader. The pair consisting of a section and its reader constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the reader associates a closed cylinder with each prefix, preserves nesting, and makes its diameter tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (3.16)$$

The concrete construction of these cylinders and the determination of their boundaries are developed in Section 4. Definition (3.15) does not by itself assert that the image of (3.16) is the entire line: a coverage claim requires its own theorem. Nor does equality of values establish equality of histories without an injectivity result. This separation allows precision to be discussed as depth while preserving the distinction between the arithmetic antecedent and its scalar realization.

3.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are its admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps come from TPK selection, transport, and updating. A local tritic fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of this composition. For each object x , a unital associative real algebra A_x , possibly noncommutative, is given. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (3.17)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for every composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (3.18)$$

These are explicit hypotheses on transports and their coefficients, not a consequence of calling the dynamics holonomic. Applying this presentation to a TPK sector requires identifying its ρ_u and $\Omega(v, u)$. Transitions expressed through bimodules require preserving that structure and are not replaced by (3.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{t(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (3.19)$$

when the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. Addition gathers histories with coefficients; multiplication concatenates histories while preserving the record multiplier.

Proposition 3.9 (Sectorial associativity with a central multiplier). *Under (3.17) and (3.18), product (3.19) is associative. If the object set is finite, its identity is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to move to the right of that factor; (3.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero because of their initial and terminal states. Bilinearity completes the proof. Local identities and normalization of Ω verify the stated identity element. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a + b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

The two carry sums of a triple are $\lfloor (a + b + c)/9 \rfloor$; hence (3.18) holds. The formal variable z records the turn without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphase record of calendar (3.5); retaining z without that relation distinguishes successive turns. The example realizes this calendar's memory without identifying it with all TPK memory.

3.10 Double variance of refinement and conservation of unity

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (3.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the whole route algebra: prolonging transport operators also requires their compatibility with refinement.

Lemma 3.10 (Identity and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (3.21)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (3.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global identity is the constant function one on all possibilities; an individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement may increase the number of extensions and the depth of each history while preserving the identity and all evaluations of its antecedents.

3.11 Phase return, Weyl representation, and two readings

Law (3.6) acquires an operator realization in the bilateral suspension constructed in Section 4. There, T is one-step advance, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the turn in this representation, the Weyl relations (4.14) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{3.22}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns while irrational rotation and memory distinguish the next state. Characters and parameter δ are realized subsequently from the numerical construction specified there: they do not select the seeds retrospectively. This representation describes suspension observables and translations; it does not replace all TPK coordinates or operators.

The double reading extends the same distinction between history and observable. On its Archimedean domain, (3.16) applies; on the calibrated incidence domain, the maps Q_n of Section 9.2 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readings, $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$ can be considered jointly: the former retains the value; the latter, the transported incidence preserved by its codomain. The section precedes both. This articulation makes it possible to pass from the arithmetic of histories to constants and exceptional reading without turning their realizations into new initial data.

4 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its function is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The word *generation* designates this operation; *genealogy* designates the reconstruction of its dependencies. Both notions enter the account, but serve different functions.

4.1 Orbits, regions, and characters of closure, propagation, and self-scaling

On a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; hence these permutations realize the dihedral group D_3 . The action preserves the catalog and multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fiber predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is a single orbit with these properties in the catalog. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature preserves this class and the *ES* or *NO* orientation. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the size-six orbits, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, equal to that of closure, uniquely selects propagation; the census $\nu = (2, 2, 2)$ uniquely selects self-scaling. The diagonal support cannot be omitted: without it there are four balanced singleton-fiber orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (4.1)$$

These symbols first designate regions of the catalog. The scalar identification will be proved through their readers; they are not chosen because they match digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position in the support would eliminate the distinction between numerical region and electronic structure.

The microfiber also has a precise bipartite morphology. Each emission is a product of an additive cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions together have another 18×24 ; and the return has a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition by connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

4.2 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary-word space act before the Archimedean realization. Their determination begins with the transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite selection, transport, and update step, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each six-digit string represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In this row-vector chart, the resulting matrices are

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The calendar (L_0, L_0, L_1, L_1) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary calendars are documented in the technical supplement. The indicated calendar is the only one of those sixteen that jointly reproduces the three five-block sequences.

4.3 Five closure regions and angular compensation

The closure region contains a transverse component and four accompanying components. The symmetric reader acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1, 1)^\top.$$

The invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^\top \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among reader positions; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The reader's elliptic chart realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule passing from a normalized coordinate to a slope, rather than identifying the two notions without a map. The reader composes the four accompanying positions through the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (4.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. The composition of slopes is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 4.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and gives $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second gives $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, so $q = 239$. $\square$

The resulting closure character has realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (4.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The meaning of the construction is precise: the slope and number of compositions are declared by the penta-fiber reader; the compensation equation forces 239. Once this character has been produced, its evaluation through alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (4.3), they provide rational endpoints with explicit error. The numerical value is thus computed as a consequence of the exact character, without introducing a target prefix.

4.4 Propagation reader and normalization of the unit

The propagation region is realized through a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and the oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not merely its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

the normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (4.4)$$

The character at one parameter unit is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

The normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the parameter unit in the reader is therefore not omitted. This condition precedes the decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 4.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric majorant of the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

The strict inequality follows from the decrease in successive ratios. $\square$

4.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. In modes Σ, Π , the rule is $+1 : m \mapsto \Sigma, -1 : m \mapsto \Pi, 0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi, \Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Hence its incidence, in that order of modes, is

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (4.5)$$

The four entries are obtained from the calendar, not from an equation previously supplied with the value of φ . Repeating the calendar gives counts $3F_{\text{av}}, 9F_{\text{av}}, 36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred eight. These counts are not powers of the matrix.

Theorem 4.3 (Positive self-scaling character). *The incidence (4.5) has a unique strictly positive eigenray. Normalizing it as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and equals $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ produces the polynomial. Only one of its two roots is positive. Since F_{av}^2 has all entries positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in producing F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

The measure on all compatible histories does not coincide with the frequency of modes in the three-event calendar. The chronological frequency is $(1/3, 2/3)$. The maximal-entropy measure of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . The latter enters the areal-volumetric reading; it is not obtained simply by replacing one counting frequency with another.

After the characters have been generated, the two orientations of the reader

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (4.6)$$

The second expression enters the electronic exponent developed in Article I of the series. The first is also obtained as the limit of the marked reading $\pi F_{n-1}/F_{n+1}$. Exchanging the arguments relates the two, but does not turn a reparameterization of the electronic equation into a second independent derivation of all its coefficients.

4.6 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not participate in selecting the prefixes that produce them.

4.6.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, consisting of integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. On residue evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. Quotients of the integer lift are retained in the corresponding arithmetic record.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (4.5) gives

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent-transport plane, and the modal-incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 4.4 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 4.5 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\boxed{\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.} \quad (4.7)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotency removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

Thus the exponential equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of one common exponential collection.

In this collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , previously obtained from its region, is recognized here as the unit evaluation of the exponential; this identity does not replace its coinductive genealogy.

4.6.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

Using Q keeps this operator square distinct from the dodecaphase record K .

Proposition 4.6 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

One thus obtains

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \tag{4.8}$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation gathers the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows recognition of their types without replacing those dynamics by an identification of fibers.

4.7 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A block sequence defines integer prefixes through $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting at $N_0 = m$, where m is the integer part determined by the reader. The preceding brackets place the closure, propagation, and self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; hence their initial values are $m = 3, 2, 1$. The following blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \tag{4.9}$$

The half-open convention is used to select a representative when a value lies on a boundary. For passage to the limit, the closures of these intervals are used, and the two equivalent boundary expansions are identified. This qualification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 4.7 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (4.9). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending by one block preserves the preceding prefix through Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints are Cauchy and have a unique limit. For a fixed decimal length, a value separated from the partition boundaries has positive distance from them. A sufficiently narrow bracket lies within a unique cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; that decision does not follow from numerical width alone. $\square$

The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π , e , φ guarantees their separation from the rational boundaries at any finite decimal precision. Arbitrary depth means *for every finite horizon there exists a computation determining its prefix*; it does not require a finite machine to store infinitely many digits at once. The mathematical object is the full compatible collection, and an execution produces a finite projection of it.

4.8 Specular involution and synchronization of the regions

The nonadic connection jointly transports the three characters. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under the specular exchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same trajectory system.

The census coincidences illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality of π , e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine, the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a displacement of -16 per channel. The closure leading to α must preserve both pieces of information: relative agreement and the evolution of memory.

4.9 Sturmian helical suspension

The relation between bases nine and ten in the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (4.10)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic reading of refinement has an exact consequence.

Lemma 4.8 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three, whereas that of the right-hand side contains only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and its sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, event separations are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first, not a second free parameter.

4.9.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record constructed later.

Proposition 4.9 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (4.11)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. Irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting consecutive terms produces (4.11). Summation over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is incorporated into the history. Here a vacancy is a capacity-retention event, with a determined position and antecedent. The preceding balance preserves length exactly between acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = \lfloor \mathcal{C}_t \rfloor_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to be preserved while the state changes.

The relation with TRIT can be followed in the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j \delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$ and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (4.12)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one in the negative direction. The canonical selector is applied to the positive phase $r_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(r_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the vacancy's binary encoding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their separations 6 and 7 determine the lengths of plateaus of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (4.11) and (4.12) make explicit the link between refinement, vacancies, and regime selection. The quadratic reading of TRIT subsequently realizes these types through their respective generators; the temporal event and its geometric representation thus retain an identifiable dependence.

Inverting the monotone index supplies another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks needed to gain m units of capacity is

$$\mathcal{T}(k + m) - \mathcal{T}(k) = m + \lfloor (k + m)\sigma \rfloor - \lfloor k\sigma \rfloor \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, whereas return time measured by the other index admits two values. This distinction belongs to refinement dynamics and is not an a posteriori correction of its numerical values.

Proposition 4.10 (Tritic renewal and marked capacity). *Let K_{ph} be the reduced-memory representation following the holonomy Γ_9 , and let $Q(u, r) = (u, [r]_3)$ be the phase quotient $C_9 \rightarrow C_3$. In every block of three regimes, the conditional expectations $I, E_+, E_\times$ satisfy*

$$\bar{K}^3 = E_+ E_\times =: \Pi, \quad \text{rank}(I, E_+, E_\times, \Pi) = (468, 26, 18, 1).$$

If $a_t = \mathcal{C}_{t+1} - \mathcal{C}_t$ and $\mathcal{R}_t = K_{\text{ph}}^{a_t}$, the enlarged transport

$$(t, \mathcal{C}_t, v_t) \mapsto (t + 1, \mathcal{C}_t + a_t, v_t + 1 - a_t)$$

preserves $t = \mathcal{C}_t + v_t$. From the first event $p_1 = 21$, the first time at which the composition becomes independent of the initial microstate is $T_{\text{ren}} = 24$, and

$$(\mathcal{C}_{p_1}, T_{\text{ren}}, T_{\text{ren}} + 1, \mathcal{C}_{T_{\text{ren}}}) = (20, 24, 25, 23). \quad (4.13)$$

The direct composition has ranks $468 \rightarrow 26 \rightarrow 1$; the conjugate one has $468 \rightarrow 18 \rightarrow 1$.

Proof. The strong quotient follows by summing the kernel over each fibre of Q . Each three-phase block contains one occurrence of every expectation, and these expectations are idempotent, self-adjoint, and commuting; their tensor ranks give the four integers. The capacity increments from 21 to 24 apply $I, E_+, E_\times$, so the first rank-one composition occurs at 24. The conjugate reading reverses the two expectations and preserves the same final product. $\square$

Regime return is not state reset: the quotient and memory continue to advance. The subsequent marked sections use the localisation rule

$$s_j = \min\{t \geq 24 : \mathcal{C}_t = 23 + 3j\},$$

which preserves threshold 23, the tritic residue, and the unbounded quotient.

Theorem 4.11 (Complexity of the phase readings). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. Retaining the phase $n \bmod 9$ gives complexity $p_9(m) = 9(m + 1)$; retaining only its ternary class gives $p_3(m) = 3(m + 1)$. All three topological entropies are zero.*

Proof. For factors of length m , the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: the coincidence between shifted endpoints removes duplicates, and irrationality prevents further coincidences. These points determine $m + 1$ intervals with distinct constant itineraries. Every irrational orbit visits each interval, giving $p(m) = m + 1$.

Fix a phase modulo nine. The positions of that phase follow a rotation of step 9δ , also irrational. By density, every factor occurs in each phase, and the first coordinate distinguishes its nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has frequency characters

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization of (4.10).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n + 1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (4.14)$$

The first two relations are Weyl covariances; the third expresses that one nonadic turn increases the memory degree. The base rotation is abelian, whereas the algebra of observables and translations is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward motion with backward motion. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

For this count we fix the intercept $\theta = 0$ and cumulative endpoints

$$(N_0, \dots, N_4) = (0, 729, 972, 999, 1000),$$

corresponding to the cubic order $(729, 243, 27, 1)$. The lower counts are $[N_j \delta] - [N_{j-1} \delta]$, giving $(33, 11, 1, 0)$. The upper counts are $[N_j \delta] - [N_{j-1} \delta]$, with the same initial boundary zero, giving $(34, 11, 1, 0)$. These values depend on the intercept and the order of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in the same count before any units chart; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The term *aperiodic crystal* is used here for this symbolic order, with complexity, spectrum, and return specified. Zero topological entropy does not by itself imply zero physical dissipation: a thermodynamic realization requires a defined energetic dynamics and erasure operation. Likewise, this Sturmian reading does not replace all TPK operators or by itself prove periodicity or aperiodicity of the complete α output. Its function is to make explicit a structure of order that the mere return of the calendar does not describe.

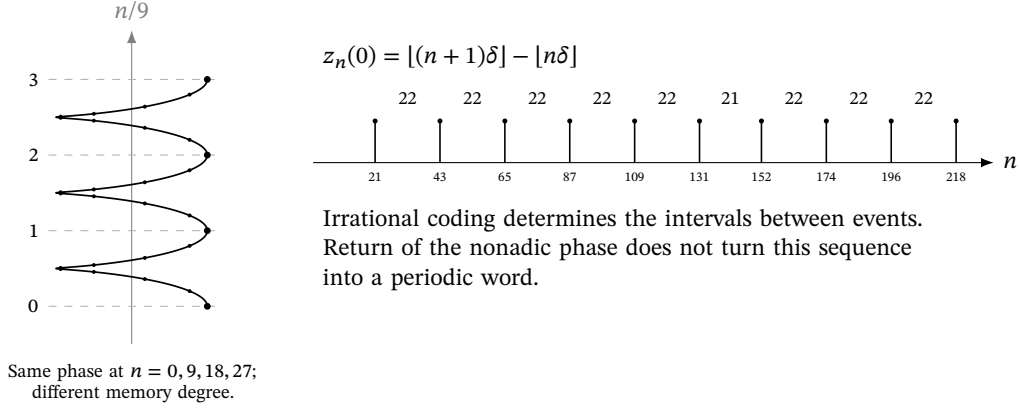


Figure 5: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact separations 21 or 22. The helical representation introduces no additional physical metric.

4.10 Nonadic connection, holonomy, and monodromy representation

The regional structure requires distinguishing the transport producing a history, observation of its phase, and representation of its returns. APP supplies the two arithmetic evaluations with remainder and quotient; TRIT determines regime and orientation; the Topological Phase Kernel composes selection, transport, and update. The initial conditions of cardinality 104 976, their catalog of 468 emissions, and the 243 visible words in the ambient space of 729 elements belong to different stages of that construction. The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

leads to a joint connection over the three regions. Its nine phases are positions of a single transport, and the regional coordinates π, φ, e , together with the dodecaphasic determination of α , arise from its compatible readings, developed in Sections 4, 7, and 8.3.

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k collects the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry record, Σ_k the boundary signature, and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete turns. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphasic position, whereas the integer m_k distinguishes its successive turns.

Definition 4.12 (Transport through one nonadic turn). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the admissible extension relations of the TPK. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation through s turns is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individualized history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the function of memory and boundary conditions in determining the trajectory.

Proposition 4.13 (Iteration of return and accumulated memory). *In every admissible composition of s turns,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-turn rule preserves the phase and increments the counter by one. Applying it to the already transported state repeats exactly these two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness of the difference between turns. After 12 turns the pair of finite phases is restored, but the counter increases by 12. The nonsplit extension (3.5) expresses the composition of the calendar through its cocycle; the trajectory also preserves the integer lift of that datum. Preservation here means transport of prior contributions, not immobility of all coordinates.

4.10.1 Bilateral representation of the return index

The geometric realization of the cyclic graph admits an especially simple operator representation. Here we number the phases from 0 to 8, shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ allows the unit shift to be expressed as

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one when $r > 0$, and sends $|0, m\rangle$ to $|8, m-1\rangle$. Passing through the phase origin therefore incorporates the exact unit of quotient.

Proposition 4.14 (Representation of cycle monodromy). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the group of turns of the cycle.

Proof. Nine shifts preserve the remainder of division by 9 and increment the quotient. The powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the covering of the cycle, whose group of turns is $\mathbb{Z}$, and represents the phase with its counter. The TPK state also contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the covering; a trajectory beginning at an initial condition uses its admissible half-axis. An operator inverse of the index is thus available without turning every extension relation into a bijection of the full state.

With $D_M|k\rangle = [k/9]|k\rangle$, the identity $[D_M, S] = S$ expresses the memory increment on the dense domain of finitely supported vectors. If $\mathcal{J}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one also obtains

$$\mathcal{J}T\mathcal{J} = T^{-1}, \quad \mathcal{J}D_M\mathcal{J} = -D_M - (I - P_0). \quad (4.15)$$

Indeed, for $k = 9m + r$, $[-k/9] = -m$ if $r = 0$, and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

4.11 Block clock and capacity clock

The index of a refinement map and the number of available decimal positions obey related but different laws. To express them, reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

Here $\mathcal{C}_n$ denotes capacity, not the dodecaphasic record K . At the canonical phase origin, the preceding mechanical word provides

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 4.15 (Exact balance between capacity and vacancies). *For $n \geq 0$, and in the second identity for $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. Irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities concludes the proof. $\square$

An event $z_n = 1$ adds a block to the history while preserving capacity $\mathcal{C}_n$; an event $z_n = 0$ increments both counters. Over any interval, capacity increments and vacancies sum exactly to its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

The uniform block phase and capacity phase are related through the integrated memory of vacancies. In particular, a capacity hold and a complete turn of the connection have distinct mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time needed to gain b units of capacity is

$$T_{\text{cap}}(a+b) - T_{\text{cap}}(a) = b + \lfloor (a+b)\sigma \rfloor - \lfloor a\sigma \rfloor.$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. This yields 9 or 10 blocks to gain 9 capacity units, and 113 or 114 to gain 108. These are inverse-clock times measured from a first arrival; the turn counter is always interpreted in the index to which it belongs.

4.12 Symbolic suspension and aperiodic time crystal

Rotation coding combines phase periodicity, local recurrence, and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0,1\}^{\mathbb{Z}}$ be the closure, in the product topology, of shifts of the bilateral word $z_n(\theta)$. Denote by σ_{sh} the shift of its sequences. Retaining a nonadic phase, the step is

$$F(r, z) = ([r+1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 4.16 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t+1) \sim (Fy, t),$$

with flow $\Phi_\delta[y, t] = [y, t+s]$. This model of temporal order, together with its vacancy coding and turn-counter representation, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ : the uniform discrepancy bound of less than one, obtained by telescoping over any window, persists under closure. A periodic sequence would have rational frequency. Therefore, $\mathcal{X}_\delta$ contains no periodic orbits. Nor does the suspension have closed orbits: a return of the flow would require an integer time and periodicity of the itinerary.

Finite words do recur. Length- m itineraries come from a partition of the circle by $m+1$ points; every interval is visited recurrently by irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases and explains the already established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

The realization through observables V_δ, W_δ and the shift T expresses the same order through the covariances (4.14). The frequencies of the uniform clock belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[C_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by the uniform phase when returns are interpreted. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specifying an energetic dynamics, its temporal observables, and its stability, data distinct from the present symbolic construction.

4.13 Specular symmetry of the regions and dual realization

Specular exchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$\mathfrak{c}B = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixedness concerns the involution in a fiber; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$, and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the allocation that the aggregate preserves without distinguishing. Time-index inversion $\mathcal{I}$ and the channel involution $\mathfrak{c}$ therefore act on different data.

Phase return has an explicit witness:

$$\begin{aligned} B_1 &= (012222 \mid 121221 \mid 112202), \\ B_{10} &= (101222 \mid 112221 \mid 112002). \end{aligned}$$

Both positions have phase 1; the block, axis, and memory have changed. Likewise, census synchronization in phases 8 and 9 annihilates their relative differences, while the diagonal component passes from (59, 59, 59) to (43, 43, 43). Equality of a relative observable does not exhaust the state determining it.

The dual realization preserves this architecture. In section 9.2, the same prefix determines nested cylinders on one side and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_mP_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two maps. The subsequent incidence construction produces codes, lattices, and the Moonshine realization in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

4.14 Temporal transport and tritic exponential composition

The Euler relations in section 4.6 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation sends J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau)\exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is the unit. In the elliptic type it is realized through $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, through $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

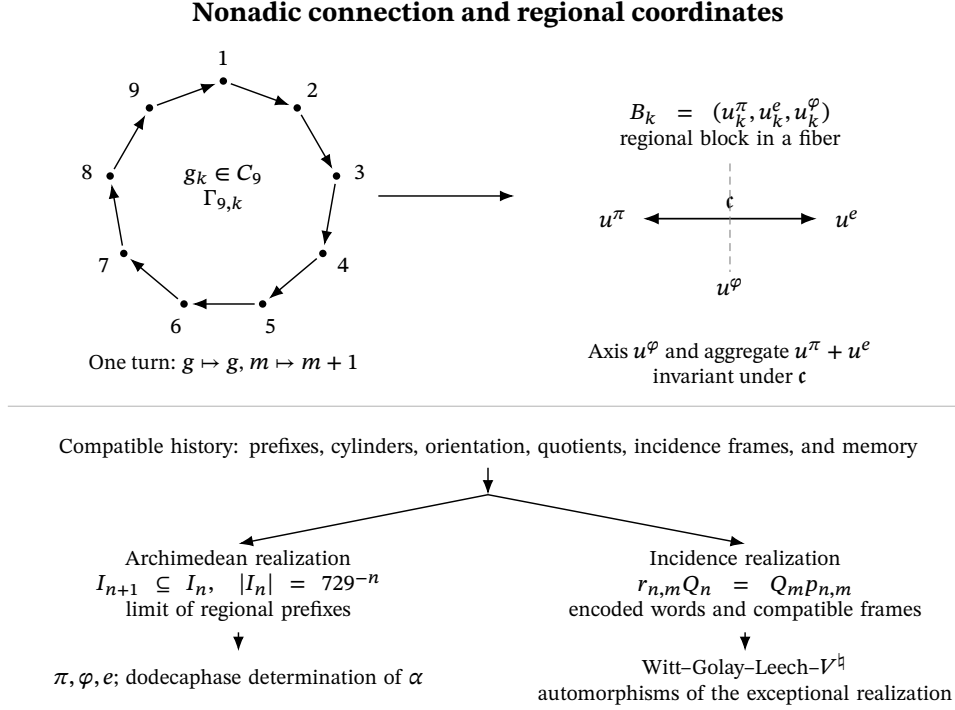


Figure 6: Nine phases of a connection and two realizations of a compatible history. The cycle represents phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the self-scaling axis in a fiber. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, while the other preserves words and incidence frames. The maps of the second realization are specified in Section 9.2; the exceptional construction is developed afterwards.

The elliptic realization closes a turn; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the regional parameters already generated,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \phi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unit of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transporting a local realization through an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. The identity transports the regime and its norm. A transition between different quadratic types, by contrast, belongs to TPK sheet and regime selection; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and the information at their endpoints.

This articulation distinguishes the closure of a local representation from the holonomy of a history. One may have $\exp(2\pi J_e) = I$ at the same time as $S|r, m\rangle = |r, m + 1\rangle$. The first equality restores the regime's orientation; the second records another turn. The discrete structure preserves both: the local return and the accumulated provenance of the state realizing it.

4.15 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. APP's local product contains both residue and quotient; iterating it constructs normal forms and factorizations before their decimal periods are examined. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, rather than to an isolated coincidence of periods. Numeration and the product are constructed below.

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is bijective onto the nonnegative integers: for a positive integer, division of $n - 1$ by 9 uniquely determines the first digit and continuation. This is bijective nonadic numeration; it retains the boundary representative 9.

Addition $\boxplus_\#$ is obtained by applying APP's additive sheet to the digits at each position and propagating their quotients. If there is only one digit, it is the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated by

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

Start with $c_0 = 0$ and incorporate the final quotient when nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. Local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the complete product is constructed through the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_\# a_{j+1}) \boxplus_\# (v_j \odot_\# u), \quad u \boxtimes_\# v = a_0.$$

The rule composes both local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_\# v) = \text{ev}_9(u) \text{ev}_9(v). \quad (4.16)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (4.16). Uniqueness of normal form transports associativity and commutativity to the constructed product; its identity is (1).

Definition 4.17 (Reduced multiplicative coproduct). On the finitely supported vector space generated by nonempty normal forms, define

$$\tilde{\Delta}_\# e_w = \sum_{\substack{u \boxtimes_\# v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A non-unit normal form is irreducible when its image under $\tilde{\Delta}_\#$ is zero.

Factorization fibers are finite by (4.16). In the completion $\mathcal{H}_\# = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, images of distinct forms have disjoint supports. If $d_\#(w)$ counts ordered non-unit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_\#}) = \left\{ x \in \mathcal{H}_\# : \sum_w d_\#(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is this weighted sum. Hence the operator

$$A_\# = (\overline{\tilde{\Delta}_\#})^* \overline{\tilde{\Delta}_\#}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered non-unit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_\#)$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to ordinary primes. Selection depends on factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{lcm}_{p^{a_p} \parallel n} \text{ord}_{p^{a_p}}(10).$$

Composite return synchronizes the returns of prime powers. For a prime $p \notin \{2, 3, 5\}$,

$$\tau_p \mid p-1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses nonadic closure of decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for every n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This differentiation specifies the relation with aperiodic refinement. The vacancy sequence determines increments of the capacity index; factorization determines multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while quotient and factorization retain information that this closure does not distinguish. The connection to the treatise's spectral constructions resides in these operators and their intertwining. Proof of their subsequent analytic results retains its own development; it is not replaced by periodicity of a residue observation.

4.15.1 Vacancy discrepancy and Dirichlet readings

The connection also admits an exact analytic expression. Retain the same coding $z_n = z_n(0)$, for $n \geq 1$, and consider two subsequent observables: the series over all indices and the product over indices selected by irreducibility. For $\text{Re } s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in this half-plane. Their common factor is density δ , but their discrepancies differ.

Proposition 4.18 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\text{Re } s > 0).$$

In $\text{Re } s > 1$, with logarithms defined by absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centered partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centered term by $s \int_1^\infty A_{\lfloor x \rfloor} x^{-s-1} dx$. The integral converges locally uniformly in $\text{Re } s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition reveals what the reading on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Controlling discrepancy over all integers gives the first holomorphic continuation; the second observable preserves its own discrepancy on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic kernel without transferring the proof of the treatise's terminal results to this article.

4.16 Prime-vacancy observer and modal resolution

In the causal chain of the already generated HMT state, the nonadic product, its reduced multiplicative coproduct, and the spectral projection P_{irr} produce the irreducible forms; the evaluation ev_9 subsequently recognizes them as ordinary primes. The 729/1000 boundary produces the vacancy calendar. The observer $\mathcal{C}_{\varepsilon, X}$ composes both internal objects with the logarithmic clock. A conventional sieve enters only as an external reader for numerical reproduction.

4.16.1 Calendar and equivalence of the two representations

Let

$$\varepsilon_{\text{vac}} = 1 - \log_{10} 9 = \log_{10} \left(\frac{10}{9} \right), \quad z_n = \lfloor (n+1)\varepsilon_{\text{vac}} \rfloor - \lfloor n\varepsilon_{\text{vac}} \rfloor, \quad n \geq 1. \quad (4.17)$$

The number ε_{vac} is irrational. Indeed, if $\varepsilon_{\text{vac}} = a/b \in \mathbb{Q}$, then $(10/9)^b = 10^a$, contradicting unique prime factorization because of the nonzero power of 3. Consequently, neither $n\varepsilon_{\text{vac}}$ nor $(n+1)\varepsilon_{\text{vac}}$ is an integer for $n \geq 1$, and

$$z_n = \lfloor (n+1)\varepsilon_{\text{vac}} \rfloor - \lfloor n\varepsilon_{\text{vac}} \rfloor \in \{0, 1\}. \quad (4.18)$$

The restriction $n \geq 1$ is necessary for this equivalence: at $n = 0$, the ceiling and floor conventions do not agree.

4.16.2 Typed chain and correlation observer

For $X \geq 2$, the causal chain can be written without loss of type as

$$\text{Ran}(P_{\text{irr}}) \xrightarrow{\text{ev}_9} \mathbb{P} \xrightarrow{\iota} \{0, 1\} \times \mathbb{R}_{>0}, \quad ((z_p, \log p))_{p \in \mathbb{P}_{\leq X}} \xrightarrow{\mathcal{C}_{\varepsilon, X}} \mathbb{R}, \quad (4.19)$$

where

$$\iota(p) = (z_p, \log p), \quad \mathcal{C}_{\varepsilon, X} : \prod_{p \in \mathbb{P}_{\leq X}} (\{0, 1\} \times \mathbb{R}_{>0}) \longrightarrow \mathbb{R}, \quad \mathcal{C}_{\varepsilon, X}((a_p, \ell_p)_{p \leq X}) = \sum_{p \leq X} (a_p - \varepsilon_{\text{vac}}) \ell_p.$$

It follows that

$$\theta_{\text{vac}}(X) = \sum_{p \leq X} z_p \log p, \quad \theta(X) = \sum_{p \leq X} \log p, \quad R_{\text{vac}}(X) = \theta_{\text{vac}}(X) - \varepsilon_{\text{vac}} \theta(X). \quad (4.20)$$

The exact centering of the observer is $\varepsilon_{\text{vac}} \theta(X)$; the expression $\varepsilon_{\text{vac}} X$ belongs to a different normalization. Two prefixes containing the same number of vacancies can yield different values of R_{vac} , because the observer retains both prime position and logarithmic weight.

4.16.3 Dirichlet readers and analytic domains

The finite observer and the two Dirichlet readers constitute distinct, compatible realizations:

$$\begin{aligned} Z_{\text{vac}}^+(s) &= \sum_{n \geq 1} \frac{z_n}{n^s} = \varepsilon_{\text{vac}} \zeta(s) + H_{\text{vac}}(s), & H_{\text{vac}} &\in \mathcal{O}(\text{Re } s > 0), \\ Z_{\text{vac}}^\times(s) &= \prod_p (1 - p^{-s})^{-z_p} = \zeta(s)^{\varepsilon_{\text{vac}}} G_{\text{vac}}(s), & \text{Re } s &> 1, \\ \log G_{\text{vac}}(s) &= \sum_p (z_p - \varepsilon_{\text{vac}}) \log(1 - p^{-s})^{-1}, & \text{Re } s &> 1. \end{aligned} \quad (4.21)$$

The first series and the Euler product converge absolutely for $\text{Re } s > 1$. The partial discrepancy of the mechanical word is bounded, and Abel summation extends H_{vac} holomorphically to $\text{Re } s > 0$. The power $\zeta(s)^{\varepsilon_{\text{vac}}}$ is defined here through the Euler logarithm in the half-plane of absolute convergence.

4.16.4 Fourier resolution and reality of the observer

For $k \in \mathbb{Z} \setminus \{0\}$, define

$$c_k = \frac{e^{2\pi i k \varepsilon_{\text{vac}}} - 1}{2\pi i k}, \quad S_k(X) = \sum_{p \leq X} (\log p) e^{2\pi i k p \varepsilon_{\text{vac}}}. \quad (4.22)$$

Proposition 4.19 (Modal decomposition of the prime correlation). *For every $X \geq 2$,*

$$R_{\text{vac}}(X) = \lim_{K \rightarrow \infty} \sum_{0 < |k| \leq K} c_k S_k(X) = \lim_{K \rightarrow \infty} \sum_{0 < |k| \leq K} \left(1 - \frac{|k|}{K+1}\right) c_k S_k(X). \quad (4.23)$$

The first limit is defined through symmetric partial sums and the second through Fejér means; these are exactly the two asserted notions of convergence. Absolute convergence in k lies outside both the hypotheses and the conclusion.

Proof. Let

$$f_\varepsilon(x) = \mathbf{1}_{(1-\varepsilon_{\text{vac}}, 1)}(\{x\}) - \varepsilon_{\text{vac}}.$$

Its nonzero Fourier coefficients are precisely c_k . The irrationality of ε_{vac} prevents $p\varepsilon_{\text{vac}}$ from reaching an endpoint of the interval; by (4.18), $f_\varepsilon(p\varepsilon_{\text{vac}}) = z_p - \varepsilon_{\text{vac}}$. Dirichlet's theorem gives convergence of the symmetric sums at these continuity points, and Fejér's theorem gives convergence of the means. Finiteness of the set $p \leq X$ permits interchange of the limit with the weighted prime sum under symmetric and Fejér convergence. The estimate $|c_k| = O(|k|^{-1})$ does not imply absolute summability and therefore does not permit an absolute rearrangement.

Finally, $c_{-k} = \overline{c_k}$ and $S_{-k}(X) = \overline{S_k(X)}$. The modes k and $-k$ contribute $2 \text{Re}(c_k S_k(X))$, which directly recovers the reality of $R_{\text{vac}}(X)$. $\square$

4.16.5 Numerical witnesses and independent reproduction

Table 7 presents the six available finite cutoffs. The values of R_{vac} reproduce every stored digit, and the normalized column retains the rounding of the numerical source.

X	$\pi(X)$	$\pi_{\text{vac}}(X)$	$R_{\text{vac}}(X)$	$R_{\text{vac}}(X)/\sqrt{X}$
10^3	168	7	-4.553590011317908	-0.143997159663565
10^4	1 229	61	41.334631595982586	0.413346315959826
10^5	9 592	427	-141.1579619980301	-0.446380669781268
10^6	78 498	3 595	32.21713602110386	0.032217136021104
10^7	664 579	30 384	-418.47581825715594	-0.132333673139529
10^8	5 761 455	263 841	4021.0180144347314	0.402101801443473

Table 7: Prime-vacancy observer at six finite cutoffs.

The complete thirty-mode table is stored in `S_k_1_30_X1e8.csv`. Its five largest values of $|S_k(10^8)|$, sorted by modulus, are:

k	$\text{Re } S_k$	$\text{Im } S_k$	$ S_k $	$ S_k /\sqrt{X}$	$ S_k /(\sqrt{X} \log X)$
19	30646.583055876723	-39786.84285255695	50221.56824686792	5.0221568246867925	0.2726368745267788
24	-1325.0060816425494	47631.087438932576	47649.51344695589	4.764951344695589	0.25867400944234675
21	1214.7582244307505	-44824.22356680968	44840.68081453668	4.484068081453668	0.24342575303172864
26	-26879.47488364538	33364.6769815865	42845.161221614406	4.284516122161441	0.23259271368502904
1	40391.94841735065	13384.21089257329	42551.693246765084	4.255169324676508	0.23099956965887428

Table 8: Five largest modes in the sample $1 \leq k \leq 30$ at $X = 10^8$. On this sample domain, $|S_{19}(10^8)|$ is the maximum.

The independent calculation of the first four rows, obtained from (4.17) with a sieve used as a verification reader, reproduces the values through 10^6 with absolute errors below 2.1×10^{-11} . The rows at 10^7 and 10^8 , together with the thirty modes, are finite numerical witnesses of the historical calculation. The projection P_{irr} in (4.19) retains the role of internal generator.

4.16.6 Analytic condition for the critical bound

The estimate

$$R_{\text{vac}}(X) = O(X^{1/2} \log^B X) \quad (4.24)$$

would control the centered prime term of the factor G_{vac} . Its proof requires estimates for $S_k(X)$ that are uniform in k and compatible with symmetric or Fejér summation. The tables characterize a finite window and do not provide that uniformity. The proved result is the exact modal identity for each finite X ; the critical asymptotic bound remains conditional on uniform modal estimates.

5 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

5.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (5.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 5.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (5.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 5.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (5.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (5.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (5.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 5.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (5.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (5.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (5.5) and its residual bound. $\square$

5.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (5.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (5.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 5.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (5.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 5.2. For the second, apply (5.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (5.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (5.8) determines its compatibility with every subsequent extension exactly.

Example 5.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (5.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

5.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (5.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$X_0 = \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, \quad Y_0 = \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix},$$

$$X_1 = \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, \quad Y_1 = \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}.$$

Theorem 5.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}.$$

Both matrices are invertible.

Proof. The matrices

$$A_0 = \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, \quad A_1 = \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 5.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

5.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (5.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 5.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (5.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (5.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (5.13)$$

Lemma 5.8 (Integer criterion for intersection). *The cylinders in (5.10) have a nonempty intersection if and only if (5.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (5.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (5.13). $\square$

Definition 5.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 5.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (5.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (5.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 5.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 5.2 places v_c simultaneously in both intervals in (5.10); the intersection criterion gives compatibility. The update identities are (5.5) in each base. Recovery follows from (5.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 5.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

5.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 5.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (5.16)$$

Theorem 5.14 (Uniqueness of the prolongation compatible with the reader). *Condition (5.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (5.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (5.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

5.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (5.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (5.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (5.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (5.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 5.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (5.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (5.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (5.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (5.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (5.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (5.23). $\square$

Example 5.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^{\text{W}} &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 5.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (5.24)$$

Proposition 5.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (5.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

5.7 Composition theorem for the block, cylinder, and signature

Theorem 5.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (5.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (5.14) and (5.15);*
2. *its six trits produce the signature $\mathcal{S}(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (5.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 5.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 5.11 to be applied. The deterministic application of (5.6) produces the bands used by (5.17)–(5.20). The recovery identities are those of Theorem 5.15. Finally, Theorem 5.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

5.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \tag{5.26}$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl, pr, au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \tag{5.27}$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (5.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

6 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

6.1 Regional bands, encoding, and integer signatures

6.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (6.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (6.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (6.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

6.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (6.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (6.4)$$

$$\alpha_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (6.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (6.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 6.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

6.2 Conversion clock and first zero advance

6.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (6.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 6.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (6.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 6.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

6.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (6.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

6.3 Oriented construction of the twenty-four-trit descriptor

6.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative–positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

6.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are (5, 7, 7, 7, 7, 7); this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (6.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \pmod{3}$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase–charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (6.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

6.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (6.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 6.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (6.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (6.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

6.4 Functional panel and terminal repertoire

6.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (6.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

6.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (6.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its $8!$ orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (6.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

6.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 9: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 9,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 6.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 10, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 10: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

6.5 Comparative readers, affine reader, and temporal incidence

6.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (6.14)$$

Lemma 6.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (6.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

6.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (6.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

6.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \text{ in } \mathbb{F}_3^6. \quad (6.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

6.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (6.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (6.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 6.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (6.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

6.6 Enumeration of cylinders and decimal candidates

6.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (6.19)$$

6.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 6.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (6.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 6.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 6.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (6.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

6.7 Exceptional incidence and heterotypy condition

6.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (6.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (6.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (6.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

6.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (6.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (6.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (6.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

6.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 \rightarrow 4, & \quad 1 \rightarrow 5, & 2 \rightarrow 6, \\ 3 \rightarrow 7, & \quad 4 \rightarrow 8, & 5 \rightarrow 9, \\ 6 \rightarrow 10, & \quad 7 \rightarrow 11, & 8 \rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

6.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \quad (6.26)$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 6.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_9$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (6.27)$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (6.22); the destination bands T_8, T_{12} and residues $R_{4,8}, R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (6.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (6.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

6.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_{72} = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

$$\begin{array}{cccc} 020022 & 222111 & 211201 & 021101 \\ 002001 & 020111 & 200110 & 121012 \\ 010122 & 001001 & 221110 & 222110 \\ 121020 & & & \end{array}$$

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

6.8 Regional composition and continuity of the realizations

6.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{6.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (6.13).

The register is then formed by the twelve Euclidean divisions in (6.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

6.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

6.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

6.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

6.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in the module

`TerminalReaders.lean`.

Enumeration and counts are formalized in the module

`TerminalOrbitalSelection.lean`.

The regional panel is linked through `RegionalPanelBridge.lean`, and composition (6.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduce Bool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

7 Dodecaphasic record and rational phase constant

The generation of regional coordinates does not exhaust the information in the APP-TRIT-TPK state. One Archimedean reading determines a value through compatible cylinders; another reading preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphasic record belongs to this second reading before entering the closure of alpha. It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block word from the catalog, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphasic record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide those maps, and the loss of signs is not a mere simplification of notation.

This chapter maintains the causal cut fixed in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving remainder, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite calendar. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory reading, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence between three coordinatizations of a record previously produced by APP-TRIT-TPK. The Hadamard transformation relates the signed record to the integer record; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions, and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplication by $1000^{12} - 1$ followed by Euclidean division recovers the complete record. The proofs in §7.8, §7.9, and §7.11 establish these equivalences in their respective domains.

Two compositions subsequently act on this record. The positional composition combines its components with the closure, propagation, and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \mapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The complete record allows both compositions to be reconstructed. The support B_K specifically preserves membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (9.3) links this integer extraction to the incidence of the regional codes. The negative support of their balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional extension. Thus, the function of K is defined through an invertible transformation and precise subsequent maps; the additional information on depth and transport remains in the state from which the record originates.

7.1 Cyclic extension and memory grading

The observable calendar contains one hundred eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \dots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphasic sector and the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 7.1 (Calendar carry). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the defect of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and equals the image of ι . Reduction is surjective, and multiplication by nine is injective on the indicated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{lcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the displayed carry and verifies the section identity. $\square$

The result explains a difference that will recur in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the calendar retains only $\beta = [q_{\text{mem}}]_{12}$. After one hundred eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and vertical memory. Closing a finite chart does not turn that history into a memoryless cycle.

7.2 The oriented record and the integral 1 + 5 + 6 chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the calendar forgets precisely some of the data required by the signed reader.

An initial chart of the record can be written through oriented events. The digital reader fixes the set of event codes $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the symbol of the digital record at position t , one obtains

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. The digital symbol, cardinal direction, and TRIT sign retain their respective domains. The vector e , whose components are signed local counts, is not yet the vector U appearing below either.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The $1 + 5 + 6$ decomposition is not an arbitrary allocation of coordinates: one coordinate preserves charge, five compare pairs relative to a reference pair, and six preserve the opposition of the semicoronas.

Lemma 7.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, yielding division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly under the parity congruence. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize removing the rest of the ledger. The full record has a larger domain than this count chart. Extracting the channels feeding U uses that full record and does not follow from the list (e_i) through an identification of names.

7.3 Incidence evaluation preceding the dodecaphasic record

Recovering a record from its cyclic differences and total charge is distinct from evaluating those coordinates on the incidence ledger. The first problem is solved through the transverse extractor of Proposition 7.12. For the second, a historical presentation of the record provides three integer counts in each window. It is useful to make their composition explicit and determine exactly the information their evaluation requires.

Let $\mathcal{L}$ be a previously generated ledger of visits and oriented traces. Each visit preserves the route, APP position, sheet, instant, multiplicity, and boundary incidence. Arithmetic evaluation produces the integer and its decomposition

$$n_t = r_t + 9q_t, \quad 1 \leq r_t \leq 9.$$

The integer n_t is computed from the sum or product sheet; its remainder is not received as an independent value. The distinction between corona incidence and interior incidence remains explicit for visits with remainder 9.

In window E_m , the count presentation considers

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (7.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (7.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_{\delta}(v). \quad (7.3)$$

Here $w(v)$ is the incidence multiplicity, and $\epsilon_{\delta}(v) = 1$ on the corona, -1 in the interior. The traces counted by $\nu^{\pm}$ must be enumerated through the corresponding ledger rule. The integer sum requires finite support or an alternative construction determining its meaning. The number of chronological incidences and the reduced length $120(1 + 3j)$ have different types; their relation requires the reader's length unit.

Define $[x]_{10} \in \{0, \dots, 9\}$ as the Euclidean representative and retain the quotients

$$x = [x]_{10} + 10[x/10].$$

The incidence block is

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}. \quad (7.4)$$

Composition yields the map

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \left((z_m - z_{m+3})_{m=1}^{12}, (z_m - z_{m+4})_{m=1}^{12}, \sum_{m=1}^{12} z_m \right). \quad (7.5)$$

Indices are read cyclically modulo 12. In this subsection, $(D_j z)_m = z_m - z_{m+j}$, for $j = 3, 4$, and $Q(z) = \sum_{m=1}^{12} z_m$; thus the preceding composition is $(D_3 z, D_4 z, Q(z))$. This convention is retained in §7.9, where complete integral reconstruction is proved. This formula produces its coordinates from the counts. It does not use K , U , α , or the expected transverse coordinates as arguments. Its identification with $\mathcal{E}_{108}^{90,120}$ requires composition with the effective collection and incidences of the terminal state; it does not follow solely from invertibility of the transverse system.

Proposition 7.3 (Sufficient and necessary arithmetic information). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$, ordered by the counts (A, C, V) . For the map defined by (7.4)–(7.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

In particular, the 36 sector residues determine exactly the 25-coordinate output. Preservation of quotients and provenance constitutes additional information in the record.

Proof. Equality modulo 10 produces the same blocks and therefore the same transverse coordinates. Conversely, if the transverse coordinates agree, $D_3(z - z') = D_4(z - z') = 0$. Steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of total charges annihilates that constant. Finally, the decimal representation of an integer between 0 and 999 has three unique digits. Thus the three residues of every window are recovered. The statement is an algebraic identity for all N, N' , not an inference obtained from finite tests. $\square$

7.4 Conjugation of counts and boundary terms

Let $\rho(m) = 13 - m$. Consider an incidence conjugation that reverses the window order, preserves the arithmetic and boundary classes of visits, and exchanges the trace channels. Then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

If $c_m = [C_m]_{10}$, decimal reduction produces the identity

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (7.6)$$

Indeed, $[-C]_{10} = 0$ when $[C]_{10} = 0$, and $[-C]_{10} = 10 - [C]_{10}$ otherwise. The formula preserves exactly this difference. Negating the channel is not equivalent to negating the entire decimal block.

Applying this identity to a concrete TPK involution requires checking its action on incidences. For an occupation reader evaluated before advancement, reversing a route $\gamma = (x_0, \dots, x_n)$ satisfies

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0).$$

Endpoint terms are incorporated before reduction modulo 10. On closed routes they vanish; in open windows they can alter the residues. This correction, already present in the bidirectional transport development, specifies what the connection between route and counts must preserve.

Proposition 7.4 (Necessity of the record's nonperiodic information). *Suppose the observable of a fixed route configuration satisfies $o(t + 54) = o(t)$, multiplicity is preserved under that return, and the same local reader acts in windows of 9 instants. Then its counts and blocks satisfy*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. Translation $t \mapsto t + 54$ bijectively sends E_m to E_{m+6} , preserving each counted incidence and its multiplicity. Equality passes to the sum and decimal reduction. $\square$

Thus a presentation with 12 distinct blocks requires the reader to preserve some information that does not factor through that period-54 observable: incidence selection, effective orientation, endpoints, multiplicity, or memory. The proposition delimits an inadmissible state reduction; it does not assert that the full dynamics has period 54 or itself determine its terminal reader.

7.5 Complete cylinder census and terminal-boundary selection

The terminal reading uses two kinds of information jointly: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This subsection constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 4, not values introduced to define APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent reading of those outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The Gram, weight, distance, and margin profile is specified as a datum of the terminal reader. Exhaustive enumeration proves what this profile selects from the fixed catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

7.5.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readings. The length of one thousand is sufficient for the integers below, as verified from the two bounds of each interval; it is not a depth assumed to be infinite. Define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (7.7)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphase record K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 7.5 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1)-1}{h} \right\rfloor. \quad (7.8)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer that $\lfloor mx \rfloor$ can attain is $\lfloor m(d+1)/h \rfloor - 1$, equal to the right-hand side because the numerator and denominator are integers. Equality of the endpoints of the interval of values proves the statement. $\square$

The stability test holds in all three channels for $m = 3^{30}$, $Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of regional transport:

Channel	Thirty-trit prefix	P_χ
π	010211012222010211002111110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the extension already constructed; the decimal reading is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (7.9)$$

The catalogue includes every tail for which these two cylinders have non-empty intersection.

Proposition 7.6 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max \left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r \right), \quad (7.10)$$

$$b_\chi = \min \left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r \right). \quad (7.11)$$

For the fixed regional cylinders, neither truncation is active, giving

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(7.12)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (7.9) intersect precisely when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D / Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1) / Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (7.8) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint is also obtained directly from $\lfloor T_\chi D / Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 is the ternary representation of length six, including leading zeros. By the table, each catalogue has fewer than 729 consecutive elements, so this reduction is injective on it. The three boundary catalogues are therefore described completely, without choosing the final triples:

$$\begin{aligned}\mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}.\end{aligned}\tag{7.13}$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would be a different condition: it would omit the first cylinder intersecting the boundary and yield 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

7.5.2 Specified profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, set $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions are integer-valued; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned}(g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3).\end{aligned}\tag{7.14}$$

The words are ordered as in (7.13). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} the diagonal matrix of $[n(u) = 0][w(u) = 3]$.

Theorem 7.7 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (7.14) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331.\tag{7.15}$$

These four counts are, respectively,

$$\begin{aligned}N_g &= \text{tr}(A_{\pi e} A_{e \varphi} A_{\varphi \pi}), \\ N_{gn} &= \text{tr}(D_\pi^n A_{\pi e} D_e^n A_{e \varphi} D_\varphi^n A_{\varphi \pi}), \\ N_{gnw} &= \text{tr}(D_\pi^{nw} A_{\pi e} D_e^{nw} A_{e \varphi} D_\varphi^{nw} A_{\varphi \pi}), \\ N_{gnwh} &= \text{tr}(D_\pi^{nw} A_{\pi e}^h D_e^{nw} A_{e \varphi}^h D_\varphi^{nw} A_{\varphi \pi}^h).\end{aligned}\tag{7.16}$$

All products and traces in this formula are computed over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. Expanding a trace in (7.16), its indices run over exactly one triple of words. The summand is one if it satisfies every represented condition and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the resulting cardinality is summed. The final sum runs over $0 \leq i < 196$, $0 \leq j < 197$. Substituting the words in (7.13) and the profile in (7.14) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions for each of the 196 indices i , together with the 331 complete triples; its program evaluates these sums using exact binary indicators and checks each survivor by scalar computation. No sampling or tolerance is involved. $\square$

7.5.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (7.17)$$

where w_{col} counts the non-zero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Write $\mathcal{F}$ for the set of 331 triples in the preceding theorem.

Proposition 7.8 (Reduction of the census to the terminal pair). *The indicator sums over $\mathcal{F}$, upon successively requiring rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (7.18)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For every triple in the already enumerated set, compute the rank by elimination over $\mathbb{F}_3$, count the non-zero elements in the six columns, and sum those columns modulo three. The products of the indicators in (7.17) give (7.18). Substitution of the two final index triples into (7.13) yields the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing those two matrices as its domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are (0, 2, 0) and (2, 2, 1); only the second belongs to that line. This recovers the final step of the selector in Section 7.6, now preceded by the construction and enumeration of its complete domain. The stable reading of $[3^{1080}x_{\chi}]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same catalogue positions (129, 197, 148). This subsequent comparison neither defines the profile nor participates in the indicator sums.

7.5.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 here preserves each candidate's identity because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled margins and charge supply effective additional conditions.

A complete finite enumeration and its selector on output cylinders have therefore been recovered, with an explicit reading profile. Construction of the signed record additionally uses its memory reader and channels, as specified below; the visible matrix is not identified with that record merely because they have the same number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a statement concerning alpha or the complete continuum.

7.6 Visible reading and signed record of the terminal state

Denote by x_{term} the state produced by the composition of selection, transport, and update up to the terminal cut. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readings are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary reading; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The census of the terminal selector considers catalogs of sizes 196, 197, and 196. Their local signatures reduce the $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331. The rank, weight, and column-margin conditions $q_{\partial} = (1, 2, 2, 1, 2, 0)$, evaluated in Proposition 7.8 leave two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the census proved in Section 7.5, with the stated reader profile. It does not make the column margin a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed reader has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (7.19)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections starts exactly at this typed output. The preceding evaluation is developed in §7.3: visits and traces determine the counts, which produce the blocks through (7.4) and the differences and charge through (7.5). Proposition 7.3 exactly characterizes the arithmetic information in that evaluation; subsection 7.4 preserves its boundary terms under conjugation. Identification of the effective incidence collection of this terminal state has the scope specified there. The harmonic reconstruction below uses the typed output and does not replace the evaluation preceding it.

7.7 From the channel record to the signed vector

The harmonic reader Π_H can be written without using either K or α . To avoid confusing its orbital sums with the α blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (7.20)$$

Division by three belongs to the three-orbit reader. Its integrality is checked on compatible record outputs; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (7.21)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (7.22)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalog concatenation of period six. The difference concerns domain and preserved information, not just values.

The constructive order so far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion below produces the dodecaphasic record. The subsequent ability to recover the channels from the dodecaphasic record is a reversibility check; it does not reverse this causal order.

7.8 Hadamard transformation by orbits and integer reconstruction

On the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation into four sectors of each orbit is what, once origin and orientation are fixed, admits the angular reading of ninety degrees. The transformation does not act on three consecutive four-element sets in the natural order.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders according to the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 7.9 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 7.10 (Integral dodecaphasic transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines exactly the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves the two inverse compositions. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 7.11 (Canonical integral record). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (7.21) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural order, the dodecaphasic record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (7.23)$$

It is the unique rational preimage of U , and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (7.23). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the order-two minors are even, with at least one of absolute value two; the order-three minors are multiples of four, with at least one of absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

7.9 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second compares sectors separated by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to an identification with a physical operator from another domain.

Proposition 7.12 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphasic record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. Total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral claim, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every order-twelve minor must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in the Q row. All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The last factor is therefore twelve. $\square$

Checking (7.23) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (7.20) and (7.21) in turn reconstruct U from these differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphasic record. Equality of results does not collapse their codomains into one; it proves that the declared maps recover the relevant information.

7.10 Integral compatibility and determination of the transverse record

The domain of this construction is the representation of the record of an APP–TRIT–TPK history. Evaluation of record events, compatibility of its channels, and inversion of its coordinates are distinguished. The following result gives a complete integral criterion for the second level and an explicit factorization connecting the first to it. The proof uses the record operators and establishes their relations for every element of the indicated integer domains.

7.10.1 Characterization by adjacent increments

The indices belong to $\mathbb{Z}/12\mathbb{Z}$. Let

$$(Sz)_i = z_{i+1}, \quad D_3 = I - S^3, \quad D_4 = I - S^4, \quad Q(z) = \sum_{i=0}^{11} z_i.$$

Retain the transverse extractor

$$\mathcal{D} : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}, \quad k \longmapsto (D_3 k, D_4 k, Q(k)).$$

For a triple of integer coordinates (b, c, q) , write

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (7.24)$$

Theorem 7.13 (Integral image of the transverse extractor). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if it satisfies*

$$b_i = w_i + w_{i+1} + w_{i+2} \quad (0 \leq i < 12), \quad (7.25)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (7.26)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (7.27)$$

Its unique integer preimage is

$$k_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (7.28)$$

The twelve equations of (7.25) and equation (7.26) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$, one has

$$c_i - b_{i+1} = (k_i - k_{i+4}) - (k_{i+1} - k_{i+4}) = k_i - k_{i+1}.$$

The sum of three increments gives b_i ; their full cyclic sum is zero. Moreover, $P_i = k_0 - k_i$, so $q = 12k_0 - T(w)$. All conditions and the inverse formula follow.

Conversely, (7.27) makes formula (7.28) integral; (7.26) allows its cyclic prolongation. Its adjacent differences are w_i . Thus, $D_3 k = b$ and

$$(D_4 k)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summing the inverse gives $Q(k) = q$. The same formula proves uniqueness. For independence, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ turns the first twelve relations into $b = (I + S + S^2)w$. Each solves for a different coordinate of b . The condition $\sum w_i = 0$ is an additional independent relation. $\square$

The rational image has dimension twelve. Its integral structure adds a congruence of order twelve: this is the constructive manifestation of the final Smith factor of $\mathcal{D}$. The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$ and congruence (7.27). The original twenty-five coordinates preserve these same twelve degrees of freedom through thirteen relations.

7.10.2 Commutativity with the harmonic reading

Let $\mathcal{H}_{12}$ be the record's Hadamard transformation, with orbits $(r, r+3, r+6, r+9)$ for $r = 0, 1, 2$. Its block is

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For compatible (b, c, q) , define

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad A_0 = \frac{q + 2B_0 + B_1}{3}, \quad A_1 = A_0 - B_0, \quad A_2 = A_1 - B_1.$$

The reader Π_H of (7.21) has blocks

$$(\Pi_H(b, c, q))^{(r)} = (A_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 7.14 (Agreement of the two reconstructions). *On $\text{im } \mathcal{D}$,*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is exactly

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}.$$

Proof. For $k = \mathcal{D}^{-1}(b, c, q)$, let $a_r = \sum_{j=0}^3 k_{r+3j}$. Shifting four positions sends each orbit to the next, so $B_r = a_r - a_{r+1}$. Together with $q = a_0 + a_1 + a_2$, these identities give $A_r = a_r$. On an orbit, write (x_0, x_1, x_2, x_3) for the coordinates of k and $d_j = x_j - x_{j+1}$. Then

$$d_1 - d_3 = x_0 + x_1 - x_2 - x_3, \quad d_0 + d_2 = x_0 - x_1 + x_2 - x_3, \quad d_0 - d_2 = x_0 - x_1 - x_2 + x_3.$$

These are the remaining three rows of $H_4 k^{(r)}$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The condition $\Pi_H(b, c, q) \in \mathcal{L}_H$ controls only the harmonic output. Membership of the twenty-five coordinates in $\text{im } \mathcal{D}$ additionally requires (7.25)–(7.27). For example, adding the vector $e_0 - e_3$ to c preserves the three orbital sums and leaves Π_H unchanged. Starting from $(0, 0, 0)$, one thus obtains a triple with $\Pi_H = 0$ that violates (7.25). This is an exact negative control of channel compatibility, independent of any terminal evaluation.

7.10.3 Determination and transport on the record's orbits

Theorem 7.13 determines k once its channels and charge have been produced. As a condition on outputs not yet individualized, the same image is a group isomorphic to $\mathbb{Z}^{12}$. Fixing only a charge q , the solutions form an affine class of

$$L_0 = \{v \in \mathbb{Z}^{12} : \sum v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $100\mathbf{1} + e_0$ and $100\mathbf{1} + e_1$ are two different records of charge 1201, with components in $\{0, \dots, 999\}$. Each satisfies every image condition and every Hadamard congruence through its own channels. These witnesses compare compatibility equations; they are not attributed the status of terminal HMT histories.

Proposition 7.15 (Cyclic covariance and stabilizer of the record). *Let ρ be window translation on a record domain $\mathcal{R}$, and S the preceding translation on $\mathbb{Z}^{12}$. On the orbit of $R \in \mathcal{R}$, a covariant reader is determined by a vector*

$$k \in \text{Fix}(S^p),$$

where $p \mid 12$ is the least period of R under ρ . The charge condition $Q(k) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine class of rank $p - 1$. In particular, on a free orbit, covariance and charge leave eleven degrees of freedom.

Proof. The rule $F(\rho^j R) = S^j k$ is well defined exactly when $S^p k = k$. Such vectors repeat a word of length p exactly $12/p$ times. Their charge is $\frac{12}{p} \sum_{i=0}^{p-1} k_i$. Divisibility and rank follow from this formula. Channel covariance follows from $D_a S = S D_a$ and $Q S = Q$. On $\mathcal{L}_H$, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12} / 4$; by Proposition 7.14, both reconstructions intertwine these actions. $\square$

If the domain carries a marked origin, the action transporting that mark as well must be used. The proposition concerns this declared cyclic action; it does not assume periodicity of the enriched history. Naturality under truncations after the 108 cutoff is obtained, for any already defined reader F_{108} , through $F_N = F_{108} \circ \tau_{N,108}$. This naturality preserves the initial evaluation rule; it does not by itself choose one of the possible rules on the first one hundred eight events.

7.10.4 Event evaluation and composition of contributions

The preceding characterization provides a factorization of extraction through two effective readings of the record:

$$\omega : \mathcal{R} \longrightarrow \{w \in \mathbb{Z}^{12} : \sum w_i = 0\}, \quad q_{\text{reg}} : \mathcal{R} \longrightarrow \mathbb{Z}, \quad q_{\text{reg}}(R) + T(\omega(R)) \equiv 0 \pmod{12}. \quad (7.29)$$

These maps must be evaluated on the events, orientation, incidence, and memory produced by APP-TRIT-TPK. Once their actions on these data are given, the composition

$$R \longmapsto (w, q) \longmapsto ((I + S + S^2)w, (I + S + S^2 + S^3)w, q) \xrightarrow{\Pi_H} u \xrightarrow{\mathcal{H}_{12}/4} k$$

is determined and satisfies all inverse checks. The formula $w = c - Sb$ also allows connection to a producer already delivering both channels: it is a check on its output, prior to any comparison with the terminal witness.

Proposition 7.16 (Accumulation of contributions and prospective uniqueness). *Suppose the initial state and each event a_t of the record produce, through previously fixed rules, contributions $(\delta w_t, \delta q_t)$ with*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, the update

$$w^{(t+1)} = w^{(t)} + \delta w_t, \quad q^{(t+1)} = q^{(t)} + \delta q_t$$

produces a unique record at each stage. Its increment is

$$\delta k_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects concatenation of records, and the reading of a prefix remains unchanged under subsequent prolongations.

Proof. The compatibility conditions are preserved under addition. Formula (7.28) is additive in (w, q) and produces the indicated increment. Induction fixes each state successively. The sum over two segments is the concatenated sum, and a prefix receives exactly the same summands before and after prolonging the history. $\square$

The proposition provides a sufficient criterion, not a requirement of linearity for every TPK reader. If events are transported by nontrivial actions, their contributions can be expressed in the terminal chart through the corresponding affine cocycle. The cocycle equation then preserves the same independence from the partition of the trajectory. In both cases, event rules and the initial state constitute the constructive data that must come from the ledger: choosing them to reproduce an expected vector would replace that provenance with another task.

The result of this development is the exact determination condition (7.29), its proved inversion, and its transport identities. Its scope is the transverse record; the remaining HMT constructions retain their own domains and proofs.

7.11 Two rational readings: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart permits two distinct operations. The first reads a single turn; the second prolongs the dodecaphasic record periodically. Let $B = 1000$, and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (7.30)$$

The leading zeros of a block, such as that of 058, are part of its three-digit representation. They do not change its integer value, but allow the twelve blocks to be concatenated unambiguously.

Definition 7.17 (Finite and periodic readings of the dodecaphasic record). The one-turn reading is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 7.18 (Exact value and period of the prolonged dodecaphasic record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12} \kappa_{\text{per}}. \quad (7.31)$$

The infinite word of the dodecaphasic record has least period twelve in base thousand and least period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series yields $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the least block period is therefore twelve. The decimal expansion is canonical, since the dodecaphasic record has no tail of 999 blocks. If its least decimal period were d , it would divide 36. Grouping digits in threes would produce a block period $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve through this operation. Hence the least decimal period is 36. $\square$

Both readings are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their closeness does not permit substituting one for the other in an exact identity. In particular, the finite alpha closure uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

7.12 Reversibility of the rational constant and phase change

The rational constant κ_{per} determines the entire record K when the base, length, and reading origin are retained. Its numerical and structural functions are related by an exact inversion.

Proposition 7.19 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1) \kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros in each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000 \kappa(K) - K_1. \quad (7.32)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. Furthermore,

$$I(\rho K) = 1000 I(K) - K_1 (1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ yields (7.32). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the already constructed transformations $D_3 K, D_4 K, Q$. Rational evaluation preserves the signed record and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphasic record.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical reading, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

The subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α additionally uses its regional records and boundary normalization. The relationship between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the function of K : it provides a reversible integral coordinatization of the oriented record, whose periodic evaluation and incidence functions participate in subsequent constructions. The value, transformation, and domain are preserved together.

7.13 Periodicity and cokernel of the transport operator

The periodicity just proved belongs to the dodecaphasic record. It does not imply periodicity of the regional coordinates, the closure carries, or alpha. It is useful to clarify this separation through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it does not authorize changing the dodecaphasic record already selected by the record.

Proposition 7.20 (Quotient of periodic carry). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ is equivalent to imposing $t = b$ and leaves the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient arise from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite alpha closure will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

The next chapter is thus prepared. The dodecaphasic record has been fixed before solving the alpha recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses alpha as an input, and none yet determines the arithmetic type of the output of the complete closure.

8 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase record. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses compatibility of that oriented difference with cylinder prolongation. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in Section 4, are prior coordinates of the same arithmetic state with trajectory memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility without requiring that all four components be evaluated in a single step.

The two routes express different mathematical roles of the same parameter. The first realizes an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded prolongations. One route determines blocks and quotients; the other determines a root through an equation and a uniqueness domain. Their identification is formulated afterward in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination, and compatibility of the two.

8.1 Positional determination through integer dodecaphase closure

8.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase record by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase record is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation is

$$s_j = P_j + E_j - \Phi_j - K_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (8.1)$$

Thus $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 8.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (8.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no residual sign choice: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of closure are not anomalies; they are the integer information needed to maintain the oriented equality.

8.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining coordinate P preserve their distinct antecedents. Sharing this reading does not identify them as regions or histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, obtained block, and outgoing carry is shown. The table is a reproducible integer computation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 11: Initial closure with zero right boundary. The dependency propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\begin{aligned} \alpha_{12}^{(0)} &= \frac{7297352569283800997285105472380663}{10^{36}} \\ &= 0.007297352569283800997285105472380663. \end{aligned} \quad (8.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

8.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (8.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate gives a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 8.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (8.4)$$

Proof. Multiplying (8.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 11, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions when passing to the complete section.

Local inversion is exact as well:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , blocks A , and the complete carries, it recovers the dodecaphase record. Concatenation additionally recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe starting from alpha and manufacturing the record afterward. The direction of construction remains signed record, dodecaphase record, closure, and output.

8.1.4 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphase record and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 8.3 (Existence and uniqueness of complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (8.5)$$

converges absolutely. On the domain of the obtained blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no eventually constant 999 tail, and a unique bounded integer sequence (c_j) satisfying (8.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Furthermore, $R_j(A) \in [0, 1)$, so $-3 < c_j < 2$, placing all carries in $\mathcal{C}$. The tail-shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique, and once it is fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite normalization procedure propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it gives favorable digits, nor is all-depth compatibility inferred from observing a long run.

The initial comparison shows why this precaution matters. Stabilized prolongation gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The zero-boundary window ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

8.1.5 Operative definition and complete additive equation

Definition 8.4 (Alpha coordinate of closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (8.6)$$

where (A_j, c_j) is the compatible canonical normalization of (8.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (8.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base thousand arises from the cubic chart, and period twelve from the dodecaphase record. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determinate algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, signed record, orientation, and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the signs of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. It is useful to distinguish the two consecutive stages of positional division by means of a common notation. The balance before quotient transport is

$$Z_j := P_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (8.1) denotes the sum that already includes the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

In (9.2), the letter s_j locally denotes the balance before that contribution, namely Z_j in the present notation. The two formulas therefore express the same Euclidean division with differently named stages. The incidence configuration H_{α}^- uses the signs of the balance Z_j in the initial window; the determination of the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic record originate in the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on reading depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (8.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window with that of its continuation, both expressed in the same coordinates.

Proposition 8.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the records in the domain under consideration,*

$$\alpha_A - a_N^{(t)} = B^{-N} (R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (8.8)$$

The quotient of the complete continuation is $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^{\infty})} < B^{-N},$$

so that the resulting window is the canonical prefix of order N .

Proof. Telescoping (8.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, each incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the series decomposition proves (8.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For complete canonical normalization, $c_{N+1}^{\infty} = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. The integrality of the quotient implies $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^{\infty})} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Computation over the five boundaries preserves this information as a finite collection of possibilities until the blocks on the left stabilize. Quantitative contraction follows from depth and tail transport; its object is the normalization of the coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition exhibited in §8.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation yields exactly $T_{7;9}$; degree 9 is the final component of this domain. The identity $N^3 = 0$ explains why the resolvent is expressed as three summands. To describe a continuation to higher degrees, one must additionally specify the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus perform different mathematical functions.

8.1.6 Constructive determination and the function of the dodecaphasic record

The preceding definition makes it possible to specify what it means to determine α from the structure. The domain of the operation contains four records of known provenance: the three regional sequences and the integral phase record. Normalization additionally incorporates a boundary condition compatible with continuation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 8.3 extends this determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not participate in selecting any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations, and provenance of the coefficients; the latter would be a statement about polynomial relations. The result used here is the former. The construction determines a coordinate through records produced by APP-TRIT-TPK and subsequently permits verification of the relations it satisfies. This order makes it possible to use an already generated coordinate in a later operation without turning it into an external parameter.

The function of the record K is more precise than that of a numerical correction added to a formula. Its twelve positions originate in the reversible transformation of the signed record. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that record, while the transverse differences and total sum allow another reconstruction. In periodic evaluation, the order of the positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation therefore form part of the definition of the number occurring in (8.7).

Alpha normalization composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented balance of blocks is formed; then that balance is represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows a support to be extracted subsequently. The second allows prefixes of the real coordinate to be formed. Both readings remain related because the state preserves the data before and after normalization. The signed support and the positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. The interior quotients cancel in pairs when a prefix is evaluated, while the endpoint terms remain in the equality. This cancellation is a property of the weighted sum; it neither removes the quotients from the state nor allows them to be discarded before the sum is performed. The right boundary of a window receives the contribution of the continuation. For this reason, evaluating an isolated window and restricting a continued section are different operations. The change of the final block from 663 to 662, proved above, provides an explicit realization of this difference without altering the already stabilized blocks to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible continuations; normalization transmits the integer information at their boundary from positions of lower weight towards those of higher weight. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fibre is verified by reconstructing K from the other records and the complete quotients. This inversion certifies the consistency of the original operation; the genealogical direction remains the one that first produces the record and then the alpha coordinate.

The second determination has complementary content. In it, the immediate object is a vacancy balance, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of records; the analytic route studies the vanishing of an operator. Presenting both requires precisely this difference in function to be retained. Their relationships with the common state must be expressed through the refinement maps and the comparison of their complete realizations, as developed in the corresponding subsection. Agreement at a finite approximation constitutes a localized check; the identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains alpha's place in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electronic construction in Article I uses this coordinate as a factor in its composition. The exceptional extension receives the records and their supports, in which incidence information is preserved. These two continuations return to a common provenance and develop different mathematical functions: evaluating a central section and constructing a boundary configuration. Developing each makes it possible to follow the structure through the equation, rather than restricting the comparison to its scalar value.

8.2 Analytic determination through the vacancy structure

8.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (8.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

From its determinant one extracts

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the thousand chart gives

$$v_- = [1000\delta] = 45, \quad v_+ = [1000\delta] = 46.$$

These integers are typed antecedents of the truncation's evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch additionally coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the preceding chapter's record gives $r_{\text{dod}} = 11$, and the phase semicrown provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (8.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order prolongation. The pair 45/46 crosses the separation between the two: the upper branch occupies degree six, and the lower reappears in degrees eight and nine.

The factorization of (8.10) is formulated for a state preserving origin, channel order, and orientation. Naturality with respect to morphisms preserving those data is not equivalent to uniqueness among readers that do not preserve them. The result identifies the reader associated with the typed state and the factorization of its coefficients; its domain of naturality consists of morphisms preserving origin, channel order, and orientation.

8.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits an especially explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 8.6 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

Evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (8.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45, and 495 are not decimal fits: they arise from the semicrown, APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only cycle symmetry and replacing this operator by another symmetric operator does not necessarily preserve the coefficients.

8.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$P_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = P_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

Proposition 8.7 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. Directed rational evaluation provides the isolation*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (8.12)$$

Proof. The internal-coordinate bounds $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms exceeds 0.0626, so the operator value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. The narrower interval (8.12) comes from directed rational evaluations with opposite signs. To make them explicit, for $t > 0$ and $z = (t - 1)/(t + 1)$, we use

$$\log t = 2 \sum_{j=0}^{m-1} \frac{z^{2j+1}}{2j+1} + R_m, \quad |R_m| \leq \frac{2|z|^{2m+1}}{(2m+1)(1-z^2)}.$$

Apply this with $t = 10/9$ and $t = 10$, $m = 520$, and divide the first interval by the second to bound δ . The generated interval for π is transported through the rational operations of B_3 ; evaluation at the two endpoints of (8.12) gives a negative and a positive sign, respectively. The executable supplement reproduces these exact operations. The existence and uniqueness proof does not use that interval as data to select the coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. A truncation root may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows control of root movement under perturbations; it does not annul those perturbations.

8.3 Compatibility of the dodecaphase and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphase representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same APP-TRIT-TPK enriched state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below does not follow from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This distinction preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the record K are prior outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\hat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states extending the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves residue, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, record, memory, survival, provenance, and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$\mathcal{U}_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6,r}^{\alpha, \rightarrow} = \text{graph}(\mathcal{U}_r^\alpha).$$

The functional character of these extensions, with the seed R_{36} fixed, produces a unique compatible section: phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \hat{\Omega}_\alpha := \lim_{\leftarrow r \geq 36} (\hat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6,r}^{\alpha, \rightarrow}). \quad (8.13)$$

8.3.1 Dodecaphase closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad T_{\alpha, N}(s) := (A_1, \dots, A_N), \quad s \in \mathcal{X}_\alpha^{\text{enr}},$$

where the blocks A_j are obtained through (8.1) with the compatible boundary $c_{N+1}^\infty = [R_{N+1}(Z)]$. For $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be restriction to the first N blocks.

Theorem 8.8 (Complete dodecaphase output and projective compatibility). *The collection $(T_{\alpha, N})_{N \geq 1}$ is compatible:*

$$\rho_{N', N} \circ T_{\alpha, N'} = T_{\alpha, N} \quad (N' \geq N). \quad (8.14)$$

Its canonical cylinders

$$C_N^A(s) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_\alpha \quad (8.15)$$

are nested, have diameter at most 1000^{-N} , and satisfy

$$\bigcap_{N \geq 1} C_N^A(s) = \{\alpha_A(s)\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (8.16)$$

Consequently, the first route expresses a unique infinite section, denoted by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 8.3 provides the unique canonical expansion and the unique bounded carry sequence. The local identity (8.3) and its telescoping form (8.4) preserve the remote boundary; Proposition 8.5 proves that restricting an extended execution recovers precisely the previous prefix. This gives (8.14). Membership of the complete sum in every cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (8.7) identifies this point with the expression in (8.16). All its coefficients and signs are fixed before α is expressed. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^\infty = -1$. In particular,

$$\rho_{36, 12}(\alpha_{A, 36}) = \alpha_{A, 12} \quad (8.17)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

8.3.2 Exact analytic characterization of order nine

Let

$$\mathbb{K} := \mathbb{Q}\left(\pi_{\text{HMT}}, \log_{10} \frac{10}{9}\right), \quad E_9(x) := E_{21/22}^{(9)}(x) = B_9(x) - \delta.$$

Equations (8.10) and (8.11) construct $B_9 = P_6 + T_{7:9}$ from the state's signature and the nilpotent resolvent; Proposition 8.7 proves that E_9 has a unique simple root $\xi_9 \in I_\alpha$, isolated by the rational interval (8.12).

Finite bidirectionality can be expressed without introducing an analytic tail. Define

$$B_9(x) := \delta + \frac{7}{4}x^2 - \frac{x^4}{20} + \frac{2x^5}{21} + \frac{x^6}{46} + \frac{x^7}{120} - \frac{x^8}{45} - \frac{2x^9}{495}, \quad (8.18)$$

$$q_x(T) := xT^2 - B_9(x)T + x^3, \quad x \in I_\alpha = (0, 10^{-2}), \quad (8.19)$$

$$J_x : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \quad J_x(T) := \frac{x^2}{T}. \quad (8.20)$$

Proposition 8.9 (Finite quadratic equivalence and involution). *For every $x \in I_\alpha$ and every $T \in \mathbb{R}^\times$,*

$$\frac{q_x(2\pi_{\text{HMT}})}{2\pi_{\text{HMT}}} = B_9(x) - \delta = E_9(x), \quad (8.21)$$

$$J_x(J_x(T)) = T, \quad T^2 q_x(J_x(T)) = x^2 q_x(T). \quad (8.22)$$

Let $Z_x = \{T \in \mathbb{R}^\times : q_x(T) = 0\}$. Consequently, $J_x|_{Z_x} : Z_x \rightarrow Z_x$ is involutive. When $E_9(x) = 0$, it exchanges the roots $2\pi_{\text{HMT}}$ and $x^2/(2\pi_{\text{HMT}})$. This is an exact involution on the scalar fibre of the order-nine jet; it does not act on the enriched state.

Proof. Dividing $q_x(T)$ by T and substituting $T = 2\pi_{\text{HMT}}$ reproduces the definition of $B_9(x) - \delta$ term by term. The first identity in (8.22) is immediate. For the second,

$$T^2 q_x(x^2/T) = x^5 - B_9(x)x^2T + x^3T^2 = x^2 q_x(T).$$

Thus J_x preserves the root set and, by Viète's formula, exchanges the two roots whose product is x^2 . $\square$

The finite decimal witness thus retains its exact force:

$$\text{Tr}_9(\xi_9^{-1}) = \text{Tr}_9((\alpha_{12}^{(0)})^{-1}) = 137.035999084. \quad (8.23)$$

Equality (8.23) certifies the finite characterization and the preceding involution. It does not turn the root of a truncated polynomial into the root of every compatible extension.

8.3.3 The jet alone does not determine an analytic extension

Let

$$j_9 : \mathbb{K}[[x]] \longrightarrow \mathbb{K}[x]/(x^{10})$$

be truncation, and for $h \in \mathbb{K}[x]$ set

$$E_h(x) := E_9(x) + x^{10}h(x). \quad (8.24)$$

All these polynomials satisfy $j_9(E_h) = j_9(E_9)$.

Theorem 8.10 (Non-uniqueness of the root from the jet). *There is no map*

$$r_9 : \mathbb{K}[x]/(x^{10}) \longrightarrow I_\alpha$$

that assigns, from the jet alone, the positive root of every polynomial extension of E_9 having a unique root in I_α . In particular, E_0 and $E_{1/729}$ have the same jet and distinct positive roots.

Proof. The bound established in the proof of 8.7 gives $E'_9(x) > 6.24$ on $[0, 10^{-2}]$. For $t \geq 0$, write $E_t(x) = E_9(x) + tx^{10}$. Then

$$E'_t(x) = E'_9(x) + 10tx^9 > 6.24,$$

whereas $E_t(0) = -\delta < 0$ and $E_t(10^{-2}) > E_9(10^{-2}) > 0$. Each E_t therefore has a unique positive root $\eta(t) \in I_\alpha$. For $t = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0;$$

since $E_{1/729}$ is strictly increasing and changes sign only once, its root satisfies $\eta(1/729) < \xi_9 = \eta(0)$. However, $j_9(E_{1/729}) = j_9(E_0)$. A map that factored the root through j_9 would have to assign the same point to both polynomials, a contradiction. $\square$

The variation is not merely existential. The implicit function theorem applies because $\partial_x E_t(\eta(t)) > 6.24$, and gives

$$\eta'(t) = -\frac{\eta(t)^{10}}{E'_9(\eta(t)) + 10t\eta(t)^9} < 0. \quad (8.25)$$

Thus coefficients beyond degree nine are proved to change the root even while leaving the entire jet constructed through that degree invariant. This conclusion does not deny the tail produced by the enriched state: it proves only that the root does not factor through the bare truncation j_9 . Choosing a tail h from an external conventional root would be circular. The completion below consults neither that root nor a metrological table: it directly evaluates the regional and terminal coordinates of the state. Since that evaluation agrees algebraically with the integer representation, the result is a correlated representation, not an assertion of independence between two selectors.

8.3.4 Analytic representation of the alpha coordinate

The preceding analysis of the jet requires the tail to be exhibited: merely naming a complete reader is insufficient. Let $s \in \mathcal{X}_\alpha^{\text{enr}}$. Before decimal normalization, the same state already expresses the pre-coordinate

$$a(s) := p(s) + e_0(s) - f(s) - \kappa_{\text{per}}(s) \in I_\alpha, \quad (8.26)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$, and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. Conventional α is not introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal record. The integer closure applies the normalization in (8.1) to (8.26); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it contains no edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows the two representation names to be identified subsequently.

Write $q = 1/729$, let F_s be the ordered field generated by the coordinates of s , and set $E_{9,s}(X) = E_{21/22}^{(9)}(s; X)$. Define the linking coefficient

$$\lambda(s) := -\frac{E_{9,s}(a(s))(1 - qa(s)^3)}{a(s)^{10}}. \quad (8.27)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (8.27) must be read with its exact type: once the state's first representation has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$, set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (8.28)$$

Thus the coefficient seed has no freedom once the state and the class of q -geometric extensions are fixed. This uniqueness is relative to the specified class; it does not assert that the order-nine jet alone selects a tail from all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (8.29)$$

Consequently, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (8.30)$$

This is the rule that computes every coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (8.31)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (8.32)$$

To formulate the projective system without implicit types, fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{F}_M := \prod_{s \in \mathcal{X}_\alpha^{\text{nr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (8.33)$$

For $M' \geq M$, restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (8.34)$$

Proposition 8.11 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

For $M' \geq M$, truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (8.35)$$

Projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (8.30) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the previous polynomial exactly. Compatibility for $M' \geq M$ follows by composition. Thus the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

8.3.5 Uniform convergence, a simple root, and rational intervals

The regional cylinders and periodic record provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified nonadic prefix of one thousand digits. The datum used is not the rational P_c treated as the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}_c}^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively on these closures. If P_π, P_e, P_φ denote the three prefixes, this gives

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (8.36)$$

The natural interval extension of (8.27) then transports A_{1000} , the cylinder of π_{HMT} , and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\frac{7297352569283800997285105472380662}{10^{36}} < a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \quad (8.37)$$

$$-8.711 \times 10^{-6} < \lambda(s) < -8.709 \times 10^{-6}.$$

These inequalities use certified prefixes with their rational tail bound; they receive no alpha table. The package verifier reproduces them from the three nonadic outputs and the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (8.38)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1-x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (8.39)$$

The derivative tail beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (8.40)$$

Theorem 8.12 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (8.41)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two successive terms in the tail is bounded by $r < 1$, which gives (8.38); termwise differentiation and summation of the two geometric series give (8.39) and (8.40). The jet proof establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence the complete perturbation leaves the derivative above 6.239. Finally, substituting (8.27) into (8.32) makes the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

Continuity guarantees a root in the interval, and the second inequality ensures uniqueness. Only after this existence, monotonicity, and the invariant that the bracket contains the root have been established is the mean-value theorem fallback applied. Start with $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$, and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (8.42)$$

When $F_j^+ < 0$, retain the right half; when $F_j^- > 0$, retain the left half. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that every root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (8.43)$$

Indeed, let ξ_j be the root, and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (8.42), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (8.43). The argument includes the case $y_j = 0$: the root then lies at the midpoint, but the algorithm need not recognize that equality in order to retain it.

If the available enclosure does not satisfy (8.42), the source cylinders are first refined and the evaluation repeated; no sign is asserted and no insufficient contraction is accepted. In each of the three branches, the new interval retains the root and has diameter at most $w_j/2$. For every finite target depth, directed refinement makes it possible to satisfy the width bound without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (8.44)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule at twenty-four levels in base one thousand and verifies that the entire outer closure A_{1000} , not merely its lower endpoint, lies within each expressed cylinder. Its negative tests reject an excessively wide enclosure, a reversed enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$, the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (8.45)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and prevents an endpoint lying exactly on the right boundary from being assigned to the cell on its left.

Let $C_N^A(s)$ be the integer cylinder in (8.15). Recursively choose $m(N) > m(N - 1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (8.46)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is non-empty because both factors contain $a(s)$. Since the cylinder is half-open, u_N need not belong to $I_{B,N}$. Evaluation at this endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (8.47)$$

Bound (8.38) allows a truncation to be chosen that preserves the sign at every endpoint where the complete function is non-zero. If an endpoint coincides with the root, the identity for the complete function and the established membership are used; the same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

On the positive domain under consideration, where $\lambda(s) < 0$, evaluation at the root gives

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Accordingly, certification of the weak sign inequalities in (8.47) concerns $E_{s,\infty}$. Evaluation uses the complete function or the truncation together with its signed remainder. This covers both the included infimum and the potentially excluded supremum of the cylinder.

Proposition 8.13 (Compatibility squares and common cylinders). *Restriction of words and of analytic representations commutes:*

$$\begin{array}{ccc}
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N'}} & \mathcal{W}_{N'} \\
 \parallel & & \downarrow \rho_{N',N} \\
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N}} & \mathcal{W}_N
 \end{array}
 \qquad
 \begin{array}{ccc}
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\
 \parallel & & \downarrow \tau_{M',M} \\
 \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M.
 \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and explicitly satisfies $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square expresses compatibility of integer normalization; the second is Proposition 8.11. The inclusion follows from Definition (8.46), and non-emptiness and the sign inequalities follow from Theorem 8.12. No step identifies the root of a truncation with the root of an arbitrary tail. $\square$

We now define, with the prefix reader made explicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (8.48)$$

Theorem 8.14 (Projective equality of the two correlated representations). *For every N , the integer route and the analytic route express the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (8.49)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (8.50)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 8.12 proves that the analytic chart represents the same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Their prefixes therefore coincide for every N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The analytic function constitutes a correlated representation of the same state, not a second derivation causally independent of the integer closure. This precision does not weaken the proved equality: it identifies its exact mechanism, prevents the jet from being assigned a tail it does not determine, and avoids promoting a conventional value to a generator.

9 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves the two sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean reading contracts compatible cylinders. The reading now under study instead preserves the incidence of positions and their transport between frames. Both start from the same arithmetic state with trajectory memory. An exceptional geometry is not reconstructed from the final decimals of π , φ , e , or α .

This point clarifies the meaning of the prolongation $\alpha/K \rightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic record and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered record K can be recovered from its rational constant κ under the previously fixed base, length, and origin conventions. Projection onto a support, by contrast, preserves only the selected incidences. For its part, K_{ph} denotes a different, reduced-memory reading, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the resulting objects through the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate those recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

9.1 Structural function of K and α in exceptional incidence

The link between numerical determination and exceptional realization admits an operational description. In the arithmetic reading, K acts in the normalization of regional records. In the incidence reading, its components select the support

$$B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The same ordered record therefore participates in two specified functions: a positional composition and an incidence selection. Its rational realization preserves the record when base, length, and origin are fixed, while passage to the support retains the information needed for the combinatorial flag.

The signed record preceding normalization of α supplies the hexad H_{α}^- . The intersections of this hexad with the regional supports and APP incidence determine the marked point. That point individualizes the norm-54 vector used in the lattice neighbor. Thus the orientation entering arithmetic normalization acts again in geometric selection. The relation does not require deriving a lattice from a decimal approximation: it uses the signed data preceding its evaluation.

The exceptional realization successively preserves different types of relation. The code preserves linear combinations; the design preserves incidences between supports; gluing preserves discriminant classes; the lattice preserves its bilinear form; the vertex algebra preserves grading and the corresponding products. Their automorphism groups are defined with respect to these structures. The appearance of the Monster at the end of the construction expresses a property of the algebra $V^{\natural}$ obtained through the cited procedure, and Moonshine relates the graded traces of that action to modular functions.

This hierarchy gives a concrete mathematical meaning to the linking function attributed to K and α . The former is a reversible integral record with positional and incidence realizations; the latter is the arithmetic-compatibility coordinate whose oriented record also enters the marked incidence. The definitions acquire content through these operations and the identities presented in this chapter. The incidence development is a substantive part of the article's argument: it shows which relations persist when the construction is no longer evaluated as a real coordinate and is instead realized as an algebraic structure.

Articulation of regional coordinates and closure incidence. The role of this prolongation is understood through a distinction between three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the article's structural thesis: a numerical realization has an operator provenance, and certain data from that provenance additionally admit a combinatorial and lattice realization. The equations allow both paths to be followed with their domains and the information remaining in each.

The starting point is the terminal state with its compatible prolongations. It preserves the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scale regions are determined in that domain before their readers establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K arises from the integral transformation proved in §7.8:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (9.1)$$

The congruence defining $\mathcal{L}_H$ ensures integrality of the inverse. Thus K is a reversible coordinatization of the signed record in the relevant sublattice. Its positions preserve phase order; its differences D_3K, D_4K preserve transverse relations; the sum $Q(K)$ determines the uniform component. That structure explains why the record enters both numerical composition and the selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves membership of each position in a class defined by a threshold and groups records with the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

Alpha normalization and its signed configuration. The coordinate of α is determined in the positional route through composition of the regional blocks and the dodecaphase record. Before positional division, the oriented combination has components $s_j = P_j + E_j - \Phi_j - K_j^{\text{per}}$. Normalization satisfies

$$A_j = s_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (9.2)$$

This identity simultaneously preserves the normalized block and the contribution of the transported quotient. The limit of compatible prefixes determines the scalar coordinate of closure. The signed configuration, in turn, preserves which positions of the combination before normalization have a negative defect. In the initial window, that configuration is the vector s_0 of (9.12), and its negative support is H_{α}^- .

Two contents of the same operation are thereby defined. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the balance of records before normalization. Its incidence reading remains available because the state preserves that balance together with the normalized result. This articulation corresponds to the proved positional route; comparison with the analytic vacancy operators preserves the precise scope established in the chapter on α .

From ternary coding to the marked flag. The regions also supply the codes $w_{\pi}, w_e, w_{\varphi}$, and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transformation by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, preserves the nonzero positions. For weight-six words, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. Thus, the sign quotient produces the hexads, whereas the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (9.3)$$

The left-hand side arises from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded supports of propagation and closure. The threshold belongs to the declared block chart and remains explicit in the reader's definition. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes that tetrad to a hexad. The condition of membership in support P_π , together with the intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs bring together the two determinations: that obtained through the sign of s_0 and that obtained through incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (9.4)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π recover their internal differentiation here. They share w_π as a visible ternary code but preserve distinct emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to the lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and chart order individualizes the three positive regions. This correspondence preserves the regional roles and distribution $432 + 4 \cdot 144 = 1008$, while also realizing the incidence partition $1 + 4$.

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its role is to distinguish one component among the twelve entering the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights, and distance of the code translate, respectively, into integrality and determinant properties, parity, and norm bounds for the glue classes. Here the symbolic information of the code performs a role that the isolated support no longer performs.

The A_2 metric and positive radial chamber are fixed in the lattice development. Within that chamber, the congruence needed for an even index-three neighbor has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the declared chamber. The subscript α records the incidence provenance of the selection: its content is the frame that entered closure and now individualizes a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adjoining the class $v_\alpha/3$ produces the neighbor N_α of (9.17). This modification preserves rank and recovers unimodularity and evenness; congruence selection eliminates the preceding roots, and local bounds exclude roots in the new classes.

Identification with Leech appears after those verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as glue data.

Grading, unit, and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, central extension of the lattice, vertex product, and sector twisted by the lift of negation. The result is the algebra V_{HMT}^h of (9.18), whose automorphism group is isomorphic to the Monster. Its action domain is thus fixed by an algebraic structure with grading and product. Borchers’ Moonshine theorem subsequently determines the graded traces with insertions of those automorphisms [13, 14].

At this level, the unit acquires a precise realization: the weight-zero component is the vacuum line and contributes the coefficient 1 of q^{-1} in the graded character. At the cylinder level, refinement maps preserve the constant function through precomposition. Every conservation statement therefore has its object and its map. The development of the proofs makes this continuity of construction visible, from arithmetic operations and memory to incidence, metric, and vertex products.

This articulation places K and the closure of α in the main argument. K reversibly preserves an integral coordinate of the history; normalization of α composes the regional records with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients, and realizations that add metric or algebraic structure. A provable relation between structure, equation, and value is thus expressed, preserving the attribution of each construction and the scope of its theorems.

9.2 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript indicates that the declared initial frame and its transport are preserved, not that a parameter is fitted against an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this reading. The return of the phase resets neither the frame nor the memory of the state. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a six-trit block b , the Archimedean reading takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class belongs first to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence reading we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (9.5)$$

Words are written as row vectors. If $A^f = fAf^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{Af}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (9.6)$$

The calibrated codomain retains f together with the encoded word. We therefore define

$$Q_n(x) = ((f_j, \gamma_{f_j A_W f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (9.7)$$

Proposition 9.1 (Compatibility and prolongation of the incidence reading). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves the visible word exactly, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (9.7) agree, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery claim is the projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not by itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are also transported. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

9.3 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, with the order $\infty, 0, 1, 2, 3, 4$ declared, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ at the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is chart data, not a new numerical input into the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three yields

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (9.8)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. They are distinct realizations of a common operator relation, with their corresponding domains.

Definition 9.2 (Code of the dodecaphasic chart). The ternary code associated with (9.8) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 9.3 (Self-duality, distance, and the Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (9.9)$$

The set $\mathcal{H}$ has 132 elements and constitutes an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by the symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (9.9); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two weight-six words have the same support, one of the signs 1, -1 occurs at least three times among the six ratios of their nonzero coordinates. Subtracting the corresponding multiple would produce a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two distinct supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [15]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

9.4 Dodecaphasic flag associated with K and α

In the frame inherited from the dodecaphasic record, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (9.10) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_W} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a weight-nine word; the other three are hexads. Through its high-block reading, the record supplies

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (9.11)$$

Here the integer 729 marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in Definition 9.2. In this particular record, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$s_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (9.12)$$

Its negative support is

$$H_{\alpha}^{-} = \{4, 5, 6, 7, 9, 10\}. \quad (9.13)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are retained in the state preceding its numerical expression.

Proposition 9.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (9.13) is the unique hexad H satisfying*

$$B_K \subset H \subset P_{\pi}, \quad (|H \cap H_e|, |H \cap H_{\varphi}|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed reading and the incidence reading select the same flag $B_K \subset H_{\alpha}^{-}$.

Proof. The four hexads containing B_K are obtained by adjoining, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_{π} . For the second, the intersection with H_{φ} has size three, and that with H_{APP} has size three; it does not meet the last condition. For the third, the three intersections have sizes 4, 3, 2. This last pair produces precisely the negative support of (9.12). $\square$

It is useful to retain both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_{\alpha}^{-}), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_{\pi}, H_e, H_{\varphi}, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under a simultaneous relabeling of the design. The marked class, not the position numbers taken in isolation, is the transported object.

Lemma 9.5 (Origin selected by the marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_{\varphi}) \setminus (H_{\alpha}^{-} \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The subtracted union contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor once the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

9.5 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (9.14)$$

Let $V_{11} = \mathbf{1}^{\perp} \subset \mathbb{R}^{12}$. The two Euclidean realizations are subsequent languages for the same finite design here.

Proposition 9.6 (Isometry of the dodecaphasic screen). *We have*

$$I^{\top} I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the automorphisms of the design.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^T I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral reading. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 9.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This yields the identity. The rectangular term vanishes on V_{11} . $\square$

There is indeed a material intertwiner here: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This statement does not identify all the operators acting on these spaces solely by their trace.

9.6 The Witt star and the generation of M_{12}

Definition 9.8 (Star on a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and exchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

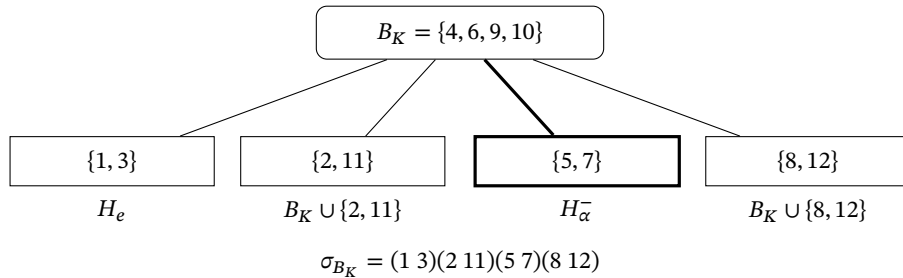


Figure 7: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 9.9 (The collection of stars). *The $\binom{12}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. The preservation claim is a finite check on the 132 hexads of Definition 9.2: for each four-element set the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In the lexicographic enumeration, it suffices to add successively six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and all 495 stars belong to the final group. The stars are constructed by Definition 9.8; the enumeration starts from the identity and, in lexicographic order of four-element sets, adds the first star outside the subgroup already generated, then takes its closure under composition. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , still less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 9.10 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (9.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

9.7 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a weight-twelve word. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 9.11 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the permitted weights, with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by selecting five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads on B give four distinct octads. The fifth cannot meet D in six points, since it would be another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A hexad-preserving chart $\ell : [12] \rightarrow D$ identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. The domain of this injectivity is the hexads, not the record K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible tritic word $w_\pi = 010211$, but not the same regional genealogy. The three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has a constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative region to $\iota_D(H_{\bar{\alpha}})$, and the three positive regions to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads on $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart order; incidence does not determine it without that datum. The precise claim is therefore a marked correspondence of regions, compatible with their roles and with the flag, not a canonical bijection inferred from the cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalog checks, not substitutes for the map ι_D or the regional chart.

9.8 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations thus remain distinct: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ with Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

Represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with this discriminant. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (9.15)$$

Theorem 9.12 (Ternary gluing). *The object (9.15) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes stable under addition. Self-orthogonality annihilates the discriminant pairing: the latter is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (9.9) are multiples of three, so the lattice is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity follows from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Hence a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No norm-two roots occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

9.9 Incidence selection of the rootless unimodular neighbor

The label $o_* = 1$ of the flag is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The least norm in the chamber is obtained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \quad (9.16)$$

The word “least” refers to this positive chamber. It does not assert minimality throughout the residual class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not an omissible conclusion.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \quad (9.17)$$

Theorem 9.13 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \pmod{3}$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , the integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

Hence it is even.

It remains to exclude roots, the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and the determinant, and eliminates roots through a local bound. Only then does it apply the classification theorem. The dependence on α is preserved in the flag selecting o_* and in the vector (9.16); the scalar value of the constant is not converted into a length of the lattice.

9.10 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the central extension of the lattice. The cocycle and the lift of the isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^{\natural} = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (9.18)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 9.14 (Prolongation to the Moonshine module). *The algebra (9.18), constructed on the neighbor (9.17), is a realization of the Moonshine algebra $V^{\natural}$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^{\natural}} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \dots, \quad q = e^{2\pi i\tau}. \quad (9.19)$$

Proof. Theorem 9.13 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. The Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice then apply. They identify the orbifold with $V^{\natural}$ and establish its character and its automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not inferred from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof of the FLM construction in this article [13]. Moreover, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It does not introduce π or e as inputs into the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes an alternating sign. Hence

$$\text{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character of $V^{\natural}$ without insertion. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

9.11 Graded Moonshine traces and representation of the Monster

For $g \in \text{Aut}(V_{\text{HMT}}^{\natural})$, the trace with insertion is

$$T_g(\tau) = \text{Tr}_{V_{\text{HMT}}^{\natural}} g q^{L_0-1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [14, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which one first guesses J and then infers the Monster.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^{\mathfrak{h}} = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A morphism of vertex algebras preserves the vacuum vector. This property provides the technical meaning of preservation of the unit at this level; the unit of cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 in the Monster representation. The other expressions have lattice-theoretic or arithmetic readings. Their equality does not automatically identify their summands with TPK regions, words, or routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (9.18), and a choice of isomorphism with $V^{\mathfrak{h}}$ transports its automorphism group. This does not construct an action of the Monster on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^{\mathfrak{h}}}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present claim does not remove the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different promotion, namely an action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, s_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_\alpha \longrightarrow V_{\text{HMT}}^{\mathfrak{h}}. \end{aligned}$$

At the incidence level, the stars generate M_{12} , and the residual chart realizes the regional 4+1 structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of this section is therefore an assembled derivation: the finite and lattice proofs make the passage from dodecaphasic incidence to the hypotheses of the recognition theorems effective. This result concerns the specified constructions and hypotheses; it does not add experimental validation to the comparison of constants.

10 Reconstruction of quantum principles from the genealogical state

The Hilbert-space formulation organizes quantum mechanics in terms of states, self-adjoint observables, unitary evolution, tensor composition, and a probability rule for outcomes [12]. HMT-MD constructs a layer prior to that representation: a compatible section, its observable route, the memory of transport, and the reader that expresses an outcome. Hilbert space is not replaced; it appears as the codomain of an explicitly specified linear realization.

The typed sequence is

$\begin{aligned} \text{genealogical section} &\longrightarrow \text{linear realization} \longrightarrow \text{phase and superposition} \\ &\longrightarrow \text{observable and evolution} \longrightarrow \text{probability and measurement} \end{aligned}$
--

Each arrow preserves its domain. In particular, the state vector does not replace the TPK section, just as the real value does not replace the numerical genealogy that produces it.

10.1 Genealogical state and Hilbert realization

Definition 10.1 (Realized genealogical state). A genealogical state is a quintuple

$$\mathfrak{G} = (x_\infty, \Gamma, \mathcal{F}, \mathcal{M}, \mathcal{R}), \quad (10.1)$$

where $x_\infty \in \Omega_\infty$ is a compatible section, Γ is its observable route, $\mathcal{F}$ is the collection of realizations, $\mathcal{M}$ is the memory of the lifts, and $\mathcal{R}$ is the specified reader. Let $\mathbf{P}_{\text{TPK}}$ be the category whose objects are prepared TPK sections, finite or coinductive, and whose morphisms are recorded routes compatible with their boundary data. In particular, depth restriction is a morphism

$$r_{n+1,n} : x_{n+1} \longrightarrow x_n.$$

Let $\text{Hilb}_{\mathbb{R},J}$ be the category whose objects are pairs $(\mathcal{H}_{\mathbb{R}}, J)$, with $\mathcal{H}_{\mathbb{R}}$ a real Hilbert space and J orthogonal, $J^2 = -I$, and whose morphisms are bounded real-linear maps T intertwining the complex structures, $TJ_1 = J_2T$. A quantum realization is a functor

$$\mathcal{Q} : \mathbf{P}_{\text{TPK}} \longrightarrow \text{Hilb}_{\mathbb{R},J},$$

and a *pointed realization* is the pair $(\mathcal{Q}, \Psi)$ assigning to each prepared section x a nonzero vector $\Psi_x \in \mathcal{Q}(x)$, compatible with the restrictions of the inverse system.

With the convention of linearity in the second argument, the associated Hermitian form is written

$$\langle u, v \rangle_{\mathbb{C}} = \langle u, v \rangle_{\mathbb{R}} - i \langle u, J_E v \rangle_{\mathbb{R}}.$$

Choosing the chart $J_E \leftrightarrow i$ realizes the complex structure that the formal kernel and its development construct as an internal square root of $-I$. For prefixes of depth n , set $\mathcal{H}_n := \mathcal{Q}(x_n)$ and

$$p_{n+1,n} := \mathcal{Q}(r_{n+1,n}) : \mathcal{H}_{n+1} \longrightarrow \mathcal{H}_n.$$

Compatibility between restriction and realization requires

$$p_{n+1,n} \Psi_{n+1}(x_{n+1}) = \Psi_n(x_n). \quad (10.2)$$

If $r_{\infty,n} : x_\infty \rightarrow x_n$ is the coinductive restriction and $p_n := \mathcal{Q}(r_{\infty,n}) : \mathcal{H}_\infty \rightarrow \mathcal{H}_n$, a limit $\mathcal{H}_\infty$ realizes the complete section when $p_n \Psi_\infty(x_\infty) = \Psi_n(x_n)$ at every depth.

Principle 10.2 (Realized quantum state). The pure state associated with a section is represented by the ray of the nonzero vector $\Psi = \Psi_\infty(x_\infty)$; a statistical preparation is represented by a positive, trace-one density operator on the same realization. Global-phase identification acts in the representation: the section and its record remain in the genealogical domain.

10.2 Evolution, Hamiltonian, and Schrödinger equation

Genealogical evolution preserves composition of transports. Its realization satisfies

$$U(t_3, t_2)U(t_2, t_1) = U(t_3, t_1), \quad U(t, t) = I.$$

When it preserves the inner product and satisfies $U(t)J_E = J_E U(t)$, it is orthogonal in the real chart and unitary in the complex chart. For a strongly continuous one-parameter unitary group, Stone's theorem provides a self-adjoint, J_E -linear generator H [11]:

$$U(t) = \exp(-J_E H t / \hbar_{\text{int}}).$$

On the domain $D(H)$, differentiation with respect to time gives

$$\hbar_{\text{int}} J_E \dot{\Psi}(t) = H \Psi(t), \tag{10.3}$$

and the chart $J_E \leftrightarrow i$ yields $i \hbar_{\text{int}} \dot{\Psi} = H \Psi$.

The calendar C_{108} realizes this correspondence in finite form: the self-adjoint Hamiltonian constructed for it has, over t_0 , the unitary one-iteration shift as its exponential. The continuous group additionally requires strong continuity; it does not follow from the periodicity of the calendar alone.

11 Planck constants and the determinantal structure of action

11.1 Determinantal action scale and decimal valuation

The absolute scale and the relative corrector originate in distinct objects. The former uses the local block of the step-three dodecaphase record,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (11.1)$$

The operation uses a local quotient. The three blocks of the full record are not replaced by a tensorization that would multiply the number of sheets of this reader once again.

Lemma 11.1 (Number of sheets of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has value -2 , so their greatest common divisor is two. Since $H_4 H_4^\top = 4I$ and $\det H_4 = -16$, the cofactors are ± 4 ; the third determinantal divisor is four. The fourth is 16. The successive quotients of the divisors 1, 2, 4, 16 are 1, 2, 2, 4. Their product gives the order of the quotient. $\square$

Definition 11.2 (Determinantal action reader). The multiplicative norm of the scalar section α , constant over the sheets of G_H , and one application of the self-scale φ produce

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (11.2)$$

The associated decimal chart determines $\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus arises from the number of sheets of the quotient, once this local reader has been fixed. The factor φ acts once on the basis of the action line. The Smith computation alone would not select any possible functional on the quotient: the multiplicative norm and the self-scale form part of the declared operation.

Theorem 11.3 (Decade of the action section). *Reader (11.2), applied to the already generated HMT coordinates, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (11.3)$$

Proof. The following rational intervals suffice; they are much coarser than the available cylinders:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto xa^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. No SI unit or reference mass enters these inequalities. $\square$

The normalized section begins with $\Sigma_H = 1.0462438381 \dots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal reader fixes the decimal order; the two return coordinates are obtained through the following readers.

11.2 Action sections and dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line, and let $\mathcal{U}_S$ be its positive basis. The fifth-order character and the regional return term have the expressions

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (11.4)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (11.5)$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (11.6)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha). \quad (11.7)$$

These are subsequent readings of the dodecaphase closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas make explicit exactly which readings are needed for the electronic stage. It is not necessary to introduce other functionals of mass theory here.

For $0 < \alpha < 1$, one has $0 < D_A < 1$, so $D_A\alpha < 1$. The rational term in (11.6) preserves the complete sum of the geometric tail

$$\frac{D_A\alpha^7}{1 - D_A\alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (11.8)$$

It is not a truncation that may be silently modified when the precision changes. After the term of index N , the remaining tail is $D_A^{N+2} \alpha^{N+8} / (1 - D_A\alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 11.4 (Action sections and return reader). The pre-return and post-return sections are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (11.9)$$

Whenever $\eta_{\text{ret}} > 0$, define

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (11.10)$$

Proposition 11.5 (Positivity and invariance of the ratio). *In the intervals used in the proof of (11.3), $0 < \eta_{\text{ret}} < H_5$, and therefore $R_{\text{act}} > 1$. The ratio does not change when the basis $\mathcal{U}_S$ is replaced by any other common positive basis.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$, and $\pi > 3$. The root in (11.4) is greater than one, whereas the sum of the negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Likewise,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. Finally, the basis change divides both action coordinates by the same positive factor, which cancels in their quotient. $\square$

Subsequent evaluation gives

$$H_5 = 1.0545718176461544696 \dots,$$

$$\eta_{\text{ret}} = 1.0545718169150390611 \dots,$$

$$R_{\text{act}} = 1.0000000006932817630 \dots.$$

The difference and the quotient are two readings of the same sections, one additive and the other multiplicative. They do not represent two independent Planck constants. The passage from angular action to action per complete turn is expressed, in each section, by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathcal{J}_S$ assigns a unit but does not act retroactively on the decimal order already proved.

Regional return can also be written in an angular chart. This reparametrization is not an additional indispensable dependency: (11.4)–(11.7) directly supply H_5 , η_{ret} , and R_{act} . If angle and counter-angle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

11.2.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object that motivates realization of the angular-action section. Its position in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the decade of the scale. The character H_5 and the regional return then determine the coordinates of the action sections. This separation allows the result needed for the electron to be presented without anticipating the full developments of other sectors of dimensional mechanics.

The provenance of the fifth-order character can be made explicit through invariants already fixed in the support. In the central chart, consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-3 orbit has length 4, and the relevant census quotient is $104976/1944 = 54$. The character coefficients satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (11.11)$$

The assignment of these invariants to degrees 2, 3, 4, 5, with the alternating signs of (11.4), belongs to the definition of the analytic character used. Making it explicit is essential: the collection of cardinalities, considered without the character and without the ordering of its degrees, would admit other expressions. The constructive content lies in the composition of the support, orientation, and declared analytic rule.

11.2.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent prolongation is organized by the transfer D_A and a unit normalization of the determinant line. These rules admit a formulation in formal series before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathcal{C}(z) = 169z^6 + R(z), \quad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating a prolongation increases the degree by one and multiplies its weight by D_A . The first strict prolongation has weight D_A . Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \quad (11.12)$$

Proposition 11.6 (Formal uniqueness of the normalized continuation). *Equation (11.12) determines a unique series, and*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic for $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the ring of formal series. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A , and each subsequent coefficient is D_A times the preceding one. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first prolongation is an indispensable operative datum. If only the degree-six impulse and the concatenation ratio were retained, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would still share both data for every λ . The unit rule fixes $\lambda = 1$ before evaluation is performed. This locates the contribution of each rule to the uniqueness of the reader, and the rational expression preserves the full continuation.

11.2.3 Reduced Planck constant and return of the action section

In the dimensional chart of (11.9), the pre-return section is the model construction intended for comparison with the reduced Planck constant:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S.$$

The post-return section additionally preserves the regional compensation $90\mathcal{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. The passage to h through 2π uses the relation between angular parametrization and a complete turn.

12 Quadratic regimes and temporal readout

12.1 Quadratic algebras of the TRIT and curvature signatures

The primary construction and proof of the three regimes are given in the construction of the quadratic algebras of the TRIT, whose defining relation is presented in (1.5). For $\tau \in \{-1, 0, +1\}$, with balanced image $\bar{\tau} \in \{-1, 0, +1\}$, we have

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau}),$$

and, if J is the class of X ,

$$\begin{aligned} \tau = +1 & : J^2 = -I, \quad \text{elliptic regime,} \\ \tau = 0 & : J^2 = 0, \quad \text{parabolic regime,} \\ \tau = -1 & : J^2 = +I, \quad \text{hyperbolic regime.} \end{aligned}$$

For all three values of τ , the quotient $\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \bar{\tau})$ satisfies precisely the three quadratic identities above. In particular, the internal square root of -1 is realized as an operator on the elliptic sheet; HMT need neither reject nor suppress the complex extension. The parabolic algebra contains a nilpotent, and the hyperbolic algebra decomposes by means of the two idempotents $(I \pm J)/2$.

These identities classify algebras and transports. They do not, by themselves, select a spacetime metric or prove that the physical universe occupies one of the three sheets.

12.2 Temporal TRIT reader on $(-1,0,+1)$: opening, threshold, and return with memory

Once an ordered parametrization of the transitions has been chosen, the active temporal reader is

$$\begin{aligned} +1 & \longrightarrow \text{return, retained memory, and past,} \\ 0 & \longrightarrow \text{evaluation threshold and present,} \\ -1 & \longrightarrow \text{bidirectional opening and future.} \end{aligned}$$

The reader is PROVED WITH AN EXPLICIT STARTING STRUCTURE: its conclusions follow once the TRIT signature, the order of the trajectory, and the transported memory have been specified. It does not turn the set $\{-1, 0, +1\}$ into an equation of motion. The future designates the space of compatible continuations; survival may select a branch or a multisection without erasing the genealogy of the others.

The frequency

$$\omega_0 = \frac{2\pi}{108 t_0}$$

assigns a type to the full-return clock through the decision entropy $I_{\text{TRIT}} = \log 3$ and the Boltzmann transducer. It does not, in isolation, establish a stable subharmonic phase. It does, however, enable construction of the finite reference Hamiltonian below; selection of a robust physical Hamiltonian requires additional conditions.

12.3 Unitary dynamics on 108 states: Hamiltonian, propagator, and period

Fix an orientation $a \in \{\text{pre}, \text{ret}\}$ of the sections constructed in (11.9) and write $\hbar_{\text{int}} = \hbar_a$. The same section is retained in the Hamiltonian, propagator, and subsequent thermal conversion.

Let

$$\mathcal{H}_{\text{clk}} := \mathbb{C}^{108}, \quad \{|j\rangle : j \in \mathbb{Z}/108\mathbb{Z}\}$$

with its canonical Hermitian inner product. Fix $\zeta := \exp(2\pi i/108)$, the unitary shift

$$U_{\text{clk}}|j\rangle = |j+1\rangle$$

and the Fourier basis

$$|\tilde{k}\rangle := \frac{1}{\sqrt{108}} \sum_{j=0}^{107} \zeta^{jk} |j\rangle, \quad 0 \leq k \leq 107.$$

With $t_0 \in \mathcal{L}_T^+$ and $\hbar_{\text{int}} \in \mathcal{L}_S^+$, define

$$H_{\text{clk}} := \hbar_{\text{int}} \omega_0 \sum_{k=0}^{107} k |\tilde{k}\rangle \langle \tilde{k}|, \quad \omega_0 = \frac{2\pi}{108 t_0}. \quad (\text{CLK1})$$

The operator has energy type $[H_{\text{clk}}] = \mathcal{L}_S \mathcal{L}_T^{-1} = \mathcal{L}_E$.

Theorem 12.1 (Exact unitary realization of the 108 calendar). *The operator H_{clk} is self-adjoint and its one-step propagator satisfies*

$$U_{\text{step}} := \exp\left(-\frac{i}{\hbar_{\text{int}}} H_{\text{clk}} t_0\right) = U_{\text{clk}}, \quad U_{\text{step}}^{108} = I. \quad (\text{CLK2})$$

If

$$Z_{\text{clk}} |j\rangle = \zeta^j |j\rangle,$$

then, for any basis state $|j_0\rangle$,

$$\langle Z_{\text{clk}} \rangle_n := \langle j_0 | U_{\text{step}}^{-n} Z_{\text{clk}} U_{\text{step}}^n | j_0 \rangle = \zeta^{j_0+n}. \quad (\text{CLK3})$$

The sequence has primitive period 108.

Proof. Expression (CLK1) is a spectral decomposition with real eigenvalues, hence $H_{\text{clk}} = H_{\text{clk}}^\dagger$. Moreover,

$$U_{\text{clk}} |\tilde{k}\rangle = \zeta^{-k} |\tilde{k}\rangle,$$

whereas the exponential of $\hbar_{\text{int}} \omega_0 k$ over t_0 gives $\exp(-2\pi i k / 108) = \zeta^{-k}$. The two operators agree on an orthonormal basis. The remaining identities follow from $U_{\text{step}}^n |j_0\rangle = |j_0 + n\rangle$ and the primitivity of ζ . $\square$

The corresponding variational formulation, for a curve $\psi \in C^1([t_a, t_b], \mathcal{H}_{\text{clk}})$, unconstrained during the variation, is

$$L_{\text{clk}}(\psi, \dot{\psi}) = \frac{i\hbar_{\text{int}}}{2} (\langle \psi | \dot{\psi} \rangle - \langle \dot{\psi} | \psi \rangle) - \langle \psi | H_{\text{clk}} | \psi \rangle. \quad (\text{CLK4})$$

The action functional is

$$S_{\text{clk}}[\psi] := \int_{t_a}^{t_b} L_{\text{clk}}(\psi(t), \dot{\psi}(t)) dt, \quad (\text{CLK5})$$

with $\delta\psi(t_a) = \delta\psi(t_b) = 0$. Varying ψ and $\bar{\psi}$ independently yields the Schrödinger equation for the finite clock

$$i\hbar_{\text{int}} \dot{\psi} = H_{\text{clk}} \psi \quad (\text{CLK6})$$

together with its adjoint equation [10]. Its propagator sampled at $t = nt_0$ is $\psi_n = U_{\text{step}}^n \psi_0$. Since H_{clk} is self-adjoint, the norm is conserved; normalized initial data remain normalized. The result is exact with respect to the specified H_{clk} , t_0 , $\hbar_{\text{int}}$; it neither selects a physical Hamiltonian nor introduces an experimental frequency.

The theorem 12.1 establishes the unitary dynamics of the calendar with respect to the specified complex completion, t_0 , and $\hbar_{\text{int}}$. Its realization as a finite periodic system is developed in Section 13, which specifies the drive, the order observable, the primitive subharmonic response of period $108T_d$, and finite spectral stability. A macroscopic material realization constitutes a distinct external identification, not a gap in the finite result.

13 Finite HMT time crystal: drive, subharmonic response, and spectral stability

13.1 Drive constructed from the 108-state clock

Let $N = 108$,

$$\mathcal{H}_{\text{clk}} = \mathbb{C}^N, \quad S|j\rangle = |j+1\rangle, \quad Z|j\rangle = \zeta^j|j\rangle, \quad \zeta = \exp(2\pi i/N).$$

In the Fourier basis $\{|\tilde{k}\rangle\}_{k=0}^{N-1}$, set

$$H_{\text{clk}} = \hbar_{\text{int}} \frac{2\pi}{Nt_0} \sum_{k=0}^{N-1} k|\tilde{k}\rangle\langle\tilde{k}|, \quad \exp\left(-\frac{i}{\hbar_{\text{int}}} H_{\text{clk}} t_0\right) = S.$$

This equality fixes the starting clock. To exhibit a nonconstant periodic drive, let $0 < \eta < 1$, $T_d = t_0$, and define, in each period,

$$H_d(t) = \begin{cases} 0, & 0 \leq t < \eta T_d, \\ (1-\eta)^{-1} H_{\text{clk}}, & \eta T_d \leq t < T_d. \end{cases} \quad H_d(t + T_d) = H_d(t).$$

Since the two segments commute, its Floquet propagator is

$$U_F = \mathcal{T} \exp\left(-\frac{i}{\hbar_{\text{int}}} \int_0^{T_d} H_d(t) dt\right) = \exp\left(-\frac{i}{\hbar_{\text{int}}} H_{\text{clk}} T_d\right) = S.$$

Define the Hermitian observables

$$Q = \frac{Z + Z^\dagger}{2}, \quad P = \frac{Z - Z^\dagger}{2i}.$$

Theorem 13.1 (Primitive subharmonic response of the finite HMT time crystal). *For $\rho_{j_0} = |j_0\rangle\langle j_0|$, the order vector*

$$\mathbf{q}_n = (\text{Tr } \rho_{j_0} U_F^{-n} Q U_F^n, \text{Tr } \rho_{j_0} U_F^{-n} P U_F^n)$$

satisfies

$$\mathbf{q}_n = \left(\cos \frac{2\pi(j_0 + n)}{N}, \sin \frac{2\pi(j_0 + n)}{N} \right).$$

Its primitive period is $NT_d = 108T_d$, whereas the drive has period T_d . Thus the system realizes a primitive subharmonic response of order 108 within the specified finite domain.

Proof. The Weyl relation $S^{-1}ZS = \zeta Z$ gives $S^{-n}ZS^n = \zeta^n Z$. Its real and imaginary parts yield the formula. If an integer $d > 0$ were a period, we would have $\zeta^d = 1$; the primitivity of ζ requires $N \mid d$. $\square$

13.2 Finite spectral stability

The minimum separation between distinct roots of unity is

$$g_N = \min_{k \neq \ell} |\zeta^k - \zeta^\ell| = 2 \sin \frac{\pi}{N} > 0.$$

Theorem 13.2 (Covariant rigidity and finite spectral stability). *The following two statements hold.*

1. *For every unitary V , the triple*

$$U_F^{(V)} = V U_F V^\dagger, \quad Z^{(V)} = V Z V^\dagger, \quad \rho^{(V)} = V \rho V^\dagger$$

preserves exactly the Weyl relation, the subharmonic response of order N , and the amplitude of the order vector.

2. If U' is unitary and $\|U' - U_F\| = \varepsilon < g_N/2$, the N discs of radius $g_N/2$ centered at $\{\xi^k\}$ uniquely separate the spectral sectors of U' . Their Riesz projectors have the same ranks as those of U_F throughout any deformation remaining in that ball. Moreover, for every observable A , every state ρ , and every $n \geq 0$,

$$|\mathrm{Tr} \rho (U'^{-n} A U'^n - U_F^{-n} A U_F^n)| \leq 2n\varepsilon \|A\|.$$

Proof. The first part follows by conjugating each identity. For the second, the separation g_N prevents two sectors from entering the same contour; continuity of the Riesz projectors preserves their ranks. The telescoping identity gives $\|U'^n - U_F^n\| \leq n\varepsilon$, and the same bound holds for inverse powers. The expectation bound follows by adding and subtracting $U'^{-n} A U_F^n$. $\square$

The resulting object is a *finite HMT time crystal* with a drive, a subharmonic observable, and quantified spectral stability. Whether a macroscopic material realizes this class, retains coherence at large volume, or exhibits prethermal or localized protection belongs to EXTERNAL IDENTIFICATION; it is not an unresolved mathematical part of the finite construction.

14 Genealogical memory, information reduction, and thermal transduction

The thermal construction uses two representations from the same APP-TRIT-TPK development. The first preserves states, admissible routes, and record information; the second provides oriented action and the duration of return. Their composition determines an energy per tritic decision. Temperature and energy coordinates are introduced afterward through explicit dimensional maps. This sequence makes it possible to identify what information is preserved, what information a reader expresses, and what quantity converts that information into dimensional entropy.

14.1 Reader, memory, and state recovery

Fix a finite level of the development, and denote by Ω the set of admissible enriched states. The record preserves sheet, orientation, residue, quotient, carry, route, phase, and memory in the domains where each coordinate enters. Let

$$q : \Omega \longrightarrow Y, \quad M : \Omega \longrightarrow \mathcal{M}$$

be the visible representation and retained memory, respectively. For a distribution p on Ω , write $H(p) = -\sum_x p_x \log p_x$, with $0 \log 0 = 0$. The logarithm is applied to dimensionless probabilities.

Definition 14.1 (Genealogical recovery of a representation). The pair $\hat{q} = (q, M)$ is a recoverable reader on support Ω_p when there exists

$$d : \hat{q}(\Omega_p) \longrightarrow \Omega_p, \quad d \circ \hat{q} = \text{id}_{\Omega_p}.$$

The fiber information of q is $H_{\text{fib}}(q; p) = H(X \mid q(X))$, where X has law p .

Proposition 14.2 (Recovery identity and cost of reduction). *For a recoverable reader,*

$$H(X \mid q(X), M(X)) = 0, \tag{14.1}$$

$$H(X) = H(q(X)) + H(X \mid q(X)), \tag{14.2}$$

$$H(X \mid q(X)) = H(M(X) \mid q(X)). \tag{14.3}$$

Proof. The map d recovers X from the pair $(q(X), M(X))$, so the first conditional entropy is zero. Since q is a function of X , the chain rule gives the second equality. For the third, expand $H(X, M(X) \mid q(X))$ in two ways. One gives $H(X \mid q(X))$, because M is a function of X ; the other gives $H(M(X) \mid q(X)) + H(X \mid q(X), M(X))$. $\square$

Euclidean division of the APP sheets displays the elementary arithmetic mechanism. With $i, j \in \{1, \dots, 9\}$, one preserves

$$i + j = 9q_{ij}^+ + R_{ij}^+, \quad ij = 9q_{ij}^\times + R_{ij}^\times, \quad 1 \leq R_{ij}^\sigma \leq 9 \quad (\sigma \in \{+, \times\}).$$

The residue–quotient pair recovers the integer value of the operation. Summing the differences over the eighty-one positions gives

$$2025 - 810 = 54 + 9 \cdot 129.$$

Quotient memory restores the contribution not expressed by residual reduction. Recoverability of the complete state additionally requires preserving the relevant position and route coordinates; the residue–quotient pair here identifies the result of the operation.

14.2 Reversible update, tritic decision, and erasure

Let $U : \Omega \rightarrow \Omega$ be the TPK update on a domain where enriched transport has an inverse. The transported law is $p'_y = p_{U^{-1}y}$. The bijection preserves the multiset of probabilities, and hence

$$H(U(X)) = H(X), \quad H(X | U(X)) = 0. \quad (14.4)$$

If the observed output is $q(U(X))$, the unpublished information is quantified by $H(X | q(U(X)))$. The complete state, its representation, and the reset of a record are thus three distinct maps.

For the three oriented TRIT states, let $p = (p_{-1}, p_0, p_{+1})$ be a normalized distribution. Its information is

$$I_{\text{TRIT}}(p) = - \sum_{a=-1}^1 p_a \log p_a, \quad 0 \leq I_{\text{TRIT}}(p) \leq \log 3. \quad (14.5)$$

The upper bound corresponds to $p_a = 1/3$. Indeed,

$$D(p || (1/3, 1/3, 1/3)) = \sum_a p_a \log(3p_a) = \log 3 - I_{\text{TRIT}}(p) \geq 0;$$

the inequality follows from convexity of $t \log t$, and equality characterizes the uniform distribution. The lower bound corresponds to a distribution concentrated on one state.

The ideal Landauer realization uses a memory with energetically degenerate logical states, an isothermal environment at temperature Θ , and a balance that preserves the logical map's output and accounts for auxiliary records. In that model, the decrease in logical entropy is $k_B[H(X) - H(F(X))] = k_B H(X | F(X))$ for a deterministic map F . Entropy balance requires transferring at least that amount to the environment; its energy expression is $Q \geq k_B \Theta H(X | F(X))$. The distinction between reversible logical evolution and physical erasure corresponds to the Landauer-Bennett realization [17]. The invertible update (14.4) has zero logical contribution; resetting a uniform triple requires transferring $\log 3$ nats to the environment and satisfies $Q \geq k_B \Theta \log 3$. Control energy and dissipation in a particular physical realization belong to its additional thermodynamic balance.

14.3 Action, frequency, and decision energy

Let $\mathcal{L}_{\text{act}}$ and $\mathcal{L}_T$ be the dimensional lines of action and duration; the energy line is $\mathcal{L}_E = \mathcal{L}_{\text{act}} \otimes \mathcal{L}_T^*$. The preceding development supplies the positive sections $\hbar_a \in \mathcal{L}_{\text{act}}$, with $a \in \{\text{pre}, \text{ret}\}$, and the elementary duration $t_0 \in \mathcal{L}_T$. The complete calendar provides

$$T_{108} = 108t_0, \quad \omega_{108} = \frac{2\pi}{T_{108}}, \quad h_a = 2\pi\hbar_a.$$

The dimensional contraction between action and frequency defines

$$\epsilon_a := \hbar_a \omega_{108} = \frac{\hbar_a}{108t_0} = \frac{\hbar_a}{t_P}, \quad t_P = \frac{54}{\pi_{\text{HMT}}} t_0. \quad (14.6)$$

The last member uses the reduced period of the same longitudinal clock; its subsequent identification with the gravitational scale is established in Section 27. The three expressions correspond to one energy section, with an orientation a maintained in all factors. The normalized information of a uniform decision produces the energy-per-nat section

$$\epsilon_a := \frac{\epsilon_a}{\log 3} \in \mathcal{L}_E^+. \quad (14.7)$$

The previously constructed action determines the energy scale once duration is fixed; the cardinality of TRIT alternatives determines its information normalization. The value $\log 3$ enters through the count and measure declared in (14.5). Up to this point the construction is general with respect to the longitudinal section. In the specialization proved later by (27.8), $t_P = L_\alpha/c$, and therefore

$$\epsilon_a = E_{P,a} = Q_{L,a} = \frac{\hbar_a c}{L_\alpha}. \quad (14.8)$$

This equality does not follow from the calendar cardinal alone: it uses compatibility of the clock with the already constructed longitudinal length.

14.4 Dimensional definition of Boltzmann and coordinate selection

Let $\mathcal{L}_\Theta$ be the oriented temperature line. The dimensional entropy space is $\mathcal{L}_{\text{ent}} = \mathcal{L}_E \otimes \mathcal{L}_\Theta^*$. A positive linear map $k_B : \mathcal{L}_\Theta \rightarrow \mathcal{L}_E$ is, equivalently, a positive element of $\mathcal{L}_{\text{ent}}$. The same object can therefore act on a temperature or multiply dimensionless information.

Definition 14.3 (Thermal normalization of the tritic decision). A thermal pair compatible with oriented section a consists of a positive transducer k_B and a positive section $\Theta_{\text{clk},a} \in \mathcal{L}_\Theta$ such that

$$k_B(\Theta_{\text{clk},a}) = \epsilon_a, \quad k_B \Theta_{\text{clk},a} \log 3 = \epsilon_a. \quad (14.9)$$

The Boltzmann constant serves as the energy–temperature transducer of tritic information in this realization.

In the longitudinal specialization of (14.8), the same definition is written without introducing another speed factor:

$$k_B \Theta_{\text{clk},a} \log 3 = Q_{L,a} = E_{P,a}. \quad (14.10)$$

Theorem 14.4 (Uniqueness relative to the thermal section and dimensional covariance). *With a positive section $\Theta_{\text{clk},a}$ fixed, there is a unique positive linear map satisfying (14.9). For positive bases u_E, u_Θ , if*

$$\epsilon_a = \epsilon_a^{\text{num}} u_E, \quad \Theta_{\text{clk},a} = \theta_a u_\Theta, \quad k_B(u_\Theta) = b u_E,$$

its coordinate is

$$b = \frac{\epsilon_a^{\text{num}}}{\theta_a \log 3}. \quad (14.11)$$

Under $u'_E = s_E u_E$, $u'_\Theta = s_\Theta u_\Theta$, with $s_E, s_\Theta > 0$, one has

$$b' = \frac{s_\Theta}{s_E} b, \quad \theta'_a = \frac{\theta_a}{s_\Theta}, \quad (\epsilon_a^{\text{num}})' = \frac{\epsilon_a^{\text{num}}}{s_E}.$$

Proof. A nonzero section spans a line. Prescribing its image determines the unique linear map; positivity of both sections proves its positivity. Substituting the coordinates into (14.9) gives $b \theta_a \log 3 = \epsilon_a^{\text{num}}$. The three transformation laws follow by expressing the same sections and the same map in the new bases. $\square$

The same Boltzmann constant must act in both orientations. Since t_P is common, simultaneous compatibility is equivalent to

$$\frac{\Theta_{\text{clk},\text{pre}}}{\Theta_{\text{clk},\text{ret}}} = \frac{\epsilon_{\text{pre}}}{\epsilon_{\text{ret}}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}}. \quad (14.12)$$

Indeed, an isomorphism of lines preserves quotients of sections. Conversely, if the preceding equality holds, the isomorphism determined in one orientation also satisfies the normalization of the other. Thus return modifies the thermal section of the cycle in the same proportion as energy, while the transducer remains common to both readings.

The preceding equation makes explicit the choice required to express an individual value. If the thermal section has not yet been selected by an independent reader, then for every $\lambda > 0$ the pair

$$(k_B, \Theta_{\text{clk},a}) \longmapsto (\lambda k_B, \lambda^{-1} \Theta_{\text{clk},a})$$

preserves ϵ_a . Consequently, presenting an internally selected individual coordinate requires providing the reader of $\Theta_{\text{clk},a}$, the bases, and their dimensional transport. The action and information equations remain determined before that specification. Subsequent identification of the bases with joules and kelvins belongs to the metrological representation of the result.

14.5 Dimensional entropy and information of the reduced state

For a distribution p of admitted routes, or a reduced state with density matrix ρ , let

$$s(p) = -\sum_x p_x \log p_x, \quad s(\rho) = -\text{Tr}(\rho \log \rho).$$

Their dimensional representations are

$$S(p) = k_B s(p), \quad S(\rho) = k_B s(\rho) \quad \text{in } \mathcal{L}_{\text{ent}}. \quad (14.13)$$

Evaluation at Θ produces an energy $\Theta S = k_B \Theta s$. For the uniform triple at the cycle temperature, (14.9) gives exactly

$$S_{\text{TRIT}} = k_B \log 3, \quad \Theta_{\text{clk},a} S_{\text{TRIT}} = \epsilon_a.$$

If ρ_A is the reduction of the bipartite purification constructed in the areal section, $s(\rho_A)$ measures the entanglement of that bipartition. Unit transport preserves its eigenvalues, orientation, and sectorial occupations. The areal scale $\gamma a_0 \ell_P^2$ converts occupations into area; the transducer k_B converts information into dimensional entropy. Both readings use the same record through different operators.

14.6 Spectral normalization of thermal histories

Let $\mathcal{G}$ be an already constructed finite graph of admissible extensions, with adjacency matrix A . Each edge e preserves its route coordinates and receives a dimensionless action increment $a_*(e) > 0$ from the corresponding reader. With energy section ϵ_* chosen, put $t = \beta \epsilon_*$, where $\beta = (k_B \Theta)^{-1}$. The thermal operator is

$$(L_t f)(y) = \sum_{e: x \rightarrow y} \exp[-t a_*(e)] f(x). \quad (14.14)$$

All its exponents are dimensionless. The condition $\rho(L_t) = 1$ normalizes the weighted growth of histories.

Proposition 14.5 (Existence and uniqueness of normalized pressure). *Suppose that A is irreducible, $\rho(A) > 1$, and $0 < m \leq a_*(e) \leq M$ for all edges. There is a unique $t_* > 0$ such that $\rho(L_{t_*}) = 1$, and*

$$\frac{\log \rho(A)}{M} \leq t_* \leq \frac{\log \rho(A)}{m}. \quad (14.15)$$

Proof. Entrywise comparison gives $e^{-tM}A \leq L_t \leq e^{-tm}A$ for $t \geq 0$. Monotonicity of the spectral radius of nonnegative matrices implies the corresponding bounds for $\rho(L_t)$. The spectral radius is continuous, equals $\rho(A) > 1$ at zero, and tends to zero at infinity. For $s > 0$, moreover, $L_{t+s} \leq e^{-sm}L_t$, so $\rho(L_{t+s}) \leq e^{-sm}\rho(L_t) < \rho(L_t)$. Continuity and strict decrease prove existence and uniqueness. Substituting the value one into the initial bounds proves (14.15). $\square$

The local realization of three equally weighted alternatives, represented by a vertex with three loops and $a_*(e) = 1$, satisfies $\rho(L_t) = 3e^{-t}$. Its normalization gives $t_* = \log 3$. With $\epsilon_* = \epsilon_a$, one obtains $\beta_{\text{clk}} = \log 3 / \epsilon_a$, exactly relation (14.9). This computation identifies the local tritic confluence. For a history graph with additional constraints, confluence with the same clock requires checking

$$\rho(L_{(\log 3 / \epsilon_a) \epsilon_*}) = 1$$

with its effective increments. Local normalization and the pressure of the complete system thus preserve their respective domains.

The relation obtained brings together structure, value, and meaning. The record determines recoverable information; probabilities on continuations determine its entropic reading; action and period determine ϵ_a ; the energy–temperature transducer expresses dimensional entropy. Area and entanglement are related through the reduced state and its boundary operators, keeping visible which datum produces each quantity.

15 Spectral capacity, energy memory, and thermal transport

The positive map between temperature and decision energy is constructed here on the graded carrier and its marked reference. This section retains the energy origin invisible to normalisation, and composes that reading with the calendar, area, gravitation, and the thermal distributions of this article. The causal chain remains

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\text{enr}} \longrightarrow \mathcal{C}_{\text{cont}}^{\text{disc}} \longrightarrow \text{thermal carrier} \longrightarrow \text{physical recognition}.$$

The mathematical and physical constants used here are HMT-MD outputs or later representations of that state. No metrological number selects seeds, sectors, or spectrum. In particular, the scalar 1.380649×10^{-23} is first obtained as a marked dimensionless intensity; its composition with the energy and thermal charts is proved only afterwards.

15.1 Graded carrier and two preserved coordinates

Let $\mathcal{H}_{N,L}$ be five finite spaces, with

$$\dim \mathcal{H}_{0,0} = 1, \quad \dim \mathcal{H}_{1,2} = 380, \quad \dim \mathcal{H}_{2,0} = 1, \quad \dim \mathcal{H}_{2,1} = 24, \quad \dim \mathcal{H}_{2,2} = 624. \quad (15.1)$$

Their direct sum $\mathcal{H}_{\text{th}}$ simultaneously retains the prefix degree N and the incidence-memory degree L . Its bivariate Hilbert polynomial is

$$Z(u, v) = 1 + 380uv^2 + u^2(1 + 24v + 624v^2). \quad (15.2)$$

Thus,

$$Z(u, 1) = 1 + 380u + 649u^2, \quad Z(1, v) = 2 + 24v + 1004v^2.$$

The equality $649 = 1 + 24 + 624$ describes the L -fibres in degree $N = 2$; the independent identity $649 = 1 + 2 \cdot 324$ describes the involutive decomposition of the carrier. These meanings are not interchanged.

The dimensions 380 and 649 arise from distinct operations. The first is the off-diagonal part of the endomorphism on the capacity-20 carrier; the second joins the scalar direction to two orientations of the 324-route sector. To make the latter construction explicit, let

$$X_{\times} = (\mathbb{Z}/9\mathbb{Z})^2 \times \{N, E, S, O\}, \quad |X_{\times}| = 324,$$

and let $\mathfrak{e} : X_{\times} \rightarrow \mathcal{S}_{\times}$ be the six-window emitter, with 26 signatures. Its fibres have size 8 for 24 signatures, 24 for 555555, and 108 for 000000. With $I_{24} = \{1, \dots, 24\}$, construct the microscopic tree

$$\tilde{\mathcal{T}} = \{\Omega\} \sqcup \{a_i : i \in I_{24}\} \sqcup \{b_{i,y} : i \in I_{24}, y \in X_{\times}\}, \quad |\tilde{\mathcal{T}}| = 7801. \quad (15.3)$$

The reading $p(b_{i,y}) = b_{i,\mathfrak{e}(y)}$, fixing Ω and the a_i , has image $\mathcal{T}$ of size $1 + 24 + 24 \cdot 26 = 649$. If $m_s = |\mathfrak{e}^{-1}(s)|$, the measure

$$\nu_{\text{mic}}(\Omega) = \nu_{\text{mic}}(a_i) = 1, \quad \nu_{\text{mic}}(b_{i,y}) = m_{\mathfrak{e}(y)}^{-1} \quad (15.4)$$

gives every fibre of p mass one. Rates are $r > 0$ in both directions between Ω and a_i , $r/m_{\mathfrak{e}(y)}$ from a_i to $b_{i,y}$, and r in the reverse direction.

On $\ell^2(\mathcal{T})$ with counting measure and $\ell^2(\tilde{\mathcal{T}}, \nu_{\text{mic}})$, the lift

$$J_p g = g \circ p \quad (15.5)$$

is an isometry from a class or signature to the weighted microscopic tree. Indeed,

$$\|J_p g\|_{\nu_{\text{mic}}}^2 = \sum_{z \in \mathcal{T}} |g(z)|^2 \sum_{x: p(x)=z} \nu_{\text{mic}}(x) = \|g\|^2, \quad J_p^* J_p = I.$$

Its adjoint is the weighted average

$$(J_p^* f)(z) = \sum_{x: p(x)=z} \nu_{\text{mic}}(x) f(x), \quad (15.6)$$

not the lift J_p : their domains are reversed, and one copies a class function whereas the other averages its fibre. Detailed balance $\nu_{\text{mic}}(a_i)r/m_{e(y)} = \nu_{\text{mic}}(b_{i,y})r$ and the rate sums prove, for the microscopic and aggregated generators, $\tilde{L}J_p = J_p L_{\mathcal{J}}$. Balance within a fibre does not by itself determine weights between grades; the boundary currents constructed below do so.

The variable

$$Q := N - L \quad (15.7)$$

separates states that one marginal may identify. For example, knowing $Z(u, 1)$ does not recover the distribution of L . The five cells in (15.1) and their multiplicities reconstruct the specified joint carrier; they do not reconstruct coherences discarded by a compression. This is the exact domain of the reading.

15.2 Positive lift of energy memory

On the five cells, Q takes the values 0, -1 , 2, 1, 0. Introduce a memory register R with levels 0, 1, 2, 3 and the isometry

$$V_{\text{mem}} : \mathcal{H}_{N,L} \longrightarrow \mathcal{H}_{N,L} \otimes \mathbb{C}^4, \quad V_{\text{mem}} \xi = \xi \otimes e_{N-L+1}. \quad (15.8)$$

The unit shift turns the signed charge into a nonnegative level without losing it. If $Re_j = je_j$, then

$$V_{\text{mem}}^*(L \otimes I + I \otimes R)V_{\text{mem}} = N + I. \quad (15.9)$$

On each cell this is simply $L + (N - L + 1) = N + 1$, but the operator form retains degeneracies, superpositions within each fibre, and every purification on an auxiliary system.

For an energy scale $\epsilon_0 > 0$,

$$V_{\text{mem}}^* \epsilon_0 (L \otimes I + I \otimes R) V_{\text{mem}} = \epsilon_0 (N + I) \quad (15.10)$$

proves exact energy conservation in the lift. A filter retaining one branch must also retain its success marker; otherwise the conditioned density hides the preparation cost. The lift is reversible on its image and has zero logical cost; discarding R is a distinct operation governed by the Landauer balance of the preceding section.

15.3 Decimal spectrum produced by the memory loop

The 8:1 incidence decomposition determines the coupling

$$\mathcal{Q} = \frac{1}{3} \begin{pmatrix} \sqrt{8}I & -I \\ I & \sqrt{8}I \end{pmatrix}.$$

The oriented return loop is

$$H_1 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad H_1 H_1^T = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}.$$

In the declared positive Gaussian preparation, the normalised visible covariance is

$$W = \frac{8I + H_1 H_1^T}{9}, \quad \det W = \frac{121}{81}, \quad \nu_W = \sqrt{\det W} = \frac{11}{9}. \quad (15.11)$$

The one-mode spectral theorem therefore gives

$$r = \frac{\nu_W - 1}{\nu_W + 1} = \frac{1}{10}, \quad \rho_r = (1 - r) \sum_{j \geq 0} r^j |j\rangle\langle j|. \quad (15.12)$$

The decimal ratio is obtained from the loop covariance; it is not entered as a target numerical base. The symbol ν_W is the Williamson symplectic eigenvalue and is distinct from the microscopic measure ν_{mic} in (15.4). The control $H_0 = I$ gives $r = 0$, so the spectrum depends effectively on the selected holonomy.

Euclidean division $j = 3n + b$, $0 \leq b < 3$, defines the unitary

$$U_3|j\rangle = |n\rangle \otimes |b\rangle, \quad U_3\rho_r U_3^* = \rho_{r^3} \otimes \frac{\text{diag}(1, r, r^2)}{1 + r + r^2}. \quad (15.13)$$

The proof is termwise: $(1 - r)r^{3n+b} = (1 - r^3)(r^3)^n r^b / (1 + r + r^2)$. Hence,

$$q = r^3 = 10^{-3}, \quad \sigma_r = \frac{\text{diag}(100, 10, 1)}{111}. \quad (15.14)$$

The residue retains the three intermediate positions. If $S|j\rangle = |j + 1\rangle$, its transport is

$$U_3 S U_3^* = I \otimes (|1\rangle\langle 0| + |2\rangle\langle 1|) + S_n \otimes |0\rangle\langle 2|,$$

so the third advance returns the residue and increments the quotient. Phase return and state return are not conflated.

With $H_W = \epsilon_a(N + \frac{1}{2})$,

$$U_3 H_W U_3^* = \epsilon_a \left(3N_n \otimes I + I \otimes R + \frac{1}{2}I \right). \quad (15.15)$$

The quotient factor has spacing $3\epsilon_a$, while temperature is fixed:

$$e^{-\beta_W \epsilon_a} = 10^{-1}, \quad e^{-\beta_W 3\epsilon_a} = 10^{-3}, \quad k_B T_W = \frac{\epsilon_a}{\log 10}.$$

15.4 Marked section and complete evaluation

The renewal in (4.13) supplies origin 23, and threefold refinement supplies 23, 26, 29. With $\lambda_j/\lambda_0 = r^j$ and the carrier multiplicities,

$$\mathcal{B} = \frac{\lambda_{23} + 380\lambda_{26} + 649\lambda_{29}}{\lambda_0} = 10^{-23} \left(1 + \frac{380}{1000} + \frac{649}{1000^2} \right) = 1.380649 \times 10^{-23}. \quad (15.16)$$

The residue $b = 2$ alone would select 2, 5, 8, ... and does not determine threshold 23; the latter comes from renewal time and the capacity index. The mantissa is retained in the normalised density through the sector marker; the selection weight additionally retains 10^{-23} . Thus neither the mantissa nor the exponent is taken from an external table.

Let $\mathcal{D} = \bigoplus_{n=0}^2 \mathcal{D}_n$, with $(d_0, d_1, d_2) = (1, 380, 649)$, $\dim \mathcal{D}_n = d_n$, and $N|_{\mathcal{D}_n} = nI$. Define the capacity operator and spectral reference by

$$C = 23I + 3N, \quad \eta = \bigoplus_{n=0}^2 r^{23+3n} I_{\mathcal{D}_n}, \quad b_* := \text{Tr } \eta = \mathcal{B}. \quad (15.17)$$

Reference η retains grades 23, 26, 29 separately and remains fixed when a trial state varies. For $x = e^{-\epsilon_0/\tau}$, the relative partition function and state are

$$Z_N(x) = 1 + 380x + 649x^2, \quad \rho_N(x) = \frac{x^N}{Z_N(x)}, \quad p_0 = \frac{1}{Z_N(x)}. \quad (15.18)$$

The marker p_0 recovers Z_N after normalisation. At $q = 10^{-3}$, $Z_N(q) = 1.380649$, $\eta = 10^{-23}q^N$, and $\eta/b_* = \rho_N(q)$; the marked origin produces (15.16). The free energy and mean degree are

$$F_N(\tau) = -\tau \log Z_N(x), \quad \langle N \rangle_x = x \frac{d}{dx} \log Z_N(x) = \frac{380x + 1298x^2}{Z_N(x)}. \quad (15.19)$$

15.5 Boundary currents and rigidity of the sector origin

The areal incidence supplies $18 + 18$ tritic links. Let $u = N_{90}$, $v = N_{120}$, $a = u + v$, and

$$F = 3u + 4v, \quad H_R = \epsilon_0 F, \quad \epsilon_0 = 30\Delta\tau_R > 0, \quad \tau_R = \frac{\hbar c}{2\pi R}.$$

Provenance remains in (u, v) : the same area may correspond to different energies. On one link, $T^+ = |1\rangle\langle 0| + |2\rangle\langle 1|$ and $T^- = (T^+)^*$ satisfy $[n, T^\pm] = \pm T^\pm$. For $e, f = 1, \dots, 18$, define

$$J_{ef} = T_{120,f}^+ T_{90,e}^-, \quad B_c = \left(\sum_e T_{90,e}^+ \right)^3 \left(\sum_f T_{120,f}^- \right)^2.$$

The rule $[F, AB] = [F, A]B + A[F, B]$ proves

$$[F, J_{ef}] = J_{ef}, \quad [a, J_{ef}] = 0, \quad [F, B_c] = B_c, \quad [a, B_c] = B_c.$$

The occupation sequence $(0, 2) \rightarrow (3, 0) \rightarrow (2, 1)$ has primitive energies 8, 9, 10 and areas 2, 3, 3: the same energy increment retains two distinguishable areal actions.

The coefficients $c_j = [z^j](1 + z + z^2)^{18}$ count occupations in each channel: $c_0 = 1, c_1 = 18, c_2 = 171, c_3 = 1122$. The boundary character

$$P(x) = (1 + x^3 + x^6)^{18}(1 + x^4 + x^8)^{18}$$

gives $g_7 = 324, g_8 = 171, g_9 = 1122, g_{10} = 3078$. For ranks $(1, 380, 649)$, the first admissible triple of consecutive energies starts at 8. Earlier starts are excluded because, using $g_0 = 1, g_1 = g_2 = g_5 = 0, g_3 = g_4 = 18, g_6 = 171, g_7 = 324$, and $g_8 = 171$, every earlier triple fails at least one required rank.

The connected compression is constructed by selecting one $(0, 2)$ state, 380 states $(3, 0)$, and 649 distinct neighbours under the J_{ef} . Every occupation of sum three has at least two predecessors of sum two, and every predecessor has at most 18 successors. Hence the 380 states produce at least $\lceil 760/18 \rceil = 43$ distinct predecessors and $18 \cdot 43 = 774$ neighbours $(2, 1)$, from which 649 may be selected. The current B_c connects the initial state to the 380 intermediate states, and every final neighbour retains a selected predecessor; the resulting graph is connected. Thus the compressed subspace, its labels, and its currents are constructed rather than merely postulated.

The boundary compression realises degrees $N = 0, 1, 2$ at energies $8\epsilon_0, 9\epsilon_0, 10\epsilon_0$. Let $\mathcal{D}_n = \ker(N - nI)$, $P_F = \mathbf{1}_{\{N=1\}}$, $B_N : \mathcal{D}_0 \rightarrow \mathcal{D}_1$, and $J_N : \mathcal{D}_1 \rightarrow \mathcal{D}_2$ be the nonzero compressed currents; their adjoints traverse the edges downwards. Then

$$[N, B_N] = B_N, \quad [N, J_N] = J_N, \quad [P_F, B_N] = B_N, \quad [P_F, J_N] = -J_N. \quad (15.20)$$

For the affine diagonal correction

$$H_\delta = \epsilon_0(8I + N) + \delta P_F$$

the commutators give

$$[H_\delta, B_N] = (\epsilon_0 + \delta)B_N, \quad [H_\delta, J_N] = (\epsilon_0 - \delta)J_N.$$

Requiring both currents to retain the boundary jump ϵ_0 , or merely requiring both jumps to be equal, forces

$$\delta = 0. \quad (15.21)$$

The conclusion classifies this sector offset; it does not claim a classification of the entire commutant of the Hamiltonian.

Assigning rate $x r_e$ to every ascending edge and rate $r_e > 0$ to its descending mate, with the same r_e in both directions, gives detailed balance with respect to weights x^N . Connectivity makes the stationary distribution unique on the compressed carrier and recovers (15.18). This dynamics is a Markov preparation semigroup, not isolated unitary evolution.

15.6 Thermal equivalence with origin and marker

Let $(\mathcal{H}_i, H_i, \tau_i, e_i, P_i)$ be two realisations, with $\tau_i > 0$, origin e_i , support projector P_i , and an isometric isomorphism $J : P_1 \mathcal{H}_1 \rightarrow P_2 \mathcal{H}_2$. The relevant thermal equivalence is

$$J \frac{H_1 - e_1 I}{\tau_1} = \frac{H_2 - e_2 I}{\tau_2} J, \quad J P_1 = P_2 J. \quad (15.22)$$

It requires neither $H_1 = H_2$ nor $\tau_1 = \tau_2$. It implies $J f((H_1 - e_1 I)/\tau_1) = f((H_2 - e_2 I)/\tau_2) J$ for every bounded spectral function f , and transports the centred Gibbs states. For observables that depend on absolute energy, the e_i and scale must also be transported: normalisation does not license their removal.

In the preceding compression there is a based isometry ι_N such that

$$H_R \iota_N = \iota_N \epsilon_0 (8I + N).$$

The current $J_N : \mathcal{D}_1 \rightarrow \mathcal{D}_2$ and the isometry ι_N are distinct operators. For the chronological reading, set

$$C = 23I + 3N, \quad H_{\text{cap}} = \frac{\epsilon_a \log 10}{\log 3} C, \quad \tau_{\text{clk}} = \frac{\epsilon_a}{\log 3}. \quad (15.23)$$

The relative Williamson section instead uses $H_{W,\text{rel}} = \epsilon_a C$ and $\tau_W = \epsilon_a / \log 10$. Hence $H_{\text{cap}} / \tau_{\text{clk}} = H_{W,\text{rel}} / \tau_W = C \log 10$, although their energy and temperature scales differ.

With $\tau_* = \epsilon_0 / \log 1000$, the centred comparison is

$$\frac{H_R - 8\epsilon_0 I}{\tau_*} \iota_N = \iota_N \frac{H_{\text{cap}} - 23\tau_{\text{clk}} \log 10 I}{\tau_{\text{clk}}} = \iota_N (\log 1000) N. \quad (15.24)$$

The equality retains degrees, energies, marker, and origin without identifying ϵ_0 with ϵ_a : the former belongs to the boundary compression and the latter to the chronological section. The 1/10 factor between uncentred traces and the log 10 shift are section data, not defects of the density.

15.7 Area, provenance, and thermal cost

In the two areal channels, let $N_{90}, N_{120} \geq 0$ and

$$a = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}. \quad (15.25)$$

They are recovered exactly as

$$N_{90} = 4a - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3a. \quad (15.26)$$

Area may read a , while the sector Hamiltonian retains provenance:

$$H_R = \tau_R \Delta D = \tau_R \Delta (90N_{90} + 120N_{120}). \quad (15.27)$$

Two states with equal a and different channel distributions have the same total-occupation reading and different cost. Reduction to one channel, its purification, and the finite free-energy identity apply to the reduced state already constructed; relative entropy measures the deficit from the reference Gibbs state. This composition explains why origin and provenance remain necessary when relating area, entanglement, and the Barbero–Immirzi functional.

15.8 Composition with action, gravitation, and the electron

The length L_α and oriented action $\hbar_a$ fix

$$\epsilon_a = \frac{\hbar_a c}{L_\alpha}, \quad G_a = \frac{c^3 L_\alpha^2}{\hbar_a}, \quad k_B \Theta_{\text{clk},a} \log 3 = \epsilon_a. \quad (15.28)$$

Multiplying the three identities eliminates the action section and gives

$$\boxed{G_a k_B \Theta_{\text{clk},a} \log 3 = c^4 L_\alpha.} \quad (15.29)$$

The identity is obtained in one orientation a ; G_a is not taken from metrology. The symbol L_α denotes length, not the degree L of (15.1).

If $R_{\text{act}} = \hbar_{\text{pre}}/\hbar_{\text{ret}}$ is the action ratio of the electron return, the same composition gives

$$R_{\text{act}} = \frac{\Theta_{\text{clk},\text{pre}}}{\Theta_{\text{clk},\text{ret}}} = \frac{G_{\text{ret}}}{G_{\text{pre}}}. \quad (15.30)$$

This equality transports the thermal section; it neither uses k_B to select the route nor rederives the mass law in this article. For an already individualised electron energy E_e , $y_e = E_e/\epsilon_{\text{ret}}$ satisfies

$$\beta_{\text{clk}} E_e = (\log 3) y_e, \quad \frac{G_{\text{ret}} m_e^2}{\hbar_{\text{ret}} c} = y_e^2.$$

The energy origin affects mass and radius even when it cancels from normalised probabilities.

15.9 Marked thermometric bridge and effective identity

Reference η in (15.17) must remain separate from the state on which energy is measured. Let $\mathcal{H}$ be a finite-dimensional space, let ρ be a density matrix on $\mathcal{H}$, and let $H = H^*$ be a Hamiltonian with registered origin. On $\hat{\mathcal{D}} = \mathcal{D} \oplus \mathbb{C}f$, define

$$\hat{\eta} = \eta \oplus (1 - b_*)|f\rangle\langle f|, \quad \Omega_\rho = \hat{\eta} \otimes \rho, \quad (15.31)$$

and let P project onto the summand $\mathcal{D}$, extended by the identity on $\mathcal{H}$. With $s(\rho) = -\text{Tr}(\rho \log \rho)$ and logarithms restricted to the support, the two local readings are

$$\mathcal{S}_P(\rho) = -\text{Tr}[P\Omega_\rho(I \otimes \log \rho)] = b_* s(\rho), \quad (15.32)$$

$$\mathcal{E}_P(\rho) = \frac{\text{Tr}[P\Omega_\rho(I \otimes H)]}{\text{Tr}(P\Omega_\rho)} = \text{Tr}(\rho H). \quad (15.33)$$

Indeed, $P\Omega_\rho = \eta \otimes \rho$ and $\text{Tr}(P\Omega_\rho) = b_*$. Trace factorisation gives $b_* s(\rho)$ and $b_* \text{Tr}(\rho H)$ in the two numerators; dividing the latter by $b_* > 0$ proves the energy reading. Thus the fixed reference, information relative to its marker, and energy per conditioned state are all retained.

Work in transported positive bases of the energy, temperature, and entropy lines, with $u_{\text{ent}} = u_E/u_\Theta$. In coordinate formulas, H expresses energy in u_E and $\mathcal{S}_P$ expresses entropy in u_{ent} . The realisation rule comprises four operations: receive the fixed reference (15.31); evaluate (15.32); evaluate (15.33); and define temperature as conjugate to these two functionals. The bases are transported with the operators, without a target temperature input.

Theorem 15.1 (Thermometric derivative of the marked reading). *If H is nonscalar and $\rho_\beta = e^{-\beta H}/Z(\beta)$ for $\beta > 0$, then*

$$\frac{d\mathcal{S}_P}{d\mathcal{E}_P} = b_* \beta, \quad \Theta_P = \frac{1}{b_* \beta}, \quad B_*(\Theta_P u_\Theta) = \beta^{-1} u_E, \quad (15.34)$$

where $B_*(u_\Theta) = b_* u_E$ is the unique positive linear map with that image of the basis.

Proof. For $E(\beta) = \text{Tr}(\rho_\beta H)$, the finite spectral sum gives $E' = -\text{Var}_{\rho_\beta}(H) < 0$ and $(\log Z)' = -E$. From $s(\rho_\beta) = \beta E + \log Z$ one obtains $s' = \beta E'$. Since η remains fixed, $S'_P = b_* s'$ and $E'_P = E'$. Their ratio and its reciprocal prove the first two identities; the image of a positive basis fixes a unique linear map and proves the third. $\square$

The factor depends on the pair of readings. If energy and entropy are both marked, $(b_* E, b_* s)$, or both conditioned, (E, s) , the derivative is β ; for the marked-conditioned pair (15.32)–(15.33) it is $b_* \beta$.

The same rule has a variational characterisation. For $\Theta > 0$,

$$\Phi_\Theta(\rho) = \mathcal{E}_P(\rho) - \Theta \mathcal{S}_P(\rho) = \text{Tr}(\rho H) - b_* \Theta s(\rho) \quad (15.35)$$

has the unique minimiser

$$\rho_\Theta^* = \frac{e^{-H/(b_* \Theta)}}{\text{Tr} e^{-H/(b_* \Theta)}}. \quad (15.36)$$

Indeed,

$$\Phi_\Theta(\rho) - \Phi_\Theta(\rho_\Theta^*) = b_* \Theta D_{\text{rel}}(\rho \| \rho_\Theta^*) \geq 0,$$

where $D_{\text{rel}}(\rho \| \sigma) = \text{Tr}[\rho(\log \rho - \log \sigma)]$; equality in relative-entropy positivity occurs exactly when $\rho = \rho_\Theta^*$. This proves existence and uniqueness without entering a target temperature.

Apply the result to the capacity clock. With orientation and ϵ_a fixed, set $H_{\text{clk},a} = H_{\text{cap}}$ and $\beta_{\text{clk},a} = \log 3/\epsilon_a$. This Hamiltonian is positive, self-adjoint, and nonscalar, since its three distinct eigenvalues are proportional to 23, 26, 29. Then

$$\begin{aligned} \beta_{\text{clk},a} H_{\text{clk},a} &= C \log 10, \\ e^{-\beta_{\text{clk},a} H_{\text{clk},a}} &= 10^{-23} q^N = \eta, \quad q = 10^{-3}, \\ \rho_{\text{clk},a} &= \frac{q^N}{Z_N(q)} = \frac{\eta}{b_*}. \end{aligned} \quad (15.37)$$

Reference η , independent of ϵ_a , stays fixed in the first factor, while ρ_β varies on a second copy of $\mathcal{D}$ with $H_{\text{clk},a}$ fixed. Evaluation only afterwards at $\beta = \beta_{\text{clk},a}$ gives

$$\Theta_{\text{clk},P,a} = \frac{\epsilon_a}{b_* \log 3}, \quad \boxed{B_*(\Theta_{\text{clk},P,a} u_\Theta) = \frac{\epsilon_a}{\log 3} u_E = \frac{h_a}{108 t_0 \log 3} u_E.} \quad (15.38)$$

The last equality uses the already fixed action coordinate $\epsilon_a = h_a/(108 t_0)$; it does not use the value of b_* as a fit. Nor does it identify ϵ_a with the boundary scale ϵ_0 .

If B_{clk} is the chronological transducer and J_E, J_Θ transport energy and the two thermometric sections of the same reference state, respectively, uniqueness between lines gives the typed square

$$B_* J_\Theta = J_E B_{\text{clk}}. \quad (15.39)$$

The two composites agree on the positive generating section by (15.38); linearity extends the equality to the entire line. This is the bridge between the spectral evaluation in (15.16) and the dimensional transducer: domains, reference, readers, and positive section determine the map before its numerical evaluation.

15.10 Consequences for occupation, radiation, and information

The finite carrier supplies Gibbs theory with $Z_N(x)$, $\rho_N(x)$, marker p_0 , and the free energy (15.19). The symmetric and exterior completions developed below act on complete modes and produce the Bose–Einstein and Fermi–Dirac laws; the 1/380/649 carrier does not replace them. It supplies an energy section and a preparation with a recorded origin. In the dilute regime, the same coordinate $x = e^{-\beta(E-\mu)}$ enters the Maxwell–Boltzmann limit already proved.

In the radiative realisation, composing $\beta = (k_B\Theta)^{-1}$ with $k_B\Theta_{\text{clk},a} \log 3 = \epsilon_a$ substitutes the calendar scale in Planck’s density without changing the count of two polarisations or their dispersion. Its moments retain the Stefan–Boltzmann and Wien deductions; the new section determines the source of the thermal scale and the marker that must be retained when conditioning a branch. For area and entanglement, (15.26) prevents total occupation from erasing provenance cost. Thus occupation, radiation, free energy, area, and gravitation receive the same thermal information without conflating their observables.

15.11 Thermal response of the marker and preparation cost

The marker distinguishes the reference state within the complete carrier. Its probability, its response to a perturbation and the cost of preparing that sector are three calculable readings of the same distribution, with memory retained. They recover the thermal meaning of the normalizer in addition to its numerical evaluation.

Consider the carrier with degrees and multiplicities

$$(N, L; d) = (0, 0; 1), (1, 2; 380), (2, 0; 1), (2, 1; 24), (2, 2; 624).$$

For an energy gap $a > 0$, let $H = aL$, $Q = N - L$ and $\mu = -a$. The Gibbs generator is therefore

$$K = H - \mu Q = aN, \quad q = e^{-\beta a}, \quad Z_N = \text{Tr}(q^N) = 1 + 380q + 649q^2, \quad \rho_N = \frac{q^N}{Z_N}.$$

Here $\beta = (k_B T)^{-1}$, and P projects onto the marked reference of degree $N = 0$. This reference is distinct from the state of degree $L = 0$ in the sector $N = 2$. Its probability is $p = \text{Tr}(\rho_N P) = Z_N^{-1}$.

Proposition 15.2 (Susceptibility of the marked sector). *For the perturbed generator $K_u = K - uP$, with the carrier fixed, the response $\chi_P = \partial_u \text{Tr}(\rho_u P)|_{u=0}$ satisfies*

$$\chi_P = \beta p(1 - p), \quad Z_N = 1 + \frac{\chi_P}{\beta p^2}.$$

Proof. Since P commutes with K , the perturbation multiplies the marked-sector weight by $e^{\beta u}$ and leaves the other weights unchanged. Thus

$$p(u) = \frac{pe^{\beta u}}{1 - p + pe^{\beta u}}.$$

Differentiation at $u = 0$ gives $\chi_P = \beta p(1 - p)$. Substitution of $p = Z_N^{-1}$ proves the second identity. Equivalently, $P^2 = P$ yields $\chi_P = \beta \text{Var}_{\rho_N}(P)$. $\square$

Proposition 15.3 (Relative entropy of preparation). *Let $\rho_P = P\rho_N P/p$. With $s(\sigma) = -\text{Tr}(\sigma \ln \sigma)$ and $\mathcal{G}(\sigma) = \text{Tr}(K\sigma) - \beta^{-1}s(\sigma)$,*

$$D(\rho_P \parallel \rho_N) = \ln Z_N, \quad \mathcal{G}(\rho_P) - \mathcal{G}(\rho_N) = \beta^{-1} \ln Z_N.$$

Proof. On the support of P , the relation $\rho_P = \rho_N/p$ gives $\ln \rho_P - \ln \rho_N = -\ln p$. Its trace against ρ_P is $-\ln p = \ln Z_N$. Moreover, $\ln \rho_N = -\beta K - \ln Z_N$ implies, for any state σ ,

$$D(\sigma \parallel \rho_N) = \beta [\mathcal{G}(\sigma) - \mathcal{G}(\rho_N)].$$

Taking $\sigma = \rho_P$ proves the energy identity. $\square$

The potential difference expresses reversible preparation work when the corresponding control and thermal and charge reservoirs are available. The identity describes that preparation; it does not assign a fixed cost to every observation. For an independent memory η , replacing (ρ_N, P) by $(\rho_N \otimes \eta, P \otimes I)$ preserves p , the susceptibility and the relative entropy; the state η remains intact.

15.12 Entropy and heat capacity of the graded carrier

In the spectral realization $H_{\text{rel}} = 3\epsilon N$, the condition $3\beta\epsilon = \ln 1000$ fixes $q = 10^{-3}$. The normalizer and the first two degree moments are

$$Z_N = 1.380649, \quad \langle N \rangle = \frac{381298}{1380649}, \quad \text{Var}(N) = \frac{382842620000}{1906191661201}.$$

Indeed, the probabilities of degrees 0, 1, 2 are $1/Z_N$, $380q/Z_N$ and $649q^2/Z_N$. Hence $\langle N \rangle = (380q + 1298q^2)/Z_N$ and $\langle N^2 \rangle = (380q + 2596q^2)/Z_N$; subtracting the squared mean gives the stated variance.

Holding the Hamiltonian and the carrier fixed while varying temperature, the entropy $S = k_B s(\rho_N)$ and heat capacity C_V are

$$\begin{aligned} \frac{S}{k_B} &= \ln Z_N + \ln 1000 \langle N \rangle = 2.230289295827403 \dots, \\ \frac{C_V}{k_B} &= (\ln 1000)^2 \text{Var}(N) = 9.58357621855691 \dots. \end{aligned}$$

The first formula follows from $\ln \rho_N = N \ln q - \ln Z_N$; the second uses $C_V = k_B \beta^2 \text{Var}(H_{\text{rel}})$. These are observables of this thermal realization. The spectral-origin shift recorded by 23 changes preparation energy and intensity relative to the reference; when held fixed, it changes neither the conditional probabilities nor the degree variance. The normalized distribution and the origin record thus retain distinct functions within the same state.

16 Algebraic construction of quantum completions over APP

APP and TPK provide oriented states, transport, and sheet parity. Their linear realization enables construction of the symmetric and exterior algebras, with their creation, annihilation, and occupation operators. This section establishes these algebraic constructions, the cellular field sector, and reversible measurement coupling. Transport of the completions under refinement is proved in Section 18; application of their occupation spectra to counting and equilibrium is developed in Sections 20–23. Each stage retains the conditions of the representation it uses.

16.1 Linear state space associated with a particle realization

Let S be a finite set of TPK route classes at the chosen depth, and let

$$\mathcal{H}_1 = \ell^2(S)$$

be the Hilbert space with orthonormal basis $(e_s)_{s \in S}$. A choice of positive weights E_s defines the diagonal operator

$$He_s = E_s e_s.$$

The choice of S , its quotient by symmetries, and the weights are part of the model. Once these are fixed, the multiparticle completions are canonical.

16.2 Symmetric and exterior completions

Define

$$\mathcal{F}_+(\mathcal{H}_1) = \bigoplus_{n \geq 0} \text{Sym}^n \mathcal{H}_1, \quad \mathcal{F}_-(\mathcal{H}_1) = \bigoplus_{n \geq 0} \bigwedge^n \mathcal{H}_1.$$

The first is the bosonic completion; the second is the fermionic completion. In both, the degree-zero summand is the vacuum space. The exterior powers are equipped with the inner product for which ordered wedges of vectors from an orthonormal basis are orthonormal; the symmetric powers are used with the usual normalization of creation and annihilation operators.

16.3 Sheet grading and exchange algebra

Suppose that the oriented sheet defines an additive character

$$p : \text{Mor}(\mathbf{P}_{\text{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}, \quad p(\gamma\eta) = p(\gamma) + p(\eta).$$

On the corresponding graded linear space, introduce the symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Theorem 16.1 (Statistical completion determined by parity). *The quotient of the tensor algebra by the relations*

$$v \otimes w - (-1)^{p(v)p(w)} w \otimes v$$

is symmetric on the even sector and exterior on the odd sector. Two identical odd states have zero product; even states admit arbitrary occupation.

Proof. If $p(v) = p(w) = 0$, the relation is $vw = wv$. If $p(v) = p(w) = 1$, it is $vw = -wv$; setting $v = w$ and working in characteristic zero gives $v^2 = 0$. Mixed sectors commute without an additional sign. This is the universal property of the graded symmetric algebra. $\square$

16.4 Canonical operators and idempotence of occupation

The parity character determines the completion; the product of that completion then determines the operator algebra. In the exterior sector, creation $a_s^\dagger$ is exterior multiplication by e_s , and a_s is its adjoint.

Theorem 16.2 (Pauli exclusion principle in the exterior completion). *For every mode $s \in S$, the creation and annihilation operators of the exterior completion satisfy*

$$\{a_s, a_t^\dagger\} = \delta_{st}I, \quad \{a_s, a_t\} = \{a_s^\dagger, a_t^\dagger\} = 0. \quad (16.1)$$

In particular, the occupation operator $N_s = a_s^\dagger a_s$ is a projector:

$$N_s^2 = N_s, \quad \text{Spec}(N_s) \subseteq \{0, 1\}. \quad (16.2)$$

The eigenvalues 0 and 1 correspond, respectively, to the absence and presence of mode s .

Proof. On an ordered wedge, a_s removes the factor e_s , when present, with the sign determined by its position; when absent, it gives zero. Composing this operation with exterior multiplication proves relations (16.1): for $s = t$, the two compositions cover the complementary cases of absence and presence; for $s \neq t$, the exchange signs cancel. Antisymmetry implies $e_s \wedge e_s = 0$ and $a_s^2 = (a_s^\dagger)^2 = 0$. Therefore

$$N_s^2 = a_s^\dagger a_s a_s^\dagger a_s = a_s^\dagger (I - a_s^\dagger a_s) a_s = a_s^\dagger a_s = N_s. \quad (16.3)$$

The operator is self-adjoint, and every eigenvalue of an idempotent satisfies $\lambda^2 = \lambda$. Hence its spectrum is contained in $\{0, 1\}$. The vacuum and the state e_s realize both values [8, 5]. $\square$

In the symmetric sector, the bosonic operators $b_s^\dagger, b_s$ act on the normalized occupation basis with factors $\sqrt{m_s + 1}$ and $\sqrt{m_s}$, respectively. Their compositions give

$$[b_s, b_t^\dagger] = \delta_{st}I, \quad [b_s, b_t] = [b_s^\dagger, b_t^\dagger] = 0. \quad (16.4)$$

These identities are understood on the algebraic finite-particle domain, which is invariant under the operators considered.

Theorem 16.1 turns a compositional TPK parity into an exact occupation rule. To obtain the physical spin–statistics theorem, one must also construct a positive local Lorentzian theory and prove that sheet parity agrees with spin under rotations. This equality does not follow merely from both objects taking values modulo two.

16.5 Sheet holonomy and $2\pi/4\pi$ periodicity

The preceding parity admits a natural geometric realization when a surviving extension has a closed return with a nontrivial orientation cover. Let ρ be the return generator and let

$$\varepsilon : \langle \rho \rangle \longrightarrow \{-1, +1\}$$

be its holonomy character, with $\varepsilon(\rho) = -1$. The associated real bundle is then the orientation line of a Möbius strip: one turn closes the geometric projection but changes the lifted sheet; two turns restore the orientation.

To express this property in angular terms, fix a unitary representation U of the rotation cover on the graded space $\mathcal{H}_1 = \mathcal{H}_0 \oplus \mathcal{H}_1$, compatible with sheet parity in the sense that

$$U(2\pi)v = (-1)^{p(v)}v$$

for every homogeneous vector v . Compatibility is a condition of the realization: it identifies the generator of the orientation cover with the character p already constructed on the routes.

Theorem 16.3 (Restoration of orientation and exchange algebra). *Under the preceding compatibility condition,*

$$U(2\pi)|_{\mathcal{H}_0} = I, \quad U(2\pi)|_{\mathcal{H}_1} = -I, \quad U(4\pi) = I.$$

Moreover, the same character p determines the graded symmetry

$$\tau(v \otimes w) = (-1)^{p(v)p(w)} w \otimes v.$$

Thus the even-holonomy sector is completed by symmetric powers and the odd-holonomy sector by exterior powers. In the latter, the exclusion relation $v \wedge v = 0$ holds.

Proof. The first and second equalities are the compatibility condition evaluated on each homogeneous component. Iterating one turn gives

$$U(4\pi)v = U(2\pi)^2v = (-1)^{2p(v)}v = v,$$

which proves restoration. The exchange statement is Theorem 16.1: for two even vectors the sign is positive, whereas for two odd vectors it is negative. $\square$

Three returns should be distinguished. Angular closure after 2π is a return of the projection; closure after 4π restores the lifted orientation; neither, by itself, requires every coordinate of the record to return to its initial value. This separation is the dimensional analogue of the HMT distinction between phase return and state return.

16.5.1 The sum and product sheets as supports of the 2 completions

The APP table contains two evaluations on the same support. The additive sheet acts by translations and preserves the multiplicity of continuations; in this sense, it accumulates. The multiplicative sheet permutes the units and contracts fibers over noninvertible elements; in this sense, it folds. This incompatibility precedes the statistical choice and explains why TPK must preserve two genealogies rather than reduce them to a single visible word.

The statistical assignment is not obtained, however, by verbally replacing “sum” with “boson” and “product” with “fermion.” The exact map is the parity character

$$p : \text{Mor}(\mathcal{P}_{\text{TPK}}) \longrightarrow \mathbb{Z}/2\mathbb{Z}.$$

When the transport of an extension lies in the sector $p = 0$, its completion admits cumulative occupation and is symmetric. When it lies in $p = 1$, sheet reversal introduces the exchange sign and the completion is exterior. Sum and product supply the two local geometries; the record, orientation, and character p determine the sector in which a route is realized.

The 4π periodicity and fermionic exclusion are coordinated by the same sheet character. Their consequences for finite counts and equilibrium occupations are obtained, with the Hamiltonian and thermal conditions explicit, in Section 20 and in Theorem 22.1.

16.6 The cellular complex of the APP torus

The APP graph is the one-skeleton of a cellular decomposition of the torus T^2 with 9×9 vertices. Let

$$C^0 \xrightarrow{d_0} C^1 \xrightarrow{d_1} C^2$$

be its real cochain complex. Choosing the uniform inner product, define the codifferential $\delta = d^*$ and the Laplacian

$$\Delta_k = d_{k-1}\delta_k + \delta_{k+1}d_k.$$

Theorem 16.4 (Harmonic sector of the digital torus). *The space of harmonic one-forms has dimension two:*

$$\dim \ker \Delta_1 = b_1(T^2) = 2.$$

A basis can be chosen that is represented by the uniform horizontal and vertical flows. Both are closed and coclosed, and neither is the gradient of a globally defined periodic function.

Proof. The Hodge theorem for finite cellular complexes identifies $\ker \Delta_1$ with $H^1(T^2; \mathbb{R})$. The cohomology of the torus has rank two. Directly, the constant horizontal and vertical flows have zero divergence and curl; their integrals over the two fundamental cycles are independent, so they are not exact. $\square$

The space of connections modulo gauge transformations is

$$C^1 / \text{im } d_0.$$

In a finite complex equipped with the preceding inner product, each orbit admits a Coulomb representative in $\ker \delta_1$; within the orbit, the choice is unique modulo the stabilizer $\ker d_0$ of the gauge parameter. The correct cohomological quotient is

$$\ker d_1 / \text{im } d_0 \simeq \ker \Delta_1.$$

Its two harmonic classes constitute a natural mathematical candidate for two free global modes. They are not yet the two local polarizations of a photon in $3 + 1$ -dimensional spacetime: that identification requires a dynamical limit, a dispersion relation, and a physical gauge group.

16.7 Discrete Gauss law

For an oriented 1-cochain F and a set of vertices V , divergence is defined by

$$(\text{div } F)(v) = \sum_{e: v \rightarrow w} F(e) - \sum_{e: w \rightarrow v} F(e).$$

Summing over V , every interior edge appears once with each sign and cancels. Therefore,

$$\sum_{v \in V} \text{div } F(v) = \text{Flux}_{\partial V}(F).$$

This identity is an exact Gauss law on the graph. Interpreting $\text{div } F$ as electric charge requires the charge character and dimensional realization constructed in Dimensional Mechanics.

16.8 Reversible measurement apparatus

Let S be a set of states and let $r : S \rightarrow \mathbb{Z}/9\mathbb{Z}$ be a readout. Coupling to a nonadic register is defined by

$$U_r : S \times \mathbb{Z}/9\mathbb{Z} \longrightarrow S \times \mathbb{Z}/9\mathbb{Z}, \quad U_r(s, a) = (s, a + r(s)).$$

Proposition 16.5 (Reversibility of the coupling). *The map U_r is a permutation, and*

$$U_r^{-1}(s, a) = (s, a - r(s)).$$

Information loss arises only after applying a reader to the register or conditioning the state on the outcome.

Proof. The proposed inverse satisfies both composition identities by the group law of $\mathbb{Z}/9\mathbb{Z}$. $\square$

This proposition formalizes the operational definition of measurement: coupling, reading, and restricting compatible futures are three different maps. The full coupling is reversible; the observational quotient need not be.

16.9 Algebraic correspondences and physical realizations

The Fock completions, exclusion conditional on parity, the harmonic sector of APP, discrete Gauss law, and reversibility of the apparatus are complete mathematical theorems. Their physical identifications form a diagram that must commute:

$$\text{sheet parity} \longrightarrow \text{graded exchange} \longrightarrow \text{CAR/CCR} \longrightarrow \text{physical statistics,}$$

$$H^1(\mathcal{A}_g) \simeq \ker \Delta_1 \hookrightarrow C^1 / \text{im } d_0 \longrightarrow \text{photonic realization.}$$

In the first line, the first two arrows have explicit constructions; the last requires the corresponding dynamics and equilibrium conditions. In the second, the Hodge identification and inclusion of harmonic classes in the space of connections modulo gauge are the constructions of Section 16.6. The zero eigenvalue belongs to the cellular Laplacian: identifying these modes with massless excitations and the two local photon polarizations additionally requires the dynamics, dispersion relation, and physical gauge realization specified there. This distinction allows the bridges to be developed without confusing an analogy in cardinality with a representation.

17 Arithmetic–geometric coupling and bilateral recovery

The dodecaphase record of Section 7 admits arithmetic, incidence-theoretic, and operator-theoretic readings. We call the generated record K the *holographic arithmetic–geometric coupling constant*, retaining the order of its twelve windows, the phase origin, and the orientation. Its rational reading $\kappa = K_{\text{int}}/(1000^{12} - 1)$ encodes that record; it does not replace its types or the enriched history that produces it. Hadamard inversion recovers its signed coordinate; the incidence-theoretic reading determines the Witt tetrad. The following constructions make two further functions explicit: identifying operators on its cyclic carrier and preserving information between two isometric realizations. These operations follow the realization of the continuum; they are not rules that introduce conventional constants into APP–TRIT–TPK.

17.1 Exact encoding and cyclic reference

Proposition 17.1 (Recovery of the record from its rational reading). *With base $b = 1000$, length twelve, origin, and orientation fixed, the map*

$$(k_1, \dots, k_{12}) \mapsto \kappa = \frac{\sum_{j=1}^{12} k_j b^{12-j}}{b^{12} - 1}, \quad 0 \leq k_j < b,$$

is injective. From κ , one recovers the integer $n = (b^{12} - 1)\kappa$ and, by successive Euclidean divisions, the twelve blocks, including leading zeros. An absolute error strictly smaller than $1/[2(b^{12} - 1)]$ permits the same recovery by rounding.

Proof. The fixed-length integer expansion in base b is unique. Two distinct integers are separated by at least one; their rational readings are separated by at least $1/(b^{12} - 1)$. The strict bound places the approximate datum within the rounding cell of a unique integer. $\square$

Now consider $K = (234, 543, 140, 729, 659, 824, 621, 58, 914, 794, 146, 601)$, the cyclic shift S of its positions, and $O_K = (K, SK, \dots, S^{11}K)$ on $\mathbb{R}^{12}$. The shift of positions is identified neither with the entire enriched evolution nor, solely by virtue of its periodicity, with an exceptional symmetry.

Theorem 17.2 (Invertibility of the orbital reference). *The matrix O_K is invertible, and*

$$589609\|c\|^2 \leq \|O_K c\|^2 \leq 6263^2\|c\|^2.$$

Its centred orbit has rank eleven.

Proof. The Gram matrix $O_K^* O_K$ is circulant. Its first row, obtained from the twelve inner products $\langle K, S^j K \rangle$, is

$$(4251257, 2999323, 3163385, 3287920, 3216553, 3344743, \\ 2950064, 3344743, 3216553, 3287920, 3163385, 2999323).$$

Evaluation at the twelve characters of the cycle yields the eigenvalues

$$6263^2, \quad 697225, \quad 589609 \ (2), \quad 1407529 \ (2), \quad 1053157 \ (2), \\ (1248025 - 345420\sqrt{3}) \ (2), \quad (1248025 + 345420\sqrt{3}) \ (2),$$

where parentheses indicate multiplicities. The smallest value is 589609: for the only comparison that is not immediate, $658416 > 345420\sqrt{3}$, as is seen by squaring both sides. The triangle inequality for the Fourier transform and $K_j \geq 0$ give the maximum $(\sum K_j)^2 = 6263^2$. Positivity of every mode proves invertibility and the bounds. Centring the record removes only the uniform mode. $\square$

Corollary 17.3 (Identification of an operator from its responses). *A linear operator $T : \mathbb{R}^{12} \rightarrow \mathbb{R}^{12}$ is determined by $Y = (TK, TSK, \dots, TS^{11}K)$, through $T = YO_K^{-1}$. If $TS = ST$, the complete vector response $y = TK$ suffices, since $T = O_y O_K^{-1}$. For a perturbation E of Y , $\|\hat{T} - T\| \leq \|E\|/\sqrt{589609}$.*

Proof. We have $Y = TO_K$. Under commutation, $TS^jK = S^jy$, hence $Y = O_y$. The bound follows from $\|O_K^{-1}\| = 1/\sqrt{589609}$. The norms are the Euclidean norm and its induced operator norm. $\square$

The response y contains twelve components. The statement does not attribute the recovery of an arbitrary operator to a single scalar: it specifies the observation and commutation condition that make identification possible.

17.2 Conservation and reconstruction of two channels

Let H, H' be Hilbert spaces, and let $\mathcal{B}, \mathcal{C} : H \rightarrow H'$ be bounded linear isometries. For $a > 0$, define

$$p = a^2 + a + 1, \quad N = (2a + 1)p = (a + 1)^3 + a^3, \quad Q_- = a\mathcal{B} - \mathcal{C}, \quad Q_+ = (a + 1)\mathcal{B} + \mathcal{C}.$$

Theorem 17.4 (Bilateral balance and recovery). *The following identities hold:*

$$(a + 1)Q_-^*Q_- + aQ_+^*Q_+ = NI_H, \quad W_a v = \frac{1}{\sqrt{N}} \begin{pmatrix} \sqrt{a+1} Q_- v \\ \sqrt{a} Q_+ v \end{pmatrix}, \quad W_a^* W_a = I_H.$$

The data $x = Q_- v$, $y = Q_+ v$ determine

$$\mathcal{B}v = \frac{x + y}{2a + 1}, \quad \mathcal{C}v = \frac{ay - (a + 1)x}{2a + 1}, \quad v = \mathcal{B}^* \frac{x + y}{2a + 1}.$$

Proof. Setting $X = \mathcal{B}^* \mathcal{C} + \mathcal{C}^* \mathcal{B}$, we obtain

$$Q_-^* Q_- = (a^2 + 1)I - aX, \quad Q_+^* Q_+ = ((a + 1)^2 + 1)I + (a + 1)X.$$

The weights cancel X , and the remaining coefficient is $2a^3 + 3a^2 + 3a + 1 = N$. Normalization gives the isometry. Adding the two channels recovers $(2a + 1)\mathcal{B}v$; the combination $ay - (a + 1)x$ recovers $(2a + 1)\mathcal{C}v$. Finally, $\mathcal{B}^* \mathcal{B} = I_H$. $\square$

The balance also holds in infinite dimension: only bounded operators are composed. The image of W_a is closed, and its orthogonal projector is $W_a W_a^*$. Thus a normalized pair has an input precisely when it belongs to that image; its orthogonal component measures the compatibility defect.

The factorization

$$W_a = R_a E_a, \quad E_a v = \frac{1}{\sqrt{p}} \begin{pmatrix} \sqrt{a(a+1)} \mathcal{B}v \\ \mathcal{C}v \end{pmatrix}, \quad R_a = \frac{1}{\sqrt{2a+1}} \begin{pmatrix} \sqrt{a} & -\sqrt{a+1} \\ \sqrt{a+1} & \sqrt{a} \end{pmatrix}$$

is verified by block multiplication; R_a acts on $H' \oplus H'$, with each scalar entry multiplying $I_{H'}$. In the nonadic partition $a = 8$, the weights are $72/73$ and $1/73$; the factor $73 = 8 \cdot 9 + 1$ follows from the general law $a(a + 1) + 1$, rather than being an additional primitive datum. The corresponding balance is $9\|Q_- v\|^2 + 8\|Q_+ v\|^2 = 17 \cdot 73 \|v\|^2$.

17.3 Transport of observables and incoming memory

Theorem 17.5 (Observables common to both channels). *For a bounded self-adjoint operator F on H' ,*

$$W_a^* \text{diag}(F, F) W_a = \frac{a(a + 1)\mathcal{B}^* F \mathcal{B} + \mathcal{C}^* F \mathcal{C}}{p}.$$

If $\mathcal{B}$ is unitary and $\mathcal{C} = \mathcal{B}R$, with R unitary, and $G = \mathcal{B}^ F \mathcal{B}$, the observable is preserved if and only if $GR = RG$.*

Proof. Expanding $(a + 1)Q_-^* F Q_- + aQ_+^* F Q_+$ cancels the cross terms. The other two coefficients are $a(a + 1)(2a + 1)$ and $2a + 1$. Dividing by N gives the formula. In the stated case, equality with G is equivalent to $R^* GR = G$; unitarity makes this equivalent to $GR = RG$. $\square$

The norm is obtained by taking $F = I$. A general observable requires the commutation condition: preserving the norm does not mean preserving every observable without transformation.

Theorem 17.6 (Unitary extension with memory). *If $H' = H$, $\mathcal{B} = I$, and $\mathcal{C} = U$ is unitary, set*

$$A = \frac{\sqrt{a+1}}{\sqrt{N}} Q_-, \quad D = \frac{\sqrt{a}}{\sqrt{N}} Q_+, \quad \mathcal{R}_a = \begin{pmatrix} A & -D^* \\ D & A^* \end{pmatrix}.$$

Then $\mathcal{R}_a$ is unitary on $H \oplus H$, its first column is W_a , and it preserves $\|v\|^2 + \|m\|^2$ for every input (v, m) . The inverse is

$$v = A^* z_1 + D^* z_2, \quad m = -D z_1 + A z_2.$$

Proof. The operators A, D are normal and commute with each other and with their adjoints, since they are polynomials in U, U^* . The balance gives $A^* A + D^* D = I$, and normality gives $AA^* + DD^* = I$. The cross terms in both products $\mathcal{R}_a^* \mathcal{R}_a$ and $\mathcal{R}_a \mathcal{R}_a^*$ vanish; the diagonal blocks are the identity. Applying the adjoint to (z_1, z_2) yields the inverse. $\square$

Record recovery, orbital identification, and unitary extension specify three distinct levels of information. The first preserves the blocks together with their chart; the second identifies operators under the stated hypotheses; the third transports state and memory jointly. This hierarchy allows the same generated structure to be used without confusing a reduced reading with the state in its entirety.

18 Transport of Fock completions under refinement

The completions and their operator algebras are constructed in Section 16. The result of this stage is their compatibility with refinement morphisms: transport of the one-particle fiber extends to all occupation sectors.

18.1 Second quantization of refinement morphisms

Let $\text{Hol}_{\pm}^{\leq 1}$ be the monoidal category of one-particle fibers, with contractive refinement morphisms preserving holonomy parity $\epsilon \in \{+1, -1\}$. It includes, in particular, unitary intertwiners. The monoidal operation on the fibers is direct sum; on the completions it induces the tensor product, graded in the exterior sector. Second quantization uses the completions of Section 16.2:

$$\mathfrak{F}(H, \epsilon) = \begin{cases} \mathcal{F}_+(H), & \epsilon = +1, \\ \mathcal{F}_-(H), & \epsilon = -1, \end{cases}$$

and, for a contractive intertwiner $T : H \rightarrow H'$,

$$\mathfrak{F}(T) = \bigoplus_{n \geq 0} \text{Sym}^n(T) \quad \text{or} \quad \bigoplus_{n \geq 0} \bigwedge^n T.$$

18.2 Functoriality, canonical operators, and transport domains

Let $\mathcal{D}_{\epsilon}(H)$ be the algebraic direct sum of the finite-particle sectors. In the following identities, $a_H^{\dagger}(f)$ and $a_H(f)$ denote the creation and annihilation operators of the chosen completion: the operators a of the exterior sector or the operators b of the symmetric sector in Section 16.4.

Theorem 18.1 (Internal transport of statistics). *The preceding completion defines a graded monoidal functor on $\text{Hol}_{\pm}^{\leq 1}$, with bounded operator values and compatibility with refinement. In the branch $\epsilon = -1$, creation and annihilation operators satisfy CAR and occupation operators obey $N_s^2 = N_s$; in the branch $\epsilon = +1$, they satisfy CCR. Thus Pauli exclusion and fermionic and bosonic statistics descend from holonomy parity once the one-particle representation has been fixed. For every contractive intertwiner $T : H \rightarrow H'$, $f \in H$, and $g \in H'$, the following hold on $\mathcal{D}_{\epsilon}(H)$:*

$$\mathfrak{F}(T)a_H^{\dagger}(f) = a_{H'}^{\dagger}(Tf)\mathfrak{F}(T), \quad (18.1)$$

$$a_{H'}(g)\mathfrak{F}(T) = \mathfrak{F}(T)a_H(T^*g). \quad (18.2)$$

If T is isometric, the second identity, with $g = Tf$, also transports annihilation of f . For $\|f\| = 1$, the occupation operators then satisfy

$$N_{H', Tf}\mathfrak{F}(T) = \mathfrak{F}(T)N_{H, f}, \quad N_{H, f} = a_H^{\dagger}(f)a_H(f). \quad (18.3)$$

Proof. Symmetric and exterior powers preserve compositions and identities; in each degree, their norms are bounded by $\|T\|^n$. Contractivity therefore ensures that the direct sum defines a bounded operator. The decomposition of powers of a direct sum according to their degrees in each factor gives monoidal compatibility, with the graded exchange sign in the exterior powers.

The CAR relations and idempotence are Theorem 16.2; the CCR relations are (16.4). To verify their transport, it suffices to act on symmetric or exterior products of vectors. Applying T to each factor commutes with inserting the factor f , which is transformed into Tf ; this proves (18.1). Annihilation sums the contractions of each factor and uses $\langle g, Tf \rangle = \langle T^*g, f \rangle$; exterior signs and symmetric normalizations agree on both sides. This yields (18.2). If $T^*T = I$, setting $g = Tf$ and composing the two identities proves (18.3). All these operations preserve the finite-particle domain. $\square$

For noncontractive morphisms, second quantization is defined first on $\mathcal{D}_\epsilon(H)$; its bounded extension depends on a uniform bound for the norms over all degrees. Such morphisms lie outside $\text{Hol}_\pm^{\leq 1}$. In the symmetric completion, if $\|T\| > 1$, the norms $\|\text{Sym}^n T\| = \|T\|^n$ grow without bound; in the exterior completion of a finite fiber, only finitely many degrees are nonzero. The general annihilation identity retains T^*g ; its specialization to the same transported vector corresponds to the isometry condition stated above.

The distinction is already verified in a one-dimensional fiber. If $T = tI$, with $0 < t < 1$, and e is a unit vector, then, on the one-particle state e , in either completion,

$$a(te)\mathfrak{F}(T)e = t^2\Omega, \quad \mathfrak{F}(T)a(e)e = \Omega, \quad (18.4)$$

where Ω is the vacuum. The general identity (18.2) correctly gives $\mathfrak{F}(T)a(T^*te)e = t^2\Omega$. The factor t^2 identifies precisely the hypothesis allowing specialization of annihilation transport and, subsequently, occupation transport.

The result establishes internal algebraic transport. Its identification with the relativistic spin-statistics theorem retains the additional conditions of Lorentzian representation and locality specified in Section 16.5.

19 Thermodynamic realization through Gibbs states and the pressure limit

19.1 Finite Hamiltonian, partition function, Gibbs state, and cofinal pressure limit

For a compressed Hamiltonian $H_{X,V}$ on a finite volume or refinement V , the thermal state is fully defined by

$$Z_{X,V}(\beta) = \text{Tr } e^{-\beta H_{X,V}}, \quad \rho_{X,V,\beta} = Z_{X,V}(\beta)^{-1} e^{-\beta H_{X,V}}.$$

The finite specialisation constructed in Section 15 takes $H_{X,V} = \epsilon_0 N$ and $x = e^{-\beta \epsilon_0}$. Hence (15.18) is precisely the substitution

$$Z_{X,V}(\beta) = Z_N(x), \quad \rho_{X,V,\beta} = \rho_N(x),$$

and the marker $p_0 = Z_N(x)^{-1}$ makes the normalisation recoverable. It does not define another statistic: it materialises this same Gibbs state on the graded carrier before the Fock completion is taken. The thermodynamic limit exists when the pressure

$$p_X(\beta) = \lim_{V \nearrow \infty} \frac{1}{\beta|V|} \log Z_{X,V}(\beta)$$

exists and is independent of the cofinal sequence.

19.2 Condensation criterion by saturation of the excited-state density

Bosonic condensation is determined by saturation of the excited-state density; it is not asserted merely from the existence of a symmetric completion. The problem is thus expressed by a verifiable mathematical condition on the Hamiltonian and the limit, rather than by a nominal attribution.

20 Statistical application of occupation spectra

The null and material realizations provide the geometric support for occupation. The graded completion of that support determines its multiparticle spaces and operators; Theorem 18.1 establishes their transport under refinement. This stage uses those results to characterize admissible configurations, count states, and prepare the thermal evaluation. The algebraic construction, finite counting, and equilibrium state thus retain their respective functions and premises.

20.1 Spectral restriction and admissible configurations

For a mode i , write $n_i = N_i = a_i^\dagger a_i$. Theorem 16.2, with its complete proof in (16.3), establishes $n_i^2 = n_i$ and $\text{Spec}(n_i) \subseteq \{0, 1\}$. The calculation uses only CAR and the adjoint relation, and therefore also applies to any representation of those relations. Nilpotence, $(a_i^\dagger)^2 = 0$, prevents double fermionic occupation of the same mode. Consequently, a fermionic configuration of g modes is a collection $(n_1, \dots, n_g) \in \{0, 1\}^g$; in the bosonic sector, occupations belong to $\mathbb{N}_0^g$.

The fermionic representation uses the CAR algebra constructed from the exterior completion. The relativistic spin–statistics identification additionally retains compatibility of the rotation representation with parity and the locality conditions stated in Sections 16.4 and 16.5. The 4π periodicity and exclusion are coordinated by the same sheet character, with these representation conditions explicit.

20.2 Dimensions of fixed-occupation sectors

If there are g modes and N particles, the sector dimensions are

$$\Omega_F(g, N) = \binom{g}{N}, \quad \Omega_B(g, N) = \binom{g + N - 1}{N}. \quad (20.1)$$

The first formula counts subsets of modes, with zero dimension if $N > g$; the second counts weak compositions of N into g parts. These counts are exact consequences of the two completions and are obtained before introducing the thermal state. Proposition 21.1 applies both counts to the finite spectral screen, retaining its levels, multiplicities, and energy and population constraints.

20.3 Thermal readout of the graded completions

With a diagonal Hamiltonian H , modal energy ϵ , chemical potential μ , and thermal parameter $\beta > 0$, the mean occupations of the two sectors are

$$\bar{n}_{\text{par}}(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} - 1}, \quad \bar{n}_{\text{impar}}(\epsilon) = \frac{1}{e^{\beta(\epsilon-\mu)} + 1}. \quad (20.2)$$

The bosonic sector requires $\mu < \inf \text{Spec } H$. These expressions are the per-mode specialization of Theorem 22.1, whose proof constructs the grand canonical trace from the admissible occupations. The Bose–Einstein and Fermi–Dirac formulas are therefore the thermodynamic evaluations of the symmetric and exterior completions selected by transport. The same section derives their expressions by varying the entropy under the energy and number constraints; Theorem 23.1 proves their common classical limit with an explicit error bound.

21 Microcanonical enumeration of states on a finite screen

Consider the finite realization

$$\mathcal{H}_{\text{occ}} = \bigoplus_{k=0}^3 \mathbb{C}^{g_k}, \quad H_{\text{occ}}|_{\mathbb{C}^{g_k}} = E_k^{\text{occ}} I,$$

with levels and multiplicities

$$E^{\text{occ}} = (0, 3, 6, 9), \quad g = (1, 6, 12, 8), \quad (N, E) = (12, 66).$$

The sum $1 + 6 + 12 + 8 = 27$ makes $\mathcal{H}_{\text{occ}}$ a twenty-seven-mode screen. This spectrum is the data of a specified finite representation; its cardinality does not suffice to identify it with any other occurrence of 27 in HMT. The morphism relating a particular boundary screen to $(\mathcal{H}_{\text{occ}}, H_{\text{occ}})$ must specify both the fibers and the spectral operator. The fermionic and bosonic generating functions are

$$F(y, x) = \prod_k (1 + yx^{E_k^{\text{occ}}})^{g_k}, \quad B(y, x) = \prod_k (1 - yx^{E_k^{\text{occ}}})^{-g_k}.$$

Proposition 21.1 (Microcanonical cardinalities).

$$[y^{12}x^{66}]F = 2\,104\,494, \quad [y^{12}x^{66}]B = 259\,436\,430.$$

Proof. In the fermionic product, choosing r of the g_k modes contributes $\binom{g_k}{r} y^r x^{rE_k^{\text{occ}}}$. In the bosonic product, distributing r indistinguishable occupations among g_k modes contributes $\binom{g_k+r-1}{r} y^r x^{rE_k^{\text{occ}}}$. Finite convolution over the four levels yields the stated integers. $\square$

22 Bose–Einstein and Fermi–Dirac distributions

The equilibrium formulation starts from the Hamiltonian and the total occupation number,

$$H = \sum_k E_k^{\text{occ}} n_k, \quad N = \sum_k n_k, \quad K_\mu := H - \mu N. \quad (22.1)$$

For $\beta > 0$, the grand canonical state is

$$\rho_{\beta, \mu} = \frac{e^{-\beta K_\mu}}{\mathcal{Z}_{\beta, \mu}}, \quad \mathcal{Z}_{\beta, \mu} = \text{Tr}(e^{-\beta K_\mu}). \quad (22.2)$$

Theorem 22.1 (Grand canonical factorization and occupation laws). *For energy levels E_k^{occ} with degeneracies g_k , the fermionic trace factorizes as*

$$\mathcal{Z}_{\text{FD}} = \prod_k (1 + e^{-\beta(E_k^{\text{occ}} - \mu)})^{g_k}, \quad \langle n_k \rangle_{\text{FD}} = \frac{g_k}{e^{\beta(E_k^{\text{occ}} - \mu)} + 1}. \quad (22.3)$$

In the bosonic sector, under the condition $\mu < \inf_k E_k^{\text{occ}}$,

$$\mathcal{Z}_{\text{BE}} = \prod_k (1 - e^{-\beta(E_k^{\text{occ}} - \mu)})^{-g_k}, \quad \langle n_k \rangle_{\text{BE}} = \frac{g_k}{e^{\beta(E_k^{\text{occ}} - \mu)} - 1}. \quad (22.4)$$

Proof. The operator K_μ is a sum of commuting occupation operators, so its trace factorizes over modes. In each fermionic mode, the occupations 0 and 1 are summed, and the corresponding factor is $1 + u_k$, where $u_k = e^{-\beta(E_k^{\text{occ}} - \mu)}$. In each bosonic mode, all nonnegative occupations are summed, giving the geometric series $(1 - u_k)^{-1}$; the condition on μ ensures $u_k < 1$. The powers g_k incorporate the degeneracy. Finally,

$$\langle n_k \rangle = -\frac{1}{\beta} \frac{\partial}{\partial E_k^{\text{occ}}} \log \mathcal{Z}$$

yields the two occupation laws [4, 6, 7, 5, 9]. $\square$

The microcanonical derivation recovers the same result in the thermodynamic regime. For mean fermionic occupations $p_k = m_k/g_k$, the combinatorial entropy per level is

$$S_k \simeq g_k [-p_k \log p_k - (1 - p_k) \log(1 - p_k)]. \quad (22.5)$$

Variation of $\sum_k S_k$, subject to $\sum_k g_k p_k = N$ and $\sum_k g_k E_k^{\text{occ}} p_k = E$, gives

$$\log \frac{1 - p_k}{p_k} = \lambda_N + \beta E_k^{\text{occ}}, \quad \mu = -\frac{\lambda_N}{\beta}, \quad (22.6)$$

and hence the Fermi–Dirac occupation in (22.3). The multipliers β and μ are the dual coordinates of the energy and population constraints. The symbol $\lambda_N = -\beta\mu$ denotes the multiplier associated with the occupation number; throughout this article, α remains reserved for the fine-structure constant.

For mean bosonic occupations $\bar{n}_k = m_k/g_k$, the logarithm of the level multiplicity is

$$S_k \simeq g_k [(1 + \bar{n}_k) \log(1 + \bar{n}_k) - \bar{n}_k \log \bar{n}_k]. \quad (22.7)$$

Variation of $\sum_k S_k$, subject to $\sum_k g_k \bar{n}_k = N$ and $\sum_k g_k E_k^{\text{occ}} \bar{n}_k = E$, yields

$$\log \frac{1 + \bar{n}_k}{\bar{n}_k} = \lambda_N + \beta E_k^{\text{occ}}, \quad \bar{n}_k = \frac{1}{e^{\beta(E_k^{\text{occ}} - \mu)} - 1}, \quad \mu = -\frac{\lambda_N}{\beta}. \quad (22.8)$$

This is the Bose–Einstein occupation in (22.4). The condition $\mu < \inf_k E_k^{\text{occ}}$ ensures positivity of the denominator and convergence of each modal factor.

In the finite instance constructed here, the two constraints determine

$$(\beta_{\text{FD}}, \mu_{\text{FD}}) = (0.15307011353415928, 4.502865197597043),$$

$$(\beta_{\text{BE}}, \mu_{\text{BE}}) = (0.05456727960068316, -15.92569781906737).$$

If m_k^{mic} denotes the mean occupation obtained by exact microcanonical enumeration and m_k^{eq} the evaluation of the corresponding distribution, the errors $\sum_k |m_k^{\text{mic}} - m_k^{\text{eq}}|$ are

$$0.09897380039002812 \quad \text{and} \quad 0.9190625715582017,$$

respectively. The enumeration traverses all occupation quadruples, calculates their integer multiplicities, and solves the two constraint equations. Both laws are realized on the same screen. The physical choice between them requires a grading and a Hamiltonian compatible with the trajectory representation. The thermal parameter is $\beta = (k_B \Theta)^{-1}$, with the dimensional transduction developed in Section 14.

23 Quantified convergence to the Maxwell–Boltzmann distribution

The relation between the two quantum statistics and the classical regime can be formulated without recourse to an informal approximation. For each mode, let

$$u_k := e^{-\beta(E_k^{\text{occ}} - \mu)}.$$

In the graded-carrier specialisation of Section 15, the excited degrees $n = 1, 2$ carry $E_n = n\varepsilon_0$, $\mu = 0$, and $\beta = \tau^{-1}$; therefore $u_n = x^n$, with $x = e^{-\varepsilon_0/\tau}$, and their multiplicities are 380 and 649. Degree $n = 0$ remains the marked origin. Thus (15.18) supplies the finite trace before completion, whereas the Bose–Einstein and Fermi–Dirac formulae remain the occupation laws of the completed modes. In particular, the bounds below apply to the excited sectors as $x \rightarrow 0$. The occupations per state are written

$$\bar{n}_k^{\text{FD}} = \frac{u_k}{1 + u_k}, \quad \bar{n}_k^{\text{BE}} = \frac{u_k}{1 - u_k};$$

the multiplicities g_k are incorporated afterward through $\langle n_k \rangle = g_k \bar{n}_k$.

Theorem 23.1 (Common classical limit of quantum statistics). *If $0 \leq u_k \leq \delta < 1$, then*

$$\left| \bar{n}_k^{\text{FD}} - u_k \right| \leq u_k^2, \quad \left| \bar{n}_k^{\text{BE}} - u_k \right| \leq \frac{u_k^2}{1 - \delta}.$$

Consequently, in the dilute regime $u_k \rightarrow 0$,

$$\boxed{\bar{n}_k^{\text{FD}} \sim \bar{n}_k^{\text{BE}} \sim e^{-\beta(E_k^{\text{occ}} - \mu)}},$$

which is the Maxwell–Boltzmann distribution per mode.

Proof. The exact identities

$$u_k - \frac{u_k}{1 + u_k} = \frac{u_k^2}{1 + u_k}, \quad \frac{u_k}{1 - u_k} - u_k = \frac{u_k^2}{1 - u_k}$$

give the bounds. For $0 < u_k \leq \delta$, dividing by u_k and taking $u_k \downarrow 0$ yields the asymptotic equivalence. The first term of order u_k^2 retains the physical difference: exclusion decreases fermionic occupation, whereas accumulation increases bosonic occupation. $\square$

24 Areal–volumetric incidence and topological entropy

The TPK calendar induces the hinge between the additive and multiplicative modes. In the primitive block

$$\tau_3 = (+1, -1, 0),$$

the rule

$$+1 : m \mapsto \Sigma, \quad -1 : m \mapsto \Pi, \quad 0 : m \mapsto m$$

produces the cyclic orbit (Σ, Π, Π) . The geometric reader

$$\Sigma \mapsto \mathcal{H}_{\text{ar}}, \quad \Pi \mapsto \mathcal{H}_{\text{vol}}$$

transports these modes to the areal and volumetric sheets.

Theorem 24.1 (Necessary incidence of the sheets). *In the order $(\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}})$, the primitive incidence matrix is*

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (24.1)$$

Calendars of lengths 9, 27, 108 yield

$$N_9 = 3F_{\text{av}}, \quad N_{27} = 9F_{\text{av}}, \quad N_{108} = 36F_{\text{av}}.$$

Proof. The consecutive pairs of the cycle are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Each occurs once, and $\Sigma \rightarrow \Sigma$ does not occur. This fixes all four entries of F_{av} . The longer calendars contain, respectively, three, nine, and thirty-six copies of the primitive block. $\square$

Theorem 24.2 (Fibonacci law and measure of maximal entropy). *For $n \geq 1$,*

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \quad (24.2)$$

The one-step subshift defined by F_{av} has

$$h_{\text{top}} = \log \varphi, \quad \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \varphi^{-2}. \quad (24.3)$$

Proof. The matrix identity follows by induction from $F_{n+1} = F_n + F_{n-1}$. The Perron root of F_{av} is φ , hence $h_{\text{top}} = \log \varphi$. Since the matrix is symmetric, the Parry weights [16] are proportional to the squared components of a Perron vector $(1, \varphi)$, that is, to $(1, \varphi^2)$. $\square$

The expressions $\log 3$ and $\log \varphi$ are thus distinguished by their genealogy. The first quantifies a uniform local choice among three states; the second measures the growth of histories allowed by sheet incidence. The areal–volumetric relation is a property of the complete return and therefore appears with exponent two.

24.1 Stationary distribution and variational characterization

Let X_{av} be the space of two-sided sequences over $\{\text{ar}, \text{vol}\}$ whose consecutive pairs have positive incidence in F_{av} . It is the one-step-memory envelope of the mode calendar: it retains the admissible pairs while allowing all concatenations compatible with them. Its stationary measure weights that space of histories; the chronological frequency of the orbit (Σ, Π, Π) remains, separately, $(1/3, 2/3)$.

Proposition 24.3 (Explicit characterization of the measure of maximal entropy). *With $r = (1, \varphi)$, the stochastic matrix and its stationary distribution are*

$$P_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i}, \quad P = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad \pi_{\text{av}} = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (24.4)$$

The stationary Markov measure μ determined by these data assigns to each admissible cylinder of n transitions

$$\mu[i_0 \dots i_n] = \frac{r_{i_0} r_{i_n}}{(1 + \varphi^2) \varphi^n}. \quad (24.5)$$

It is the unique stationary measure on X_{av} with maximal entropy rate, and that rate is $\log \varphi$.

Proof. The identity $\varphi^2 = \varphi + 1$ implies $\varphi^{-2} + \varphi^{-1} = 1$, so the rows of P sum to one. Direct substitution verifies $\pi_{\text{av}} P = \pi_{\text{av}}$. Both cross-transitions have positive probability; the chain is irreducible, and the self-transition at vol makes it aperiodic. The stationary distribution is also obtained by solving the two linear equations and the normalization, which give the unique solution displayed above. Multiplication of $\pi_{i_0} \prod_{k=0}^{n-1} P_{i_k i_{k+1}}$ cancels the intermediate factors r_{i_k} and proves (24.5).

For maximality, let ν be any stationary measure on X_{av} , and let $q(j \mid \mathcal{P})$ be the conditional distribution of X_1 given the entire past $\mathcal{P} = (\dots, X_{-1}, X_0)$. Its support is contained in that of the row $P_{X_0, \cdot}$. For finite distributions, the relative divergence $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is nonnegative and vanishes exactly when $q = p$. This property follows from concavity of the logarithm, with the convention $0 \log 0 = 0$. On admissible pairs,

$$\log P_{ij} = \log r_j - \log r_i - \log \varphi.$$

Stationarity cancels the expectation of $\log r_{X_1} - \log r_{X_0}$. Hence

$$h(\nu) = H_\nu(X_1 \mid \mathcal{P}) = \log \varphi - \mathbb{E}_\nu D(q(\cdot \mid \mathcal{P}) \parallel P_{X_0, \cdot}) \leq \log \varphi. \quad (24.6)$$

The identity between entropy rate and entropy conditional on the past follows from the chain rule on finite blocks and the decreasing limit of their conditional entropies; the finite alphabet ensures that all of them lie between zero and $\log 2$.

The constructed measure μ is Markov with transition matrix P , so the divergence term vanishes and $h(\mu) = \log \varphi$. If another stationary measure attains that rate, the divergence vanishes almost surely: its conditional distribution of the future depends only on X_0 and agrees with P . It is therefore the same stationary chain, whose unique initial distribution is π_{av} . This proves uniqueness. $\square$

Formula (24.6) specifies exactly what is maximized: entropy per transition of the stated symbolic envelope. The ratio $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$ arises from its stationary weights. The periodic chronological orbit has zero entropy rate and retains its own temporal distribution; both objects remain defined on their respective domains.

25 Angular coordinates and boundary observables

25.1 Angular coordinates, normalization of the turn, and units of action

The quantity of action and the angular coordinate belong to distinct spaces. The section $\hbar_{\text{ret}}$ occupies the action line $\mathcal{L}_S$; η_{ret} is its coefficient in the declared basis. Angular representation acts on the dimensionless pair $(\alpha, \eta_{\text{ret}})$, after its construction:

$$\mathcal{P}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (25.1)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units makes the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$ visible. On the positive branch of the article, the representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha).$$

Proposition 25.1 (Shift of the base-thousand representation). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. The lifted coordinate $A_{\text{deg}} = 1000\alpha$ satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

The representation preserves the block sequence from the second block onward; the initial block becomes the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; the subsequent circular representation takes its class modulo 360. $\square$

Definition 25.2 (Charts of the angular coordinate). The degree chart expresses a complete turn in 360 units. The radian chart expresses it as 2π . The coordinate changes and their inverses are

$$x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} x, \quad C_{\text{deg}}^* = \frac{180}{\pi} y.$$

On a path with preserved winding number, the lifted coordinates are used; the circular quotient expresses its class modulo the corresponding turn.

The definition characterizes degrees and radians through their exact relation to the turn. The record also retains orientation, turns, and transport memory. Representation of its angular class preserves phase, whereas the lift distinguishes paths with the same final phase. The physical reading of action requires specifying which of these coordinates enters the integral.

Proposition 25.3 (Covariance of the action reader). *For $h_a = 2\pi\hbar_a$, the action of a lifted coordinate satisfies*

$$\mathcal{S}_a(\theta) = \hbar_a \theta_{\text{rad}} = h_a \frac{\theta_{\text{deg}}}{360}. \quad (25.2)$$

A positive turn expresses h_a .

Proof. Substituting $\theta_{\text{rad}} = \pi\theta_{\text{deg}}/180$ gives $\hbar_a \pi\theta_{\text{deg}}/180 = (2\pi\hbar_a)\theta_{\text{deg}}/360$. In one turn $\theta_{\text{deg}} = 360$, yielding h_a . $\square$

The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operator restoration occurs at 4π . This property of the lift coexists with the per-turn normalization of (25.2). Each preserves its own object: integrated quantity, circular phase, and represented state.

25.2 Covariance of the channels and the incidence functional

The channels are written equivalently as

$$q_{\pm} = \exp[-(x \pm y)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (25.3)$$

Proposition 25.4 (Recovery of the even and odd coordinates). *In the lifted real chart, the positive channels uniquely determine the two coordinates:*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The exchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^* .

Proof. The sum and difference of the logarithms of (25.3) are, respectively, $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$. The real logarithm is single-valued on the positive channels. The product is symmetric, and the quotient is inverted when they are exchanged. $\square$

In the twelve-position record, the antipodal involution $r \mapsto r + 6 \pmod{12}$ forms six pairs. The uniform distribution of the odd coordinate over these pairs is $C_{\text{deg}}^*/6$: summing their six contributions recovers C_{deg}^* . This sectorial coordinate is distinguished from the Planck constant h , which belongs to the action line.

The conversion is applied only once to each argument. The quantity α remains dimensionless, and the quantity $\hbar$ continues to belong to the action line; their preceding coordinates enter the map $\mathcal{P}$.

In addition to the channels, the Barbero–Immirzi functional contains a component bilinear in the angular representatives. Its transport is

$$\frac{\pi}{180} A_{\text{deg}} C_{\text{deg}}^* = \frac{180}{\pi} xy. \quad (25.4)$$

The identity follows by substituting both degree representatives with their expressions in radians. The complete formula in analytic coordinates is therefore

$$\gamma_{\text{HMT}} = \frac{180}{\pi} xy + 12[S_{90}(e^{-x+y}) - S_{90}(e^{-x-y})] - [S_{120}(e^{-x+y}) - S_{120}(e^{-x-y})].$$

Equivalence of the two charts transports the bilinear component and the exponential functions simultaneously. This transformation makes explicit the precise role of the degree in the construction and allows the evaluation to be reproduced without ambiguity in angular units.

26 Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be a marked inclusion, $|H| = 6$, $|O| = 8$. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24 \binom{6}{2}} = -\frac{1}{12}.$$

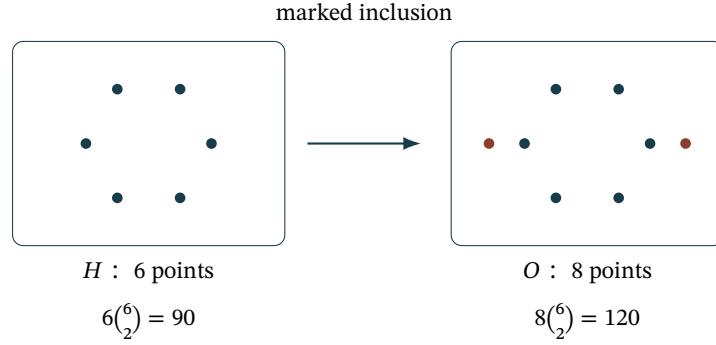


Figure 8: Marked incidence from six to eight points. The drawing represents the sets and the inclusion; it does not purport to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

26.1 The weights precede the pole coefficient

In the free module of oriented events, the source fixes the fundamental chain of the turn

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8].$$

Multiplicity twelve corresponds to the sectors, and multiplicity one to the oriented seam. The involution changes the orientation of the complete chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1.$$

The linear reader assigns S_{90} to each sector and S_{120} to the seam; consequently,

$$\mathcal{R}(c_{12}^+) = 12S_{90} - S_{120}.$$

Theorem 26.1 (Pole coefficient and conjugate evaluation). *The image of the fundamental chain has coefficient $1/24$ with respect to $(1 - q)^{-1}$ and residue $-1/24$ with respect to $(q - 1)^{-1}$. Its evaluation in the previously generated channels determines*

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (26.1)$$

Proof. The factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ implies $\lim_{q \uparrow 1} (1 - q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q - 1$ reverses the sign of the residue. Conjugate evaluation and the angular component of the source produce (26.1). $\square$

The Diophantine certificate $4a - 3b = 45$ admits the pairs $(12 + 3t, 1 + 4t)$, $t \geq 0$; it certifies minimality of $(12, 1)$, but does not replace the origin of the multiplicities in the chain. The evaluation of (26.1) is

$$\gamma_{\text{HMT}} = 0.27400767269445927474725526077398041940 \dots$$

This number is an output of the source functional; it is not an external value used to select A , C^* , or their coefficients.

26.2 Parity and structural sensitivity

Reversing C^* exchanges q_+ and q_- and changes the sign of the angular component. Hence

$$\gamma_{\text{HMT}}(A, -C^*) = -\gamma_{\text{HMT}}(A, C^*).$$

The functional preserves orientation, whereas other readers, such as q_+q_- , are even. This difference precedes their physical interpretation.

26.3 Constitutive speed and kinematic realization of the clock

Radiative propagation uses the speed section produced by the angular transport of the same HMT state. The identification is established before representation units are chosen. Let $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$ be the angular coordinates generated in Section 25.1, with the channels (25.3), and let R be the self-adjoint involution of the two sheets, with rank-one projectors P_+ and P_- . On the domain $x > |y|$, the transport and its channels are

$$T = \exp(-xI - yR) = q_+P_+ + q_-P_-, \quad q_+ = e^{-x-y}, \quad q_- = e^{-x+y}, \quad 0 < q_{\pm} < 1. \quad (26.2)$$

The response of the temporal blocks of lengths 90 and 120 is

$$V = (I - T^{90})(I - T^{120})^{-1} = r_+P_+ + r_-P_-, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \quad (26.3)$$

The construction starts from the inherited channels; the symbols x, y and $q_{\pm}$ denote their coordinates, not parameters fitted to a target speed.

Proposition 26.2 (Identity of the constitutive and kinematic realizations). *In the same dimensional chart, the clock section c_{int} and the constitutive-response section c_{vac} coincide:*

$$c_{\text{int}} = c_{\text{vac}} = \hat{c}u_C, \quad \hat{c} = (r_+r_-)^{-1} = \exp(-\mathcal{E}), \quad (26.4)$$

where

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})]. \quad (26.5)$$

The equality preserves sheet exchange.

Proof. The factors appearing in the logarithms are positive. Therefore,

$$\mathcal{E} = \log(r_+r_-) = \log \det V, \quad \exp(-\mathcal{E}) = (\det V)^{-1}.$$

The determinant is computed here in the two-sheet realization with rank-one projectors. The kinematic definition $c_{\text{int}} = \exp(-\mathcal{E})u_C$ and the constitutive definition $c_{\text{vac}} = (r_+r_-)^{-1}u_C$ therefore give the same section. Exchanging $P_+ \leftrightarrow P_-$ interchanges r_+ and r_- and preserves their product. Finally, $\det T = e^{-2x}$ corresponds to the elementary transport, whereas $\det V = r_+r_-$ corresponds to its temporal response; the two determinants have distinct roles in this composition. $\square$

26.3.1 Trajectory length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched trajectory γ of $n = |\gamma|$ transitions with a constant channel, the additive readers are

$$T_{\text{int}}(\gamma) = nu_T, \quad L_{\text{int}}(\gamma) = n\hat{c}u_L. \quad (26.6)$$

Their additivity concerns the count and realization of each transition: enriched routes additionally preserve orientation and memory.

Corollary 26.3 (Normalization of the elementary length). *For $n > 0$, $L_{\text{int}}(\gamma)/T_{\text{int}}(\gamma) = c_{\text{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\text{int}}t_0 = \hat{c}u_L. \quad (26.7)$$

In a general temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, the corresponding expression is $\ell_0 = \tau_0 \hat{c}u_L$.

Proof. Dividing the expressions in (26.6) cancels the integer n and uses $u_L/u_T = u_C$. The elementary formula is obtained by multiplying the speed section by t_0 . $\square$

Generalization to an edge-dependent channel preserves the additive formula $L_{\text{int}}(\gamma) = \sum_{e \in \gamma} \hat{c}(e)u_L$. Its quotient by nu_T is the mean of the speeds of those edges. The homogeneous radiative realization uses the constant channel specified in (26.6).

26.3.2 Change of chart and adapted normalization

Proposition 26.4 (Covariance of the identification). *Let $u'_C = du_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The same speed and the same elementary duration have coordinates*

$$\hat{c}' = d^{-1}\hat{c}, \quad \tau'_0 = \eta^{-1}\tau_0, \quad u'_L = d\eta u_L. \quad (26.8)$$

In particular, if $c_V = \hat{c}_V u_C^V$ and $u_C^V = du_C^{III}$, equality of the sections is exactly equivalent to $d\hat{c}_V = \hat{c}_{III}$.

Proof. Express each section in the two bases. This gives $\hat{c}'u'_C = \hat{c}u_C$ and $\tau'_0 u'_T = \tau_0 u_T$. Their product satisfies

$$\tau'_0 \hat{c}' u'_L = \frac{\tau_0}{\eta} \frac{\hat{c}}{d} d\eta u_L = \tau_0 \hat{c} u_L = \ell_0.$$

The final equivalence follows by substituting the relation between the bases. $\square$

The adapted choice $d = \hat{c}$ gives $u'_C = c_{\text{vac}}$ and $\hat{c}' = 1$. This choice expresses the already constructed section in a basis adapted to it. The spectral coefficient in the internal chart remains $(r_+ r_-)^{-1}$; spectral inversion uses that internal chart. For example, retaining the action and charge bases, the constitutive coordinates transform as

$$\hat{\mu}' = d\hat{\mu}, \quad \hat{\varepsilon}' = d\hat{\varepsilon}, \quad \hat{\mu}'\hat{\varepsilon}' = (\hat{c}')^{-2}.$$

In the adapted chart, $\hat{\mu}' = r_-/r_+$ and $\hat{\varepsilon}' = r_+/r_-$, since the internal coordinates are $\hat{\mu} = r_-^2$ and $\hat{\varepsilon} = r_+^2$.

26.3.3 Oriented Catalan moment and functional bridge

The constitutive response does not receive Catalan's constant as an external number. The dodecaphasic quarter turn produces the collection of oriented moments

$$\mathcal{C}_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad \mathcal{D} = s \frac{d}{ds}, \quad 0 < s < 1. \quad (26.9)$$

Absolute convergence on compact subsets permits termwise differentiation and gives

$$\mathcal{D}^2 \mathcal{C}_2(s) = \frac{s}{1+s^2}, \quad f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 \mathcal{C}_2(s) = \frac{1-s^3}{1-s^4}. \quad (26.10)$$

With $s_{\pm} = q_{\pm}^{30}$, the incidences $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$ produce exactly

$$r_{\pm} = f(s_{\pm}), \quad \mathcal{C}_2(1) = G_{\text{Cat}}. \quad (26.11)$$

The endpoint value is the representation of the collection; the functional collection, rather than that value alone, governs the two channels.

The bridge is reversible within the fixed regularity class. Given f , the condition $\mathcal{C}_2(s) \rightarrow 0$ as $s \downarrow 0$ uniquely selects

$$\mathcal{C}_2(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) dt. \quad (26.12)$$

Indeed, this integral satisfies $\mathcal{D}^2 \mathcal{C}_2 = s(1+s)f/(1+s+s^2)$; for (26.10), the integrand reduces to $\log(s/t)/(1+t^2)$. Two homogeneous solutions differ by $a \log s + b$, and null regularity at the origin forces $a = b = 0$.

The same dilogarithmic collection organizes a joint potential. Let $\mathcal{D}_2(z) = \sum_{n \geq 1} z^n/n^2$ and

$$\mathcal{F}(x, y) = \sum_{\sigma=\pm} \left[\frac{\mathcal{D}_2(q_{\sigma}^{120})}{120} - \frac{\mathcal{D}_2(q_{\sigma}^{90})}{90} \right]. \quad (26.13)$$

Since $d\mathcal{D}_2(z)/d \log z = -\log(1-z)$ and $q_{\pm} = e^{-x \mp y}$,

$$\partial_x \mathcal{F} = \log \hat{c}, \quad \partial_y \mathcal{F} = \log \hat{Z}, \quad \hat{Z} = \frac{r_-}{r_+}. \quad (26.14)$$

Thus speed and impedance are the two conjugate derivatives of a single potential, whose additive constant is fixed by $\mathcal{F} \rightarrow 0$ as $q_{\pm} \rightarrow 0$.

26.3.4 Joint reconstruction from speed and Barbero–Immirzi

Let $\psi = f^{-1}$ and define

$$\mathcal{H}_B(s) = \frac{12s^3}{1-s^9} - \frac{s^4}{1-s^{12}} - \frac{(\log s)^2}{20\pi}. \quad (26.15)$$

The formula in (26.1), together with (26.11), factorizes exactly as

$$\gamma_{\text{HMT}} = \mathcal{H}_B(\psi(r_-)) - \mathcal{H}_B(\psi(r_+)). \quad (26.16)$$

Theorem 26.5 (Joint reconstruction of the labelled channels). *In the normalized constitutive collection, the map*

$$\Psi : (3/4, 1)^2 \longrightarrow (1, 16/9) \times \mathbb{R}, \quad (r_+, r_-) \longmapsto (\hat{c}, \gamma_{\text{HMT}}) \quad (26.17)$$

is a diffeomorphism. Hence $(\hat{c}, \gamma_{\text{HMT}})$ reconstructs the two labelled eigenvalues; the projectors $P_{\pm}$ remain data of the initial representation.

Proof. The regular expression $f(s) = (1 + s + s^2)/(1 + s + s^2 + s^3)$ satisfies

$$f'(s) = -\frac{s^2(s^2 + 2s + 3)}{(1 + s + s^2 + s^3)^2} < 0,$$

with $f(0^+) = 1$ and $f(1^-) = 3/4$. Moreover,

$$\mathcal{H}'_B(s) = \frac{36s^2(1 + 2s^9)}{(1 - s^9)^2} - \frac{4s^3(1 + 2s^{12})}{(1 - s^{12})^2} - \frac{\log s}{10\pi s} > 0. \quad (26.18)$$

The ratio of the first positive term to the second before subtraction is

$$\frac{9}{s} \frac{1 + 2s^9}{1 + 2s^{12}} \left(\frac{1 - s^{12}}{1 - s^9} \right)^2 > 9,$$

and the last summand is also positive. Therefore $F_B = \mathcal{H}_B \circ \psi$ is strictly decreasing, with limits $F_B(3/4^+) = +\infty$ and $F_B(1^-) = -\infty$.

Fix $c_0 = \hat{c}$ and write $z = \log \hat{Z}$. Then $r_{\pm} = c_0^{-1/2} e^{\mp z/2}$ and

$$|z| < a(c_0) := \min \left\{ \log c_0, \log \frac{16}{9c_0} \right\}.$$

The function $\Gamma_{c_0}(z) = F_B(r_-) - F_B(r_+)$ satisfies

$$\Gamma'_{c_0}(z) = \frac{1}{2} [r_- F'_B(r_-) + r_+ F'_B(r_+)] < 0$$

and ranges over $\mathbb{R}$ between the two admissible endpoints. There is a unique solution for every γ_{HMT} ; the nonzero derivative and the implicit function theorem provide the smooth global inverse. $\square$

On the image produced by the HMT generator, reconstruction preserves causal order. Once $r_{\pm}$ have been recovered, one obtains $s_{\pm} = \psi(r_{\pm})$ and

$$A_{\text{deg}} = -\frac{3}{\pi} \log(s_+ s_-), \quad C_{\text{deg}}^* = \frac{3}{\pi} \log \frac{s_-}{s_+}, \quad (26.19)$$

whence, on the already generated positive orientation, $\alpha_{\text{HMT}} = A_{\text{deg}}/1000$ and $\eta_{\text{ret}} = C_{\text{deg}}^*/2 - \alpha_{\text{HMT}}$. This path reconstructs representations of the state; it does not turn $(\hat{c}, \gamma_{\text{HMT}})$ or conventional values into inputs of the generator.

26.3.5 Transport to the radiative sector

The dispersion relation of the homogeneous radiative realization is written $\omega_{\mathbf{k}} = c_{\text{vac}}|\mathbf{k}|$. With $h = 2\pi\hbar$ and $\beta = (k_B\Theta)^{-1}$, its spectral density and Stefan–Boltzmann coefficient have the expressions

$$u_\omega = \frac{\hbar\omega^3}{\pi^2(\hat{c}u_C)^3} \frac{1}{e^{\beta\hbar\omega} - 1}, \quad \sigma = \frac{\pi^2 k_B^4}{60\hbar^3(\hat{c}u_C)^2}. \quad (26.20)$$

These substitutions transport precisely the same speed section. The conditions of the radiative realization are retained: a free three-dimensional bosonic sector, two transverse polarizations, zero chemical potential, and positive temperature. Identity (26.4) establishes its constitutive antecedent; mode counting and the thermodynamic limit belong to the specific proofs for that realization.

Normalization under amplification. If refinement acts by $V_m = V \otimes I_{9^m}$, the normalized trace $\tau_m = \text{Tr} / (2 \cdot 9^m)$ satisfies

$$\exp[-2\tau_m(\log V_m)] = \exp[-\log r_+ - \log r_-] = \hat{c}.$$

Indeed, each eigenvalue appears 9^m times. The multiplicity cancels in the normalized trace. The ordinary amplified determinant is $(r_+r_-)^{9^m}$: the two-sheet normalization, or equivalently the normalized trace, preserves the constitutive reader under that refinement. This amplification of the representation preserves the enriched transport from which it arises.

27 Action, structural length, gravitation, and thermal information

27.1 Registered return and construction of length

Write α_{HMT} for the already generated dodecaphasic coordinate —denoted α in the preceding sections—. No metrological value of G or tabulated Planck length enters here. The marked record comprises

$$N = 12 \cdot 4 \cdot 120 = 5760 \quad (27.1)$$

positions. Let $u_\Omega = N^{-1/2} \mathbf{1}_\Omega$, $Q_\Omega = I - u_\Omega u_\Omega^*$, and let $B : V_U \rightarrow V_\Omega$ be the typed Hadamard return, with

$$B^*B = BB^* = \frac{1}{4}I, \quad C = Q_\Omega B = BQ_U. \quad (27.2)$$

For the diagonal pinching $\Delta_\Omega(A) = \sum_{i \in \Omega} E_i A E_i$, one obtains

$$\Delta_\Omega(CC^*) = \frac{N-1}{4N}I = r_N I, \quad r_N = \frac{5759}{23040}, \quad (27.3)$$

whereas the complementary channel retains $1/(4N)$. Indeed, $CC^* = Q_\Omega/4$, and every diagonal entry of Q_Ω equals $1 - 1/N$. Diagonal pinching determines the longitudinal response; the enlarged record and its complement retain the memory that the diagonal readout does not distinguish.

Let $\mathcal{L}_L$ be the oriented length line and let $L_* > 0$ be its reference section transported by the longitudinal cochain. This section is not silently identified with the native edge basis: its relation to the clock is fixed later by the circular realization. The already constructed H_4 -cokernel norm supplies $N_{H_4}(\alpha_{\text{HMT}}) = \alpha_{\text{HMT}}^{16}$. Hence the linear composition of the registered return determines

$$L_\alpha = \alpha_{\text{HMT}}^{16} r_N L_* = \alpha_{\text{HMT}}^{16} \frac{5759}{23040} L_*. \quad (27.4)$$

The power, the incidence coefficient, and the dimensional section have separate and explicit origins. L_α is a longitudinal output prior to the gravitational readout.

27.2 Operator composition and unique determination of gravitation

Let $\mathcal{H} \neq \{0\}$ be a finite Hilbert fiber and let $Y = UY_0 U^* > 0$ be the positive invertible character of a route, with its transport, sheet, orientation, and memory retained. For an action section $\hbar_a > 0$, $a \in \{\text{pre}, \text{ret}\}$, and the speed $c = c_{\text{int}} = c_{\text{vac}} = \hat{c} u_C > 0$ already constructed in (26.4), set

$$Q_{L,a} = \frac{\hbar_a c}{L_\alpha}, \quad H_a = Q_{L,a} Y, \quad M_a = \frac{H_a}{c^2}, \quad \Lambda_a = L_\alpha Y^{-1}, \quad R_a = L_\alpha Y. \quad (27.5)$$

The null sector is kept separate and is not inverted.

Theorem 27.1 (Uniqueness of the gravitational coefficient in the radial realization). *The preceding readers satisfy*

$$H_a \Lambda_a = \hbar_a c I, \quad R_a \Lambda_a = L_\alpha^2 I. \quad (27.6)$$

If the declared physical reading of the reduced radius is $R_a = (G_a/c^2)M_a$, then there is a unique compatible coefficient, namely

$$G_a = \frac{c^3 L_\alpha^2}{\hbar_a}, \quad \frac{G_a \hbar_a}{c^3} = L_\alpha^2. \quad (27.7)$$

Proof. The definitions give $H_a \Lambda_a = Q_{L,a} L_\alpha I = \hbar_a c I$ and $R_a \Lambda_a = L_\alpha^2 I$. Eliminating Λ_a yields

$$R_a = \frac{L_\alpha^2}{\hbar_a c} H_a.$$

On the other hand, $(G_a/c^2)M_a = G_a H_a/c^4$. Since H_a is invertible on the nonzero fiber, the two expressions agree if and only if $G_a = c^3 L_\alpha^2/\hbar_a$. The same cancellation shows that every other G'_a preserving both relations satisfies $G'_a = G_a$. $\square$

The proof uses the two readers jointly on the same character; it does not identify a compression of Y^{-1} with the inverse of a compression. A route variation or memory elimination acting identically on H_a and R_a preserves their proportionality and introduces no new scalar factor. Dimensional analysis confirms the exponents $c^3 L_\alpha^2 \hbar_a^{-1}$, but it is (27.6), rather than that dimensional check, that fixes the dimensionless coefficient to one.

27.3 The circular expression as a corollary

The calendar determines $T_{108} = 108t_0$ and $\omega_{108} = 2\pi_{\text{HMT}}/T_{108}$. In the circular realization of the same longitudinal reader,

$$\omega_{108} = \frac{c}{L_\alpha}, \quad R_{108} = \frac{c}{\omega_{108}} = \frac{54}{\pi_{\text{HMT}}} ct_0 = L_\alpha, \quad t_0 = \frac{\pi_{\text{HMT}} L_\alpha}{54c}. \quad (27.8)$$

This equality declares compatibility between the lifted clock advance and the longitudinal section; it does not use G_a to select L_α . Substitution into (27.7) gives

$$G_a = \left(\frac{54}{\pi_{\text{HMT}}} \right)^2 \frac{c^5 t_0^2}{\hbar_a}. \quad (27.9)$$

Consequently, in the parametrized expression $G_a(\vartheta) = \vartheta c^5 t_0^2 / \hbar_a$, the value $\vartheta = (54/\pi_{\text{HMT}})^2$ is a corollary of the constructed length, not an independent circular selector.

With $E_{108,a} = E_{P,a} = Q_{L,a} = \hbar_a \omega_{108}$ and $m_{108,a} = E_{108,a}/c^2$, one also retains

$$\frac{\hbar_a}{m_{108,a} c} = R_{108} = L_\alpha, \quad \frac{\hbar_a}{m_{108,a} c} = 2\pi_{\text{HMT}} L_\alpha.$$

Finally, the two action orientations satisfy

$$\frac{G_{\text{pre}}}{G_{\text{ret}}} = \frac{\hbar_{\text{ret}}}{\hbar_{\text{pre}}} = R_{\text{act}}^{-1}, \quad \ell_P^2 := L_\alpha^2 = \frac{G_a \hbar_a}{c^3}. \quad (27.10)$$

The areal unit thus originates in L_α^2 ; the later reciprocity between action and gravitation preserves that area under a change of orientation.

28 Area operator and reconstruction of sectorial information

Regrouping six trits constructs a bijection between $(\mathbb{Z}/9\mathbb{Z})^3$ and $(\mathbb{Z}/27\mathbb{Z})^2$, both of cardinality 729. The bijection is not presented as an additive isomorphism. On a 9×9 boundary block, the source decomposes 36 links into two sets of 18, labeled by channels 90 and 120.

28.1 Tritic regrouping and boundary incidence

The correspondence preserves an ordered word, in addition to its cardinality. Write each coordinate $x_j \in \mathbb{Z}/9\mathbb{Z}$, for $j = 1, 2, 3$, through its unique positional representatives $x_j = r_{2j-1} + 3r_{2j}$, with $r_k \in \{0, 1, 2\}$. The map is

$$\beta_{729}(x_1, x_2, x_3) = (r_1 + 3r_2 + 9r_3, r_4 + 3r_5 + 9r_6) \in (\mathbb{Z}/27\mathbb{Z})^2. \quad (28.1)$$

Proposition 28.1 (Positional correspondence between interior and boundary). *The map β_{729} is bijective and preserves the order of the six trits. Its inverse extracts the two triples of digits and regroups them into three pairs.*

Proof. The positional expansion of the representatives $0, \dots, 8$ and $0, \dots, 26$ is unique. Extracting the six digits, regrouping them, and extracting them again returns the same word. The inverse operation thus restores each of the three initial coordinates. $\square$

Carry preserves its position in the enriched word. The additive operations of the two packagings have different boundaries: for example, $\beta_{729}(0, 2, 0) = (18, 0)$ and $\beta_{729}(0, 1, 0) = (9, 0)$, whose sum in the order-twenty-seven torus is $(0, 0)$, whereas $\beta_{729}(0, 3, 0) = (0, 1)$. This distinction allows the positional correspondence to be reused while keeping the operations typed.

In the torus $(\mathbb{Z}/27\mathbb{Z})^2$, the block $B_\partial = \{0, \dots, 8\}^2$ has the oriented link sets

$$\partial B_{90} = \{((-1, j), (0, j)), ((8, j), (9, j)) : 0 \leq j \leq 8\}, \quad (28.2)$$

$$\partial B_{120} = \{((i, -1), (i, 0)), ((i, 8), (i, 9)) : 0 \leq i \leq 8\}. \quad (28.3)$$

All coordinates are read modulo twenty-seven. Each collection contains eighteen links, and the two are disjoint. Their labels preserve the incidence channels; the number of links is $18 + 18$.

Theorem 28.2 (Minimum capacity of the cubic cut). *In the cubic graph with vertices $\{0, \dots, N-1\}^3$, with $N \geq 2$, links between neighbors, and unit capacity per link, the minimum cut between two opposite faces has capacity N^2 . For $N = 9$, a maximally mixed product state on the trits of a minimum layer has information*

$$\text{cap}_{\min} = 81, \quad s_{\text{corte}} = 81 \log 3. \quad (28.4)$$

Proof. The N^2 lines parallel to the normal of the faces form edge-disjoint paths. Every separating cut intercepts each path, so it contains at least N^2 links. The links crossing a transverse layer constitute a cut of that cardinality. Each uniform trit contributes $\log 3$ nats; entropy is additive for the product state of the eighty-one trits of the layer. $\square$

The capacity of the cubic cut and the state of the thirty-six links in (28.3) belong to observables defined on different cuts. The genealogy preserves both domains and their measures.

28.2 Transport of the collective incidence modes

The spaces $\mathcal{F}_6$ and $\mathcal{F}_8$ of Section 26 provide the uniform vectors

$$u_{90} = \frac{1}{\sqrt{90}} \sum_{f \in \mathcal{F}_6} (\delta_f, 0), \quad u_{120} = \frac{1}{\sqrt{120}} \sum_{f \in \mathcal{F}_8} (0, \delta_f)$$

in $\ell^2(\mathcal{F}_6) \oplus \ell^2(\mathcal{F}_8)$. In $\ell^2(\partial B_{90}) \oplus \ell^2(\partial B_{120})$, construct

$$v_{90} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{90}} (\delta_e, 0), \quad v_{120} = \frac{1}{\sqrt{18}} \sum_{e \in \partial B_{120}} (0, \delta_e).$$

The two pairs are orthonormal. Denote their respective spanned planes by $\mathcal{H}_{\text{inc}}^{\text{coll}}$ and $\mathcal{H}_{\partial}^{\text{coll}}$.

Theorem 28.3 (Collective intertwining of the sectorial labels). *There is a unique surjective linear isometry between these planes such that*

$$\mathcal{J}_{6 \rightarrow 8} u_m = v_m \quad (m = 90, 120). \quad (28.5)$$

For the operators

$$\begin{aligned} D_{\text{inc}} &= 90|u_{90}\rangle\langle u_{90}| + 120|u_{120}\rangle\langle u_{120}|, \\ D_{\partial} &= 90|v_{90}\rangle\langle v_{90}| + 120|v_{120}\rangle\langle v_{120}|, \end{aligned}$$

the following identity holds:

$$\mathcal{J}_{6 \rightarrow 8} D_{\text{inc}} = D_{\partial} \mathcal{J}_{6 \rightarrow 8}. \quad (28.6)$$

The sectorial projectors and their oriented combination with coefficients 12 : −1 are also transported.

Proof. The image of a basis fixes the unique linear map. Norms and inner products are preserved because both bases are orthonormal. Identity (28.6) holds on each vector u_m , since its two sides are mv_m . The projectors are expressed as $(120I - D)/30$ and $(D - 90I)/30$; their linear functional calculus and any combination of the two preserve intertwining. $\square$

This transport acts on the two uniform modes and their spectral labels. Individual incidences and fluctuations within each sector remain in their complete spaces. The link preserves exactly the collective datum that the areal weighting will use.

28.3 Angular character and area operator

Let $N_e = \text{diag}(0, 1, 2)$ on each link, and let N_{90}, N_{120} be the sector sums. With the scale

$$\theta_{\text{clk}} = \frac{C^*}{6} \frac{\pi}{180}, \quad a_0 = \frac{1 + 2 \cos(10^\circ)}{3} \theta_{\text{clk}},$$

fixed, the angular factor is the normalized real character of the triplet

$$\chi_{10} = \frac{1}{3} \text{Tr} \text{diag}(1, e^{i\pi/18}, e^{-i\pi/18}) = \frac{1 + 2 \cos(10^\circ)}{3}, \quad a_0 = \chi_{10} \theta_{\text{clk}}. \quad (28.7)$$

The sum of the conjugate phases proves the equality. On the positive branch considered, $\theta_{\text{clk}} > 0$, $\chi_{10} > 0$, and $\gamma_{\text{HMT}} > 0$; hence a_0 and the areal scale are positive. The angular triplet and its normalization precede the entropic distribution on the links. With these quantities, the normalized area operator is

$$\hat{\mathcal{A}}/\ell_P^2 = \gamma_{\text{HMT}} a_0 (N_{90} + N_{120}).$$

Its spectrum has values $n\gamma_{\text{HMT}}a_0$, $0 \leq n \leq 72$, with multiplicities given by $(1 + z + z^2)^{36}$. The proof consists in summing the eigenvalues 0, 1, 2 of the 36 tensor factors.

Theorem 28.4 (Recovery of the boundary sectors). *Let*

$$N = N_{90} + N_{120}, \quad D = 90N_{90} + 120N_{120}.$$

Then

$$N_{90} = 4N - \frac{D}{30}, \quad N_{120} = \frac{D}{30} - 3N. \quad (28.8)$$

Thus the total and the weighted reader jointly recover the two occupations.

Proof. The matrix $\begin{pmatrix} 1 & 1 \\ 90 & 120 \end{pmatrix}$ has determinant 30. Its inversion gives (28.8). On the domain of integer occupations, the image satisfies $D \in 30\mathbb{Z}$ and $90N \leq D \leq 120N$. $\square$

The modular Hamiltonian in the source is $K_A = -\log \rho_A = cI + \Delta_{\text{mod}}D$. For $\Delta_{\text{mod}} \neq 0$ and known normalization, $D = (K_A - cI)/\Delta_{\text{mod}}$ expresses the second observable. This operator describes the spectral structure of the reduced state; it is not identified by notation with the generator of temporal evolution. Area preserves total occupation, and the pair adds provenance. For example, $(N_{90}, N_{120}) = (1, 0)$ and $(0, 1)$ share $N = 1$, but have $D = 90$ and $D = 120$.

28.4 State normalization and the meaning of the modular Hamiltonian

The second observable is made explicit through the state that defines it. The preceding tensor decomposition is retained: eighteen qutrits per sector, each with number operator $\text{diag}(0, 1, 2)$; N_m is the sum of these operators, and the sectors act on distinct factors. Let

$$z_m = 1 + e^{-m\Delta_{\text{mod}}} + e^{-2m\Delta_{\text{mod}}}, \quad m \in \{90, 120\}.$$

The state collection in the source has the product realization

$$\rho_A = Z^{-1} e^{-\Delta_{\text{mod}}(90N_{90} + 120N_{120})}, \quad Z = z_{90}^{18} z_{120}^{18}.$$

Commutativity of the number operators and tensor factorization prove the expression for Z . The state has full rank for real, finite Δ_{mod} , and its spectral logarithm gives

$$K_A = -\log \rho_A = (\log Z)I + \Delta_{\text{mod}}D.$$

For $\Delta_{\text{mod}} \neq 0$, K_A weights the excitations of the two sectors differently. The joint reading with N recovers the two occupations through equation (28.8).

The need for the joint reading is seen in two complementary examples. The occupation states $(1, 0)$ and $(0, 1)$ have the same total $N = 1$ and different modular weights. Conversely, $(4, 0)$ and $(0, 3)$ have the same $D = 360$ and different totals. The pair (N, D) separates both cases. This recovery concerns sectorial occupations, not each microscopic vector within a degeneracy of those occupations.

When $\Delta_{\text{mod}} = 0$, the state is proportional to the identity and K_A is scalar: that reading no longer distinguishes the sectors. For $\Delta_{\text{mod}} \neq 0$, the known normalization makes it possible to recover D and compose it with area. This defines the role of the modular Hamiltonian: it characterizes the reduced state and its information distribution. Its inclusion in the article is motivated by that mathematical role, whereas the evolution Hamiltonian belongs to the dynamical development of action.

28.5 Reduced state, Schmidt decomposition, and entanglement entropy

If $\rho_A = \sum_\nu p_\nu |\nu\rangle\langle\nu|$, the purification

$$|\Psi\rangle = \sum_\nu \sqrt{p_\nu} |\nu\rangle_A |\nu\rangle_B$$

satisfies $\text{Tr}_B |\Psi\rangle\langle\Psi| = \rho_A$. Thus $-\text{Tr}(\rho_A \log \rho_A)$ is the entanglement entropy of this particular pure bipartition. The enlarged state declares which system carries that interpretation.

For a two-sheet involution with one eigenvector of each sign,

$$\rho_y = \frac{e^{-yR}}{2 \cosh y}, \quad R^2 = I,$$

diagonalization gives

$$S(\rho_y) = \log(2 \cosh y) - y \tanh y, \quad D(\rho_y \| \rho_{-y}) = 2y \tanh y.$$

Entropy is even in y ; the distance explicitly compares the two orientations. This two-level entropy is not identified with the capacity $81 \log 3$ of the triadic cut.

28.6 Equation of state for sectorial information

The incidence already constructed provides eighteen tritic factors in each sector. This multiplicity allows the relation between area, the occupation distribution, and the entropy of the preceding purification to be written explicitly. We retain the reduced state defined there and abbreviate $\Delta = \Delta_{\text{mod}}$. For $m \in \{90, 120\}$, put

$$x_m = e^{-m\Delta}, \quad z_m = 1 + x_m + x_m^2, \quad f_m = \frac{x_m + 2x_m^2}{z_m}, \quad v_m = \frac{x_m + 4x_m^2}{z_m} - f_m^2. \quad (28.9)$$

Here f_m and v_m are, respectively, the mean and variance of the occupation of one factor in sector m . The parameter Δ preserves the status and normalization of the previously specified reduced state.

28.6.1 Explicit purification of the two sectors

The preceding Schmidt decomposition is made concrete on each link. For a finite real weight w , define

$$z(w) = 1 + e^{-w} + e^{-2w}, \quad \rho(w) = \frac{1}{z(w)} \sum_{n=0}^2 e^{-wn} |n\rangle\langle n|.$$

The sectorial weights are $w_m = m\Delta$, with $m = 90, 120$; hence $z(w_m) = z_m$.

Proposition 28.5 (Thermofield-double purification of sectorial incidence). *The normalized vector*

$$|\text{TFD}(w)\rangle = \frac{1}{\sqrt{z(w)}} \sum_{n=0}^2 e^{-wn/2} |n\rangle_A \otimes |n\rangle_{\bar{A}} \quad (28.10)$$

purifies $\rho(w)$. The product

$$|\Psi_\Delta\rangle = |\text{TFD}(90\Delta)\rangle^{\otimes 18} \otimes |\text{TFD}(120\Delta)\rangle^{\otimes 18} \quad (28.11)$$

has reduction $\rho_A(\Delta)$ on the thirty-six original links. Its entanglement entropy for the bipartition $A | \bar{A}$ is $s(\Delta)$.

Proof. The squared norm is $z(w)^{-1} \sum_{n=0}^2 e^{-wn} = 1$. In the partial trace, orthogonality of the second factor eliminates terms with distinct indices; the remaining terms form $\rho(w)$. The partial trace commutes with the tensor product. One thus obtains $\rho(90\Delta)^{\otimes 18} \otimes \rho(120\Delta)^{\otimes 18}$, which coincides with the normalized state defined previously. Equality between the entropy of a reduction and the entanglement of this purification is read directly from its Schmidt coefficients. $\square$

The two factors of the bipartition are determined within the construction. The physical realization of the second factor preserves that subsystem identification and its algebra of observables.

28.6.2 Occupation moments and information content

Proposition 28.6 (Mean area and entropy of the sectorial purification). *In this finite collection of states,*

$$\langle N \rangle_\Delta = 18(f_{90} + f_{120}), \quad (28.12)$$

$$\langle D \rangle_\Delta = 18(90f_{90} + 120f_{120}), \quad (28.13)$$

$$\frac{\langle \hat{\mathcal{A}} \rangle_\Delta}{\ell_P^2} = 18\gamma_{\text{HMT}}a_0(f_{90} + f_{120}), \quad (28.14)$$

$$s(\Delta) = 18 \log z_{90} + 18 \log z_{120} + \Delta \langle D \rangle_\Delta. \quad (28.15)$$

The dimensional entropy is $S(\Delta) = k_B s(\Delta)$, with the thermal transducer of Section 14.

Proof. In each factor, the normalized weights are $(1, x_m, x_m^2)/z_m$. Their first moment gives f_m , and their second moment gives the first fraction in v_m . Tensor factorization and additivity of the number operators produce the first three identities. Finally, $\log \rho_A = -(\log Z)I - \Delta D$, with $Z = z_{90}^{18} z_{120}^{18}$. Taking the trace against $-\rho_A$ gives (28.15). The purification specified previously has this same entanglement entropy. $\square$

Loss of uniformity has its own quantitative reading. In dimension $d_\partial = 3^{36}$, let $\tau_\partial = I/d_\partial$. Define

$$I_{\text{or}}(\Delta) = 36 \log 3 - s(\Delta) = D(\rho_A(\Delta) \| \tau_\partial) \geq 0. \quad (28.16)$$

The equality follows by substituting $\log \tau_\partial = -36 \log 3 I$. If p_ν are the eigenvalues of ρ_A , convexity of $t \log t$ gives $\sum_\nu p_\nu \log(d_\partial p_\nu) \geq 0$, with equality exactly for the uniform state. For this collection, that occurs at $\Delta = 0$. The deficit is dimensionless information; the areal quantum $\gamma_{\text{HMT}}a_0$ is another reading. Each expression preserves its definition even when their evaluations may be close.

Proposition 28.7 (Information response and boundary fluctuations). *For every finite real Δ ,*

$$\frac{d\langle D \rangle_\Delta}{d\Delta} = -\text{Var}_\Delta(D) = -18(90^2 v_{90} + 120^2 v_{120}), \quad (28.17)$$

$$\frac{ds}{d\Delta} = -\Delta \text{Var}_\Delta(D), \quad (28.18)$$

$$\frac{d\langle \hat{\mathcal{A}} \rangle_\Delta}{d\Delta} = -18\gamma_{\text{HMT}}a_0\ell_P^2(90v_{90} + 120v_{120}). \quad (28.19)$$

Proof. The sum defining Z is finite. Differentiating it gives $\partial_\Delta \log Z = -\langle D \rangle_\Delta$ and $\partial_\Delta^2 \log Z = \text{Var}_\Delta(D)$. The sectors and their factors are independent in the product state; their variances therefore add. On differentiating (28.15), the terms $-\langle D \rangle_\Delta$ and $\langle D \rangle_\Delta$ cancel. For area, use $\partial_\Delta f_m = -mv_m$ in (28.14). $\square$

These equations relate three levels of the same construction. Incidence fixes multiplicities and sectorial weights; the reduced state fixes their probabilities; fluctuations of those occupations determine the response of area and entropy. The scale $\gamma_{\text{HMT}}a_0\ell_P^2$ converts occupation into area, whereas k_B converts information into dimensional entropy. Section 27 first constructs $\ell_P^2 = L_\alpha^2$ from the registered longitudinal return. The later identity $G_a\hbar_a/c^3 = L_\alpha^2$ expresses the gravitational realization of that same areal unit without reversing its causal order. The collection of reduced states and its bipartition remain specified in this development; any horizon realization must also transport both of these data.

28.7 Occupation inversion and entropic invariants

Let $\mathcal{R}$ be the unitary involution that exchanges $|0\rangle$ and $|2\rangle$ and fixes $|1\rangle$ in each of the thirty-six tritic factors. Its action on occupations preserves the entire microscopic configuration through a bijective change of its labels.

Theorem 28.8 (Preserved entropy and complementary area). *The preceding involution satisfies*

$$\mathcal{R}N\mathcal{R}^* = 72I - N, \quad \mathcal{R}D\mathcal{R}^* = 7560I - D, \quad (28.20)$$

$$\rho_A(-\Delta) = \mathcal{R}\rho_A(\Delta)\mathcal{R}^*, \quad s(-\Delta) = s(\Delta), \quad (28.21)$$

$$\langle \hat{\mathcal{A}} \rangle_{-\Delta} + \langle \hat{\mathcal{A}} \rangle_{\Delta} = 72\gamma_{\text{HMT}}a_0\ell_P^2. \quad (28.22)$$

The relative entropy between the two orientations is

$$D(\rho_A(\Delta) \parallel \rho_A(-\Delta)) = \Delta(7560 - 2\langle D \rangle_{\Delta}). \quad (28.23)$$

Proof. Each local occupation n transforms into $2 - n$. Thus, $N_m \mapsto 36I - N_m$ and $D \mapsto 36(90 + 120)I - D = 7560I - D$. Furthermore, $z_m(-\Delta) = e^{2m\Delta}z_m(\Delta)$, whence $Z(-\Delta) = e^{7560\Delta}Z(\Delta)$. Substituting both identities into the normalized state proves unitary conjugation. Spectral invariance proves equality of entropies; transformation of N proves the area sum. Finally, subtracting the two state logarithms and taking their expectation in $\rho_A(\Delta)$ yields (28.23). $\square$

At $\Delta = 0$, the reduced state is uniform and $s(0) = 36 \log 3$. In the limit $\Delta \rightarrow +\infty$, the zero-occupation configuration survives; entropy tends to zero. Inversion takes that limit to the maximum-occupation configuration. At both extremes, entropy is the same, and area distinguishes the configurations. For $\Delta > 0$, response (28.18) is strictly negative because the state has full rank and D has more than one eigenvalue. The law thus preserves the difference between reversibility of the complete configuration, equality of entropies, and change in areal reading. These properties belong to the specified sectorial collection; physical transport of the bipartition is treated in Section 28.9.

28.8 First modular law and finite area variations

The derivatives in (28.18) vary the parameter Δ of a collection of states. The first modular law, by contrast, fixes a reference state and considers variations of the observed state around it. Take $\rho_0 = \rho_A(\Delta_0)$, with finite real Δ_0 , and keep $\Delta_0, \gamma_{\text{HMT}}, a_0, \ell_P$ fixed. The branch $\gamma_{\text{HMT}} > 0, a_0 > 0$ is retained. The normalized sectorial areas are

$$\mathcal{A}_m^n = \gamma_{\text{HMT}}a_0N_m, \quad \hat{\mathcal{A}}_m = \ell_P^2\mathcal{A}_m^n, \quad m \in \{90, 120\}. \quad (28.24)$$

The logarithm of the reference state takes the form

$$K_0 = -\log \rho_0 = c_0I + \beta_{90}\mathcal{A}_{90}^n + \beta_{120}\mathcal{A}_{120}^n, \quad \beta_m = \frac{m\Delta_0}{\gamma_{\text{HMT}}a_0}, \quad c_0 = \log Z(\Delta_0). \quad (28.25)$$

In particular, $\beta_{120} = \frac{4}{3}\beta_{90}$; for $\Delta_0 \neq 0$, their quotient is $4/3$. The coefficients arise from expressing the same modular operator in areal units.

Theorem 28.9 (First modular law in areal coordinates). *Let $\sigma(\varepsilon)$ be a differentiable collection of normalized states on an interval containing zero, with $\sigma(0) = \rho_0$. Then*

$$\left. \frac{ds(\sigma(\varepsilon))}{d\varepsilon} \right|_0 = \sum_{m=90,120} \beta_m \left. \frac{d\langle \mathcal{A}_m^n \rangle_{\sigma(\varepsilon)}}{d\varepsilon} \right|_0. \quad (28.26)$$

Proof. The state ρ_0 has full rank. In its neighborhood, the derivative of the trace of $-\sigma \log \sigma$ is $-\text{Tr}[\dot{\sigma}(\log \rho_0 + I)]$; this follows by diagonalizing ρ_0 in the formula for the spectral derivative of the trace. Normalization implies $\text{Tr} \dot{\sigma} = 0$, so the derivative is $\text{Tr}(\dot{\sigma}K_0)$. Substituting (28.25) also cancels c_0I and produces the identity. $\square$

Theorem 28.10 (Finite area identity and relative-entropy deficit). *For any normalized state σ on the same finite space, define $\Delta_\sigma\langle B \rangle = \text{Tr}[(\sigma - \rho_0)B]$. The identity*

$$s(\sigma) - s(\rho_0) = \sum_{m=90,120} \beta_m \Delta_\sigma\langle \mathcal{A}_m^n \rangle - D(\sigma\|\rho_0). \quad (28.27)$$

holds. Here $D(\sigma\|\rho_0) = \text{Tr}[\sigma(\log \sigma - \log \rho_0)]$ is finite, nonnegative, and interpreted with $0 \log 0 = 0$.

Proof. By definition, $D(\sigma\|\rho_0) = -s(\sigma) + \text{Tr}(\sigma K_0)$, whereas $s(\rho_0) = \text{Tr}(\rho_0 K_0)$. Subtraction gives $s(\sigma) - s(\rho_0) = \Delta_\sigma\langle K_0 \rangle - D(\sigma\|\rho_0)$. Substituting (28.25) proves the formula because the traces of both states are one.

To see positivity, diagonalize σ with eigenvalues p_i and vectors u_i , and ρ_0 with eigenvalues $q_j > 0$ and vectors v_j . Put $r_i = \langle u_i, \rho_0 u_i \rangle$. Concavity of the logarithm gives $\sum_j |\langle u_i, v_j \rangle|^2 \log q_j \leq \log r_i$. Hence $D(\sigma\|\rho_0) \geq \sum_i p_i \log(p_i/r_i) \geq 0$; the last inequality is convexity of $t \log t$ applied with weights r_i , which sum to one. Strict positivity of the q_j ensures that all reference logarithms are finite. $\square$

The finite identity preserves information distance from the reference state; its tangent term is (28.26). For physical areas, the corresponding coefficients are β_m/ℓ_P^2 . Multiplying the equation by the common transducer k_B expresses dimensional entropy. Thus the relation between geometry and information remains exact also for finite variations, with its explicit correction and without modifying the reference-state parameters.

28.9 Spectral entropy and preservation under transport

The entropy of a reduced state depends on its spectrum and on the bipartition defining the reduction. To make the continuation toward duality precise, let $\mathcal{H}_A, \mathcal{H}_B$ be the two factors and let ρ_{AB} be a state with finite entropies. A factorized transport $U_A \otimes U_B$ produces

$$\rho'_A = U_A \rho_A U_A^*, \quad S(\rho'_A) = S(\rho_A).$$

The first identity follows from the definition of the partial trace, and the second from spectral invariance. For a general unitary transport, the equivalent formulation also transports the inclusion of the subsystem's algebra of observables. This specification identifies the geometric datum that must accompany a claim of entanglement preservation.

28.10 Recoverable energy within a fixed area distribution

The tritic boundary distinguishes the information carried by total occupation from that carried by its sectorial origin. We use the two sets of eighteen links, their occupations 0, 1, 2, the 90/120 weights, and the thermal section constructed earlier. Write

$$X = N_{90}, \quad Y = N_{120}, \quad N = X + Y, \quad M = X - Y, \quad d = 30\Delta_{\text{mod}} > 0.$$

Let $\epsilon = k_B T > 0$ be the energy scale of the specified realization. Its Hamiltonian and reference state are

$$H = \epsilon d(3X + 4Y), \quad \rho_0 = Z^{-1} e^{-H/\epsilon}.$$

The previously constructed area operator is $\hat{\mathcal{A}} = a_b N$, with $a_b = \gamma a_0 L_\alpha^2 > 0$ fixed throughout the protocol. Define W as the exchange of the two sets of links, retaining every occupation. This permutation reverses M and leaves N invariant. The complementary occupation reversal $n \mapsto 2 - n$, treated earlier, retains its own function and is a distinct operation.

Proposition 28.11 (Energy response invisible to the area reading). *The state $\rho_1 = W\rho_0 W^*$ has the same full area distribution and the same von Neumann entropy as ρ_0 , whereas*

$$\Delta E = \text{Tr}[H(\rho_1 - \rho_0)] = \epsilon \mathcal{J}(d) > 0,$$

$$\mathcal{J}(d) = D(\rho_1\|\rho_0) = 18d [\mu(3d) - \mu(4d)], \quad \mu(a) = \frac{e^{-a} + 2e^{-2a}}{1 + e^{-a} + e^{-2a}}.$$

The inverse exchange recovers exactly ΔE ; this is the maximum energy recoverable by a unitary transformation of ρ_1 , for the fixed Hamiltonian H .

Proof. Since $WNW^* = N$, every spectral projector of the area commutes with W , preserving its probability. Unitarity preserves the spectrum and entropy of the state. In the occupation basis, write p_0, p_1 for the probabilities. Exchange gives

$$\log \frac{p_1}{p_0} = -dM, \quad \langle M \rangle_{p_1} = -\langle M \rangle_{p_0}, \quad \langle M \rangle_{p_0} = 18[\mu(3d) - \mu(4d)].$$

The weighted sum of the logarithm proves the formula for $D(\rho_1 \| \rho_0)$. Moreover, $\mu'(a) = -\text{Var}_a(n) < 0$ for the three positive weights, so $\mathcal{J}(d) > 0$. The Gibbs identity, together with equality of the entropies, gives $\Delta E = \epsilon D(\rho_1 \| \rho_0)$. For any unitary V , the state $V\rho_1 V^*$ retains that same entropy and satisfies

$$\text{Tr}[H(V\rho_1 V^* - \rho_0)] = \epsilon D(V\rho_1 V^* \| \rho_0) \geq 0.$$

The choice $V = W^*$ attains equality and proves maximality. $\square$

The area distribution retains every multiplicity of N , whereas the Hamiltonian distinguishes its allocations (X, Y) . The excess energy is located precisely in that distinction. When the corresponding horizon realization is used, its scale is $\epsilon = \hbar c / (2\pi R)$. Thus, action converts $\mathcal{J}(d)$ into an energy response at fixed radius. The area operator of this preparation and the geometric area of a horizon with backreaction retain their respective domains.

28.10.1 Conditional recovery and reduced functional

The information within each fibre admits an exact formulation. On the finite configuration space, let q be a surjective extractor, $p_{0,q} = q_* p_0$, and $p_q = q_* p$. For a distribution r on the extractor, define

$$(\mathcal{R}_0 r)(x) = r(q(x)) \frac{p_0(x)}{p_{0,q}(q(x))}.$$

This recovery map satisfies $q_* \mathcal{R}_0 = I$. Its conditional distribution retains the thermal weights within each fibre.

Proposition 28.12 (Free energy of information within a fibre). *For $F_T(p) = \sum_x p(x)H(x) + \epsilon \sum_x p(x) \log p(x)$,*

$$F_T(p) - F_T(\mathcal{R}_0 p_q) = \epsilon D(p \| \mathcal{R}_0 p_q).$$

Thus, $\mathcal{R}_0 p_q$ is the unique free-energy minimizer among distributions with marginal p_q .

Proof. On the support of p , the factorization

$$\frac{p(x)}{p_0(x)} = \frac{p_q(q(x))}{p_{0,q}(q(x))} \frac{p(x)}{(\mathcal{R}_0 p_q)(x)}$$

gives, upon taking logarithms and summing, $D(p \| p_0) = D(p_q \| p_{0,q}) + D(p \| \mathcal{R}_0 p_q)$. The identity $F_T(p) = \epsilon D(p \| p_0) - \epsilon \log Z$, together with the same formula for $\mathcal{R}_0 p_q$, proves the statement. The divergence vanishes precisely when its arguments coincide; the convention is $0 \log 0 = 0$. $\square$

For $q = N$, write $g_{18}(x) = [t^x](1 + t + t^2)^{18}$ for the tritic multiplicity already constructed. The effective Hamiltonian of the area reading is

$$\frac{H_N(n)}{\epsilon} = \frac{7d}{2}n - \log \sum_{x+y=n} g_{18}(x)g_{18}(y)e^{d(x-y)/2}.$$

This expression follows by summing $e^{-H/\epsilon}$ over the fibre. The sum is invariant under reversal of the sign of its oriented term, by exchange of x and y . For the preceding preparation, $p_{1,N} = p_{0,N}$ and $\mathcal{R}_0 p_{1,N} = p_0$: all excess free energy resides in the information omitted by this area reading. By contrast, retaining (N, M) fixes the energy of each fibre and makes its conditional distribution uniform. The result specifies which observables suffice to recover this collection and its thermal response.

28.11 Quasi-free correlation and spectral occupation entropies

For a gauge-invariant, number-conserving fermionic quasi-free state defined on a finite-dimensional one-body space, let $C_{ij} = \langle a_j^\dagger a_i \rangle$ be the one-body correlation operator. Gauge invariance excludes anomalous pairing correlations, so C determines the quasi-free state. If $0 < C < I$, its one-body modular generator and the inverse reconstruction of the correlator are

$$K_1 = \log((I - C)C^{-1}), \quad C = (I + e^{K_1})^{-1}. \quad (28.28)$$

Corollary 28.13 (Logistic form of the modular spectrum). *If $K_1 u_i = \kappa_i u_i$, the occupation of mode u_i is*

$$\langle n_i \rangle = \frac{1}{1 + e^{\kappa_i}}.$$

For $\kappa_i = \beta(\varepsilon_i - \mu)$, the Fermi–Dirac law is recovered.

Proof. The spectral diagonalization of K_1 diagonalizes every function of it. Applying $x \mapsto (1 + e^x)^{-1}$ to each eigenvalue yields the formula. $\square$

If C has eigenvalues $p_i \in [0, 1]$, the occupation entropy of the fermionic quasi-free state is, with the convention $0 \log 0 = 0$,

$$S_F = - \sum_i [p_i \log p_i + (1 - p_i) \log(1 - p_i)]. \quad (28.29)$$

For a diagonal bosonic Gaussian state with finitely many modes and mean occupations $\bar{n}_i$,

$$S_B = \sum_i [(1 + \bar{n}_i) \log(1 + \bar{n}_i) - \bar{n}_i \log \bar{n}_i]. \quad (28.30)$$

The same formula extends to a countable collection when the series on the right-hand side converges. These entropies quantify occupation configurations. Entanglement entropy has a different domain: the reduced state induced by a tensor decomposition. Identifying the two requires specifying the reduction and the correlation operator that implement it.

Finally, for a reduced state ρ_A , the *von Neumann entropy*

$$S_{\text{vN}}(\rho_A) = - \text{Tr}(\rho_A \log \rho_A)$$

is a spectral property of the density operator. When ρ_A arises from a global purification, it quantifies entanglement across the cut. In the HMT boundary construction, the area operator, modular Hamiltonian, and corresponding reduction are constructed in Section 28. The product purification of Proposition 28.5 identifies the bipartition, and identity (28.27) expresses its relation to area variations.

The maps between these domains are expressed through explicit identities. Disintegration of a measure over the fibers of q satisfies the chain rule

$$H_\mu(\mathbf{X}) = H_{q_*\mu}(q(\mathbf{X})) + H_{\text{fib}}(q; \mu). \quad (28.31)$$

A finite distribution $p = (p_i)$ is realized diagonally as

$$\rho_p = \sum_i p_i |i\rangle\langle i|, \quad S_{\text{vN}}(\rho_p) = H(p).$$

If $P_{N,E}$ projects onto the microcanonical subspace of dimension $\Omega(N, E)$, then

$$\rho_{\text{mc}} = \frac{P_{N,E}}{\Omega(N, E)}, \quad S_{\text{vN}}(\rho_{\text{mc}}) = \log \Omega(N, E) = S_{\text{mc}}.$$

For a bipartite pure state,

$$\rho_A = \text{Tr}_B(|\Psi_{AB}\rangle\langle\Psi_{AB}|)$$

defines the reduction that measures entanglement. Finally, equality between $h_{\text{top}} = \log \rho(A)$ and a metric entropy is attained, for this primitive system, by the Parry measure, which realizes the maximum metric entropy; it is not attributed to an arbitrary measure on histories. The matrix F_{av} , its one-step-memory envelope, and that measure are constructed in Theorems 24.1 and 24.2, with its explicit variational characterization in Proposition 24.3.

29 Fourier duality, finite uncertainty, and ordered memory

The domain of the finite realization follows from the typed chain already constructed in the article:

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow X_{\text{enr}} \longrightarrow (\mathbb{Z}/9\mathbb{Z})^3 \xrightarrow{\beta_{729}} \Lambda := (\mathbb{Z}/27\mathbb{Z})^2 \longrightarrow \ell^2(\Lambda). \quad (29.1)$$

Here X_{enr} retains route, orientation, carry, boundary, and memory; the passage to $(\mathbb{Z}/9\mathbb{Z})^3$ is the relevant positional register. The map β_{729} , defined in (28.1) and proved bijective in Proposition 28.1, regroups the six ordered trits into two coordinates of order twenty-seven. The positional register and the group realization are thereby kept distinct: β_{729} transports the ordered incidence, whereas the Fourier transform uses the operations intrinsic to $(\mathbb{Z}/27\mathbb{Z})^2$. The genealogy thus specifies more than the common cardinality of 729 positions.

29.1 Unitary characters and mutually unbiased bases

Set $N := |\Lambda| = 27^2 = 729$. For $x = (x_1, x_2)$ and $k = (k_1, k_2)$ in Λ , let

$$\chi_k(x) := \exp\left(\frac{2\pi i}{27}(k_1 x_1 + k_2 x_2)\right). \quad (29.2)$$

The scalar product of the coordinates is read modulo twenty-seven. Character orthogonality gives

$$\sum_{x \in \Lambda} \chi_k(x) \overline{\chi_\ell(x)} = N \delta_{k,\ell}. \quad (29.3)$$

Define the unitary transform $\mathcal{F}_\Lambda$ by

$$\widehat{\psi}(k) := (\mathcal{F}_\Lambda \psi)(k) = \frac{1}{\sqrt{N}} \sum_{x \in \Lambda} \chi_k(x) \psi(x), \quad \psi(x) = \frac{1}{\sqrt{N}} \sum_{k \in \Lambda} \overline{\chi_k(x)} \widehat{\psi}(k). \quad (29.4)$$

The conjugated character in the second formula is indispensable under the positive convention of the first. Equation (29.3) proves both inversion and $\|\mathcal{F}_\Lambda \psi\|_2 = \|\psi\|_2$.

Lemma 29.1 (Mutual unbiasedness of the bases). *Let $\{e_x : x \in \Lambda\}$ be the position basis and let $f_k(x) = N^{-1/2} \overline{\chi_k(x)}$ be the Fourier basis. Then*

$$|\langle e_x, f_k \rangle| = N^{-1/2} = 1/27 \quad (x, k \in \Lambda). \quad (29.5)$$

Thus the two bases are mutually unbiased.

Proof. The orthonormality of $\{f_k\}$ follows from (29.3); moreover, $\langle e_x, f_k \rangle = f_k(x)$, and every character has modulus one. $\square$

29.2 Exact support uncertainty

For a function $u : \Lambda \rightarrow \mathbb{C}$, write $\text{supp } u = \{x : u(x) \neq 0\}$.

Theorem 29.2 (Support bound on the triadic torus). *Every nonzero $\psi \in \ell^2(\Lambda)$ satisfies*

$$|\text{supp } \psi| |\text{supp } \widehat{\psi}| \geq N = 729. \quad (29.6)$$

Proof. Set $S = \text{supp } \psi$ and $T = \text{supp } \widehat{\psi}$. The correct inversion formula in (29.4) yields, for every $x \in S$,

$$|\psi(x)| = \left| \frac{1}{\sqrt{N}} \sum_{k \in T} \overline{\chi_k(x)} \widehat{\psi}(k) \right| \leq \sqrt{\frac{|T|}{N}} \|\widehat{\psi}\|_2,$$

by Cauchy–Schwarz and $|\chi_k(x)| = 1$. Squaring and summing over S gives

$$\|\psi\|_2^2 \leq \frac{|S||T|}{N} \|\widehat{\psi}\|_2^2 = \frac{|S||T|}{N} \|\psi\|_2^2.$$

The norm is positive, so cancellation proves (29.6). This is the specialization to the realized finite group of the Donoho–Stark support uncertainty principle [23]. $\square$

The chain (29.1) fixes the domain before the inequality is formulated: APP-TRIT-TPK supplies the positional register, and β_{729} supplies its finite realization as the group Λ .

29.3 Annihilators and classification of the saturation cases

For $H \leq \Lambda$, define its annihilator by

$$H^\perp := \{k \in \Lambda : \chi_k(h) = 1 \text{ for every } h \in H\}. \quad (29.7)$$

For divisors $a, b \mid 27$, consider

$$H_{a,b} := \{(x_1, x_2) \in \Lambda : x_1 \equiv 0 \pmod{a}, x_2 \equiv 0 \pmod{b}\}. \quad (29.8)$$

Then

$$|H_{a,b}| = \frac{27}{a} \frac{27}{b}, \quad H_{a,b}^\perp = \{(k_1, k_2) : k_1 \equiv 0 \pmod{27/a}, k_2 \equiv 0 \pmod{27/b}\}, \quad |H_{a,b}^\perp| = ab. \quad (29.9)$$

Indeed, the geometric sum in each coordinate vanishes unless $27 \mid ak_1$ and $27 \mid bk_2$. Consequently, for the normalized indicator

$$\psi_{a,b} := |H_{a,b}|^{-1/2} \mathbf{1}_{H_{a,b}}, \quad (29.10)$$

one obtains exactly

$$\hat{\psi}_{a,b}(k) = \sqrt{\frac{|H_{a,b}|}{N}} \mathbf{1}_{H_{a,b}^\perp}(k). \quad (29.11)$$

Thus the transform is uniform on the annihilator and $|H_{a,b}| |H_{a,b}^\perp| = N$.

Table 12: Four exact saturations, with distinct domains.

(a, b)	$ H_{a,b} $	$ H_{a,b}^\perp $	product	$(H_{\text{pos}}, H_{\text{F}})$
$(3, 3)$	81	9	729	$(\log 81, \log 9)$
$(9, 9)$	9	81	729	$(\log 9, \log 81)$
$(3, 9)$	27	27	729	$(\log 27, \log 27)$
$(1, 27)$	27	27	729	$(\log 27, \log 27)$

The last two rows have the same cardinalities but neither the same subgroup nor the same annihilator; numerical equality does not collapse their incidence data.

29.4 Entropic uncertainty by interpolation and differentiation

Let $\|\psi\|_2 = 1$, and define the distributions

$$p_x := |\psi(x)|^2, \quad q_k := |\hat{\psi}(k)|^2, \quad H_{\text{pos}} := -\sum_x p_x \log p_x, \quad H_{\text{F}} := -\sum_k q_k \log q_k, \quad (29.12)$$

with natural logarithms and the convention $0 \log 0 = 0$. These are the Shannon entropies of two measurement distributions in conjugate bases.

Theorem 29.3 (Entropic bound for the Λ realization). *Every unit vector satisfies*

$$H_{\text{pos}}(\psi) + H_{\text{F}}(\psi) \geq \log N = \log 729 = 6 \log 3. \quad (29.13)$$

Proof. From $\|\mathcal{F}_\Lambda\|_{\ell^1 \rightarrow \ell^\infty} \leq N^{-1/2}$ and unitarity, $\|\mathcal{F}_\Lambda\|_{\ell^2 \rightarrow \ell^2} = 1$, the Riesz–Thorin interpolation theorem [18, 19] gives, for $1 < p \leq 2$ and $p' = p/(p-1)$,

$$\|\mathcal{F}_\Lambda \psi\|_{p'} \leq N^{\frac{1}{2} - \frac{1}{p}} \|\psi\|_p. \quad (29.14)$$

Consider

$$g(p) := \log \|\mathcal{F}_\Lambda \psi\|_{p'} - \log \|\psi\|_p - \left(\frac{1}{2} - \frac{1}{p}\right) \log N. \quad (29.15)$$

Equation (29.14) implies $g(p) \leq 0$, whereas $g(2) = 0$; hence $g'_-(2) \geq 0$.

For every unit vector u in finite dimension,

$$\frac{d}{dr} \log \|u\|_r \Big|_{r=2} = \frac{1}{4} \sum_j |u_j|^2 \log |u_j|^2 = -\frac{1}{4} H(|u|^2). \quad (29.16)$$

Zero coordinates are obtained by continuity, while positive terms are differentiable. Since

$$\frac{dp'}{dp} \Big|_{p=2} = -1,$$

left differentiation of (29.15) yields

$$g'_-(2) = \frac{1}{4} (H_{\text{pos}} + H_F - \log N) \geq 0, \quad (29.17)$$

which is equivalent to (29.13).

This derivation lies in the line of entropic uncertainty developed by Hirschman and Beckner [20, 21]; for arbitrary pairs of bases, the maximum-overlap formulation is due to Maassen and Uffink [22]. In the present realization, the constant $\log N$ follows directly by differentiating the interpolation factor in (29.14). $\square$

For the states $\psi_{a,b}$, Equation (29.11) shows that p and q are uniform on $H_{a,b}$ and $H_{a,b}^\perp$, respectively. Therefore,

$$H_{\text{pos}} = \log |H_{a,b}|, \quad H_F = \log |H_{a,b}^\perp|, \quad H_{\text{pos}} + H_F = \log 729,$$

which proves entropic saturation for all four cases in Table 12 without numerical approximation.

29.5 Typed domains of information

The quantity $H_{\text{pos}} + H_F$ measures the sum of two classical distribution entropies associated with conjugate measurements. Entanglement entropy across the boundary cut, von Neumann entropy of a reduced state, thermodynamic entropy and temperature, and the marginal entropy of a tritic histogram are defined on distinct objects and domains. Every comparison among these quantities is formulated through an explicit map specifying domain, codomain, and state. The bound (29.13) belongs entirely to $\ell^2(\Lambda)$.

29.6 Ordered memory on the dodecaphasic domain

The F044 result acts on dodecaphasic words and characterizes the ordered information retained by the enriched state. Its domain and operators are independent of the Fourier realization of N19; the relation between the two results compares marginal information with ordered information.

Let $\tau = (\tau_1, \dots, \tau_{12}) \in \{-1, 0, +1\}^{12}$, with $\text{sgn}(0) = 0$ and cyclic indices modulo twelve. Define

$$G_a := \{a, a+4, a+8\}, \quad a = 1, 2, 3, 4, \quad (29.18)$$

$$v_{120}(\tau) := \sum_{a=1}^4 \text{sgn} \left(\sum_{m \in G_a} \tau_m \right),$$

$$v_{270}(\tau) := \sum_{m=1}^{12} \tau_m \tau_{m+3}. \quad (29.19)$$

The words

$$\tau^{(a)} = (1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0), \quad \tau^{(b)} = (1, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0) \quad (29.20)$$

have the same histogram—ten zeros and two ones—and hence the same marginal alphabet entropy. Nevertheless,

$$(\nu_{120}, \nu_{270})(\tau^{(a)}) = (2, 0), \quad (\nu_{120}, \nu_{270})(\tau^{(b)}) = (2, 1). \quad (29.21)$$

The second reader detects a distance-three incidence erased by the histogram.

For the involution $C_M \tau := -\text{rev}(\tau)$, reversal permutes the four classes G_a , while negation reverses their sums; the cyclic autocorrelation at lag +3 equals, after reindexing, the one at lag −3. Consequently,

$$\nu_{120}(C_M \tau) = -\nu_{120}(\tau), \quad \nu_{270}(C_M \tau) = \nu_{270}(\tau). \quad (29.22)$$

The reader ν_{120} is odd and the quadratic reader ν_{270} is even. These parities characterize the additional information retained by the ordered word.

29.7 Domain of validity and mathematical conditions

The conditions delimiting the two results are:

1. β_{729} is bijective on the positional realization of Λ used in (29.1);
2. orthogonality (29.3) and the inversion formula with the conjugated character establish unitarity;
3. every nonzero state satisfies $|\text{supp } \psi| |\text{supp } \hat{\psi}| \geq 729$;
4. every unit state satisfies $H_{\text{pos}} + H_F \geq \log 729$;
5. in all four cases of Table 12, the transform is uniform on the annihilator and the cardinality product is 729;
6. the witnesses in (29.20) share a histogram, produce (29.21), and satisfy the parities (29.22).

The first five statements belong to the typed finite realization Λ and its unitary Fourier transform; the sixth belongs to the dodecaphasic domain of ordered memory. These conditions also provide local falsifiers: failure of any corresponding identity, inequality, or witness value refutes the statement in its stated domain. The relation between N19 and F044 is complementary and keeps their domains and operators distinct.

30 Thermal radiation and the information scale of the calendar

Occupation statistics provide the link between the modes of a realization and their mean energy. In the electromagnetic realization, the dispersion relation and polarization multiplicity additionally determine the spectral density. Their composition yields the thermal moments and spectral maxima, while keeping three operations distinct: production of the action scale, representation of the modes, and evaluation of the thermal state.

30.1 Action, calendar, and temperature

Retain the clock orientation a and abbreviate $\hbar = \hbar_a$, $c = c_{\text{int}} = c_{\text{vac}}$, with the identification in (26.4) made in the same chart. The energy ε_{108} in this section is ε_a of (14.6); $\Theta_{108} = \Theta_{\text{clk},a}$. Dividing it by $\log 3$ gives the energy per nat in (14.7). The construction and covariance of the thermal pair are given in Section 14.

Write $h = 2\pi\hbar > 0$ for the action representation, $t_0 > 0$ for the temporal unit of its dimensional chart, and $\ell_0 = ct_0$ for the corresponding length. The chronological cell of length 108 determines the energy

$$\varepsilon_{108} = \frac{h}{108t_0} = \frac{\hbar c}{L_\alpha} = Q_{L,a} = E_{P,a}, \quad k_B \Theta_{108} \log 3 = Q_{L,a}, \quad \beta_{108} = \frac{108t_0 \log 3}{h}. \quad (30.1)$$

Here $\beta = (k_B \Theta)^{-1}$ is the Gibbs parameter. The equality fixes the energy-valued product of temperature and the entropy transducer. The dimensional specification of k_B and Θ_{108} separately uses the thermal section and its bases. The natural logarithm measures information per tritic branch; its factor $\log 3$ converts the cardinality of the cell into dimensionless entropy.

Also introduce the associated units

$$t_P = \frac{108}{2\pi_{\text{HMT}}} t_0 = \frac{L_\alpha}{c}, \quad \ell_P = ct_P = L_\alpha, \quad E_P := E_{P,a} = \frac{\hbar}{t_P} = Q_{L,a} = \varepsilon_{108}. \quad (30.2)$$

These equalities specify the meaning of the symbols in this section; their evaluation in another chart preserves the identities between quotients. The length $\ell_P = (54/\pi_{\text{HMT}})\ell_0$ first coincides with the structural length L_α of (27.4); the subsequent gravitational identity $\ell_P^2 = G_a \hbar_a / c^3$ is the corollary (27.7), not the origin of that scale.

The centred equivalence (15.22) and the transducer identity (15.38) fix the energy scale and its thermal reading before the electromagnetic realisation. The marker p_0 in (15.18) retains normalisation and origin when a branch is conditioned. This substitution determines β and the energy scale; it does not retrospectively select the dispersion relation or the two polarisations specified below. The Planck, Stefan–Boltzmann, and Wien laws therefore remain downstream consequences of the declared modal realisation.

30.2 Mode density and spectral law

Proposition 30.1 (Thermal realization of transverse modes). *Consider a free realization in a three-dimensional periodic box, whose excitation space contains the wave vectors $\mathbf{k} \neq 0$, with two transverse modes per vector, dispersion $\omega = c|\mathbf{k}|$, excitation energy $\hbar\omega$, zero chemical potential, and $\beta > 0$. The constant mode is kept fixed as background data. In the thermodynamic limit, the energy density per angular frequency is*

$$u_\omega(\Theta) = \frac{\hbar\omega^3}{\pi^2 c^3} \frac{1}{e^{\beta\hbar\omega} - 1}, \quad \omega > 0. \quad (30.3)$$

The energy is that of the excitations above the vacuum of this representation.

Proof. For a box of volume V , the wave vectors have density $V/(2\pi)^3$. In a radial shell of thickness dk , the density per unit volume, including the two polarizations, is

$$2 \frac{4\pi k^2 dk}{(2\pi)^3} = \frac{k^2}{\pi^2} dk = \frac{\omega^2}{\pi^2 c^3} d\omega.$$

This is the limiting density of the sums over the periodic lattice with the constant mode omitted. This choice defines the partition product before taking the limit: each excitation mode has positive energy. Excluding a single point preserves the limiting radial density. The previously obtained bosonic occupation is $\bar{n}_\omega = (e^{\beta\hbar\omega} - 1)^{-1}$. Multiplying the density by $\hbar\omega\bar{n}_\omega$ proves (30.3). Near zero, the energy integrand is of order ω^2 , and at infinity it decays exponentially, so its integral converges. $\square$

The density arises from the specified three-dimensional representation. The transverse multiplicity used here has that physical domain; the dimension of a harmonic space on the APP torus describes, separately, the cohomology of its cellular complex.

30.3 Thermodynamic limit of occupation sums

The preceding radial density can be obtained as an effective limit of the sums in the periodic realization. The estimate must retain the fixed constant mode and uniformly control both the origin and the spectral tail. The action and calendar representations remain prior to this realization operation.

Lemma 30.2 (Convergence of the three thermal observables). *Let $a, b > 0$, $L > 0$, $\Delta = 2\pi/L$, and*

$$F_N(r) = \frac{1}{e^{ar} - 1}, \quad F_E(r) = \frac{br}{e^{ar} - 1}, \quad F_Z(r) = -\log(1 - e^{-ar}) \quad (r > 0).$$

For each of these functions,

$$\lim_{L \rightarrow \infty} \frac{2}{L^3} \sum_{v \in \mathbb{Z}^3 \setminus \{0\}} F\left(\frac{2\pi}{L}|v|\right) = \frac{1}{\pi^2} \int_0^\infty r^2 F(r) dr. \quad (30.4)$$

The sums and the integral are finite. The specialization $a = \beta\hbar c$, $b = \hbar c$ gives, respectively, the number density, excitation-energy density, and logarithmic partition-function density of the transverse modes.

Proof. All three functions are continuous and positive on $(0, \infty)$. There exist constants $C_0, C_1, d > 0$, depending on a, b but independent of Δ , such that

$$F(r) \leq \frac{C_0}{r} \quad (0 < r \leq 1), \quad F(r) \leq C_1 e^{-dr} \quad (r \geq 1). \quad (30.5)$$

Indeed, $e^{ar} - 1 \geq ar$ bounds F_N and F_E near the origin; moreover, $F_Z(r) = \log(1 + F_N(r)) \leq F_N(r) \leq 1/(ar)$. For $r \geq 1$, the inequality $-\log(1 - y) \leq y/(1 - y)$, with $y = e^{-ar}$, gives the exponential bound for F_Z . The other two follow from $(e^{ar} - 1)^{-1} \leq e^{-ar}/(1 - e^{-a})$, absorbing the factor r in F_E into an exponential of smaller decay rate. These bounds already prove integrability of $r^2 F(r)$.

For the discrete sums, group the lattice points by $m = \|v\|_\infty \geq 1$. Each shell contains exactly

$$(2m + 1)^3 - (2m - 1)^3 = 24m^2 + 2$$

points and satisfies $m \leq |v| \leq \sqrt{3}m$. Fix $0 < \delta \leq 1$ and $0 < \Delta \leq 1$. If $\delta < \Delta$, there are no points with $0 < \Delta|v| \leq \delta$. Otherwise, setting $N = \lfloor \delta/\Delta \rfloor$, the first bound in (30.5) gives

$$\begin{aligned} \Delta^3 \sum_{0 < \Delta|v| \leq \delta} F(\Delta|v|) &\leq C_0 \Delta^2 \sum_{m=1}^N \frac{24m^2 + 2}{m} \\ &\leq 26C_0 \Delta^2 \sum_{m=1}^N m \leq 26C_0 \delta^2. \end{aligned}$$

The second line uses $m^{-1} \leq m$ and then $N(N + 1)/2 \leq N^2$ for $N \geq 1$. The integral contribution from the same neighborhood is also of order δ^2 , since $\int_0^\delta r^2 F(r) dr \leq C_0 \delta^2/2$.

For $R \geq 1$, every point with $\Delta|\nu| > R$ belongs to a shell $m > R/(\sqrt{3}\Delta)$. The second bound in (30.5) gives

$$\begin{aligned} \Delta^3 \sum_{\Delta|\nu| > R} F(\Delta|\nu|) &\leq C_1 e^{-dR/(2\sqrt{3})} \Delta^3 \sum_{m \geq 1} (24m^2 + 2) e^{-d\Delta m/2} \\ &\leq C_2 e^{-dR/(2\sqrt{3})}, \end{aligned}$$

where C_2 is independent of $0 < \Delta \leq 1$. To verify this uniformity, set $q = e^{-d\Delta/2}$ and use

$$\sum_{m \geq 1} q^m = \frac{q}{1-q}, \quad \sum_{m \geq 1} m^2 q^m = \frac{q(1+q)}{(1-q)^3}.$$

The first identity is the geometric series, and the second follows by applying $q d/dq$ twice, which is valid for $|q| < 1$. Moreover, $1 - e^{-d\Delta/2} \geq \Delta(1 - e^{-d/2})$ by concavity. The factor Δ^3 therefore compensates for the denominators of both expressions. The integral tail likewise tends to zero by the exponential bound. For each $\Delta > 0$, the same estimates prove convergence of the infinite sum.

On the compact annulus $\delta \leq |k| \leq R$, the function $F(|k|)$ is continuous and bounded. Riemann sums of mesh Δ converge to its integral; the two boundary spheres have measure zero. First let $\Delta \rightarrow 0$, then $\delta \rightarrow 0$, and finally $R \rightarrow \infty$, using the preceding uniform bounds. The result is

$$\Delta^3 \sum_{\nu \neq 0} F(\Delta|\nu|) \longrightarrow \int_{\mathbb{R}^3} F(|k|) d^3k = 4\pi \int_0^\infty r^2 F(r) dr.$$

Since $2/L^3 = 2\Delta^3/(2\pi)^3$, the final radial coefficient is $2 \cdot 4\pi/(2\pi)^3 = 1/\pi^2$, proving (30.4). $\square$

Control at the origin is carried out over positive-energy excitations. The constant mode remains fixed before the partition function is formed: removing it afterward from an integral would not replace that condition on the finite representation. The proof establishes the limit of the specified realization and preserves the distinction between that realization and the prior construction of its action and calendar parameters.

30.4 Thermal moments and information

Lemma 30.3 (Bosonic moment). *For every real $s > 1$,*

$$\int_0^\infty \frac{x^{s-1}}{e^x - 1} dx = \Gamma(s)\zeta(s). \quad (30.6)$$

Proof. For $x > 0$, $(e^x - 1)^{-1} = \sum_{m \geq 1} e^{-mx}$. All summands are nonnegative; Tonelli's theorem allows the sum and integral to be interchanged. The substitution $y = mx$ gives $\int_0^\infty x^{s-1} e^{-mx} dx = m^{-s} \Gamma(s)$. The sum converges for $s > 1$, yielding the stated equality. $\square$

Proposition 30.4 (Number, energy, and entropy densities). *In the realization of Proposition 30.1,*

$$n_\gamma = \frac{2\zeta(3)}{\pi^2 c^3 (\beta \hbar)^3}, \quad (30.7)$$

$$u_\gamma = \frac{\pi^2}{15 c^3 \hbar^3 \beta^4}, \quad (30.8)$$

$$s_\gamma = \frac{4\pi^2 k_B}{45 c^3 (\beta \hbar)^3}. \quad (30.9)$$

At the calendar temperature, these relations become

$$n_\gamma \ell_P^3 = \frac{2\zeta(3)}{\pi^2 (\log 3)^3}, \quad \frac{u_\gamma \ell_P^3}{E_P} = \frac{\pi^2}{15 (\log 3)^4}, \quad \frac{s_\gamma \ell_P^3}{k_B} = \frac{4\pi^2}{45 (\log 3)^3}. \quad (30.10)$$

Proof. Integrating occupation against the mode density uses (30.6) with $s = 3$, and the energy integral uses it with $s = 4$. This gives $\Gamma(3) = 2$ and $\Gamma(4)\zeta(4) = 6\pi^4/90 = \pi^4/15$. For entropy, start from the partition product of the modes:

$$\frac{\log Z}{V} = -\frac{1}{\pi^2 c^3} \int_0^\infty \omega^2 \log(1 - e^{-\beta \hbar \omega}) d\omega = \frac{\pi^2}{45 c^3 (\beta \hbar)^3}.$$

The last equality follows by integration by parts after the substitution $x = \beta \hbar \omega$. The boundary terms vanish: $x^3 \log(1 - e^{-x})$ tends to zero at both endpoints. The Gibbs identity $s_\gamma/k_B = \log Z/V + \beta u_\gamma$ proves the third expression. Finally, $\beta_{108} \hbar = t_P \log 3$ and $\ell_P = ct_P$ give the three calendar normalizations. $\square$

30.5 Photonic composition of Bendersky–Glaisher level A_2

The number density in Proposition 30.4 has a second exact reading that retains its spectral provenance. Let $\mathcal{Z}_3$ denote the normalized scalar functional produced by the tritic spectral reader and subsequently recognized as the zeta function in the Archimedean realization. Let

$$H_k = \sum_{j=1}^k \frac{1}{j}, \quad H_0 = 0,$$

and let B_j be the Bernoulli numbers under the convention $B_1 = -1/2$. To avoid any collision with lattice notation, define the Bendersky–Glaisher levels by

$$A_k^{\text{BG}} = \exp\left(\frac{H_k B_{k+1}}{k+1} - \mathcal{Z}'_3(-k)\right), \quad k \in \mathbb{N}_0. \quad (30.11)$$

The Bernoulli term fixes the normalization of the regularized product; it does not introduce a physical constant. The coordinate π_{HMT} entering the functional equation has already been expressed by the enriched APP–TRIT–TPK state before this reading.

Proposition 30.5 (Photonic composition at level two). *In the scalar normalization $\mathcal{Z}_3 = \zeta$,*

$$\log A_2^{\text{BG}} = -\mathcal{Z}'_3(-2) = \frac{\mathcal{Z}_3(3)}{4\pi_{\text{HMT}}^2}. \quad (30.12)$$

Consequently, in the calendar thermal chart,

$$n_\gamma \ell_P^3 = \frac{8 \log A_2^{\text{BG}}}{(\log 3)^3} = -\frac{8 \mathcal{Z}'_3(-2)}{(\log 3)^3}, \quad (30.13)$$

$$\frac{u_\gamma}{n_\gamma k_B \Theta_{108}} = \frac{\pi_{\text{HMT}}^2}{120 \log A_2^{\text{BG}}}, \quad (30.14)$$

$$\frac{s_\gamma}{n_\gamma k_B} = \frac{\pi_{\text{HMT}}^2}{90 \log A_2^{\text{BG}}}. \quad (30.15)$$

Proof. The completed form satisfies

$$\pi_{\text{HMT}}^{-s/2} \Gamma(s/2) \mathcal{Z}_3(s) = \pi_{\text{HMT}}^{-(1-s)/2} \Gamma((1-s)/2) \mathcal{Z}_3(1-s).$$

At $s = -2$, the simple zero of $\mathcal{Z}_3$ cancels the pole of $\Gamma(s/2)$. Comparing constant terms and using $\Gamma(3/2) = \sqrt{\pi_{\text{HMT}}}/2$ after Archimedean recognition gives

$$\mathcal{Z}'_3(-2) = -\frac{\mathcal{Z}_3(3)}{4\pi_{\text{HMT}}^2}.$$

Because $B_3 = 0$, Definition (30.11) gives the first equality in (30.12). Substitution of $\mathcal{Z}_3(3) = 4\pi_{\text{HMT}}^2 \log A_2^{\text{BG}}$ into (30.10) proves (30.13). Finally, $k_B \Theta_{108} = E_P / \log 3$; dividing the energy and entropy densities by the number density proves (30.14) and (30.15). $\square$

The evaluation $A_2^{\text{BG}} = 1.0309167521973921 \dots$ is a numerical test of the three identities, not generative input. The Bendersky–Glaisher spectral development contains the complete hierarchy of regularized derivatives and its correspondence with odd spectral values; the present section establishes the reciprocal composition used specifically by dimensionless photon-number density. It neither redefines k_B , identifies a density with a regularized operator, nor alters any of the previously proved $\zeta(3)$ densities.

30.6 Thermal flux and the Stefan–Boltzmann constant

Proposition 30.6 (Isotropic flux and action normalization). *The hemispherical flux of the isotropic realization is*

$$j_*(\Theta) = \sigma \Theta^4, \quad \sigma = \frac{\pi^2 k_B^4}{60 \hbar^3 c^2} = \frac{2\pi^5 k_B^4}{15 h^3 c^2}. \quad (30.16)$$

At the calendar temperature,

$$j_*(\Theta_{108}) = \frac{2\pi^5}{15 \cdot 108^4 (\log 3)^4} \frac{h}{\ell_0^2 t_0^2}. \quad (30.17)$$

Proof. The isotropic energy per unit solid angle is $u_\gamma/(4\pi)$. Flux normal to a surface integrates $c \cos \theta$ over the hemisphere:

$$\frac{cu_\gamma}{4\pi} \int_0^{2\pi} d\phi \int_0^{\pi/2} \cos \theta \sin \theta d\theta = \frac{cu_\gamma}{4}.$$

The formula for u_γ , $\beta^{-1} = k_B \Theta$, and $h = 2\pi \hbar$ prove (30.16). Substitution of (30.1) and $c = \ell_0/t_0$ gives (30.17). $\square$

Within this realization, the constant σ is the coefficient combining bosonic occupation, transverse-mode density, and angular projection of the flux. The fourth-power dependence arises from the cubic frequency moment together with the integration differential. The calendar readout expresses that same flux through action, length, time, and tritic information.

30.7 Spectral maxima and displacement law

Lemma 30.7 (Positive root for spectral maxima). *For every integer $n > 1$, the equation $n(1 - e^{-x}) = x$ has a unique solution $x_n > 0$. The function $f_n(x) = x^n/(e^x - 1)$ has its unique positive maximum at x_n .*

Proof. Let $g_n(x) = n(1 - e^{-x}) - x$. We have $g_n(0) = 0$, $g'_n(x) = ne^{-x} - 1$, and $g''_n(x) = -ne^{-x} < 0$. The maximum of g_n is attained at $\log n$, where its value is $n - 1 - \log n > 0$, whereas $g_n(n) = -ne^{-n} < 0$. Strict monotonicity on either side of the maximum proves existence and uniqueness of the positive root. The sign of $f'_n(x)$ agrees with that of $g_n(x)$; moreover, f_n tends to zero at zero and infinity. This proves uniqueness of the maximum. $\square$

Proposition 30.8 (Frequency and wavelength readouts). *The maxima of radiance per frequency and per wavelength are*

$$\nu_{\max} = \frac{x_3 k_B \Theta}{h}, \quad \lambda_{\max} = \frac{hc}{x_5 k_B \Theta}. \quad (30.18)$$

In particular,

$$\nu_{\max}(\Theta_{108}) = \frac{x_3}{108 t_0 \log 3}, \quad \lambda_{\max}(\Theta_{108}) = \frac{108 \log 3}{x_5} \ell_0. \quad (30.19)$$

Proof. The conversion $\omega = 2\pi\nu$ and the isotropic angular factor give

$$B_\nu = \frac{2h\nu^3}{c^2} \frac{1}{e^{\beta h\nu} - 1}.$$

With $\nu = c/\lambda$, the transformed density satisfies $B_\lambda = B_\nu |d\nu/d\lambda|$; hence

$$B_\lambda = \frac{2hc^2}{\lambda^5} \frac{1}{e^{\beta hc/\lambda} - 1}.$$

The respective changes of variable lead to f_3 and f_5 . The preceding lemma proves (30.18); the calendar scale gives (30.19). The density Jacobian explains why the two maxima correspond to distinct roots: $\nu_{\max} \lambda_{\max} = cx_3/x_5$. $\square$

Wien displacement is structurally defined by the stationary point of the chosen spectral density. A change of variable transforms both the argument and the measure. This precision locates each spectral constant in its generating operation and preserves its physical meaning when passing between frequency and wavelength.

31 Conclusions

The Bose–Einstein and Fermi–Dirac distributions are obtained as the equilibrium laws of symmetric and exterior completions of states carrying trajectory and memory. Algebraic Pauli exclusion and the dilute Maxwell–Boltzmann limit belong to the same derivation. The electromagnetic realization of bosonic occupation then yields the Planck spectrum, the Stefan–Boltzmann flux, and the Wien displacement laws. This establishes the connection between state structure, admissible occupations, and radiative observables in the specified representations.

Compositional parity determines the statistical mechanism. The exterior product makes creation operators nilpotent and occupation operators idempotent, with spectrum contained in $\{0, 1\}$; the symmetric completion admits nonnegative integer occupations. Second quantization transports contractive intertwiners, with the stated additional domain for morphisms outside that class. The relativistic identification of parity and spin retains rotation-representation and locality conditions: algebraic exclusion and this identification have distinct hypotheses.

Grand-canonical factorization turns these occupations into the characteristic Bose–Einstein and Fermi–Dirac denominators. The bosonic chemical-potential condition ensures convergence of the modal traces. In the thermodynamic regime, variation of the combinatorial entropy under energy and population constraints provides a complementary characterization of the same laws. On the declared finite screen, micro-canonical counts reproduce the fermionic and bosonic cardinalities by convolution in integer arithmetic. Dilute-regime bounds quantify their common Maxwell–Boltzmann approximation.

The Planck law results from composing three elements of the free electromagnetic realization: bosonic occupation with zero chemical potential, the density of two transverse modes, and energy per excitation. The constant mode remains fixed as background. Uniform estimates near the origin and in the spectral tail justify the passage from discrete cavity sums to continuous densities. Bose moments determine number, energy, and entropy densities. Angular integration of the isotropic flux gives the Stefan–Boltzmann law; spectral extremization gives the Wien maxima in frequency and wavelength.

The Stefan–Boltzmann constant is the coefficient of the composition of bosonic occupation, mode density, and angular projection of flux. Its expression in calendar units relates action, duration, and tritic information. The Wien constants characterize the stationary points of two densities connected by a change of measure, whose Jacobian explains the difference between their maxima. These constructions distinguish structural coefficients, dimensional units, and the experimental quantities with which they are compared.

The Boltzmann constant acts as a transducer between dimensionless information and dimensional entropy. Decision energy arises from action and period; the specified thermal section individuates the transducer and temperature, with their change-of-base laws. The calendar’s thermal relationship retains the action unit, time unit, and Boltzmann–temperature pair. The origin of energy, the measure of continuations, and their thermal reading are thereby connected through operations of distinct types.

The renewal–quotient–memory–loop chain realises that connection. The renewal (20, 24, 25, 23) and the tritic quotients produce the graded carrier; positive memory retains $N - L$; the 8:1 loop forces $\nu_W = 11/9$ and $r = 1/10$; and the grouping $j = 3n + b$ produces $q = 1/1000$ without losing the residue. The thresholds 23, 26, 29 and the multiplicities 1, 380, 649 yield exactly 1.380649×10^{-23} . The upward and downward currents force zero sectoral lag, while the marker restores the partition function after normalisation. Centred equivalence preserves the thermal response, and the origin retains the absolute cost in area, mass, and radius. Finally, $G_a k_B \Theta_{\text{clk},a} \log 3 = c^4 L_a$ and the transducer identity compose the spectral output with the dimensional section.

The dynamics preceding occupation has an explicit unitary realization. The calendar Hamiltonian has an exponential that reproduces the shift exactly; its variational action yields the corresponding Schrödinger equation. The general formulation retains strong continuity and the domain of the self-adjoint generator. In the finite periodic realization, the Floquet propagator reproduces the shift on 108 states. For the basis states $\rho_{j_0} = |j_0\rangle\langle j_0|$, the expectation vector of the two order observables has a primitive period of 108 drive cycles. Unitary conjugation transports state, observable, and propagator together. General perturbations retain the spectral estimate and time-dependent error bound, distinguishing periodicity, covariance, and quantified stability.

The interpretation of entropy depends on the object on which it is evaluated. For a number-conserving, gauge-invariant fermionic quasifree state in finite dimension, the one-body correlator $0 < C < I$ and its modular generator K_1 determine each other through the logistic law. Occupation entropy, density-operator entropy, and the entropy of an entangled reduction are related by explicit maps and retain their respective domains. Existence of the thermodynamic limit requires independence from the cofinal sequence; condensation is characterized by saturation of the excited sector.

The stationary measure on the areal–volumetric envelope is fixed by incidence and normalization. Its maximal rate $\log \varphi$ is accompanied by a relative-divergence identity quantifying the deficit of any other stationary measure. The weight ratio φ^{-2} belongs to this weighting of histories; the chronological orbit retains its own temporal distribution. This distinction identifies the dynamics producing each distribution and the meaning of the corresponding entropy.

Total area and the modular operator provide two complementary observables: the former preserves joint occupation, while the latter assigns different weights to the two incidences when $\Delta_{\text{mod}} \neq 0$. Under that condition and with known normalization, their simultaneous reading recovers the sectoral occupations; for $\Delta_{\text{mod}} = 0$, the modular Hamiltonian is scalar and this recovery is unavailable. The finite identity expresses entropy variation as the weighted sum of area variations minus a nonnegative relative-entropy deficit. Occupation inversion preserves entropy and transforms area into its complement; both properties are determined without identifying entropic equality with structural equality.

Memory preservation supports these relationships. The arithmetic core retains both evaluations, addition and multiplication, their quotients, and the history of transitions, distinguishing the transported state from its observables. Bilateral coupling preserves the norm and recovers the input from both channels; invariance of other observables follows the established commutation condition. State recovery and preservation of a particular reading are therefore properties determined by their respective operators.

The Fourier realization of the positional register completes this conservation thesis with two exact boundaries. Every nonzero state on $\Lambda = (\mathbb{Z}/27\mathbb{Z})^2$ has support product at least 729, and every unit vector satisfies $H_{\text{pos}} + H_F \geq \log 729$. Normalized indicators of four rectangular subgroups simultaneously saturate both bounds. The entropic proof follows from Riesz–Thorin interpolation and differentiation at $p = 2$. These distribution entropies retain their own domain, distinct from the domains of entanglement, von Neumann, and thermodynamic entropy. The complementary dodecaphase readers further prove that two words with identical histograms may retain different ordered correlations.

The prime-vacancy observer retains another datum discarded by a marginal count: it combines z_p with the weight $\log p$ to produce $R_{\text{vac}}(X)$. Its modal resolution is an identity under symmetric or Fejér convergence, and conjugate pairs ensure reality. The six preserved scales and the thirty modes at $X = 10^8$ are finite witnesses, with independent reproduction through 10^6 . Asymptotic promotion requires uniform modal estimates; a conventional sieve enters afterwards as a verification reader.

Photon density is also connected, without changing its generator, to the regularized level A_2^{BG} : the functional equation gives $\mathcal{Z}_3(3) = 4\pi_{\text{HMT}}^2 \log A_2^{\text{BG}}$, and substitution into the thermal moments yields exactly the dimensionless photon-number density and the mean energy and entropy ratios. This composition is the photonic reciprocal of the Bendersky–Glaisher spectral development; it keeps the HMT genealogy of π , spectral regularization, and electromagnetic realization distinct.

The complete area distribution and entropy admit preparations with different energies. Section 28.10 locates that difference in the information within each fibre: sector exchange preserves the area, and its inverse attains the maximum recoverable energy for the specified Hamiltonian. The conditional recovery map also characterizes the free-energy minimum at fixed marginal.

The combined result is an explicit constructive chain: trajectory and parity determine occupation completions; equilibrium determines Bose–Einstein and Fermi–Dirac statistics; and bosonic occupation, realized on electromagnetic modes, determines the Planck, Stefan–Boltzmann, and Wien laws. Action, calendar, and thermal transduction connect their scales, while memory and reduction maps specify the information preserved. Statistical and radiative laws are related through this structure, with the conditions of each composition stated in its own domain.

A Domains and conditions of the realizations

Algebraic exclusion, its transport under refinement, and its statistical applications are related results with distinct domains. The general annihilation identity (18.2) makes the adjoint of the transport explicit; its isometric specialization retains the necessary condition on the map between the one-particle spaces. Equilibrium occupations additionally use a grading and a Hamiltonian compatible with the trajectory representation.

The finite dynamical realization is developed in Section 13. The area operator, modular Hamiltonian, and sectoral purification are developed in Section 28 and its continuations. Sections 11 and 14 contain the constructions of action and the Boltzmann–temperature pair used by the radiative realization. The variational characterization of the history measure and convergence of the spectral sums link, respectively, incidence to entropy and discrete modes to radiation. The radiation section presents its cavity, dispersion, polarization, and equilibrium hypotheses, together with the spectral proofs it uses. Section 26.3 also establishes the identity of the kinematic and constitutive realizations of speed, its dimensional normalization, and its preservation under amplification of the two-sheet space.

Each composition retains its stated mathematical hypotheses and physical realization. In particular, the harmonic modes of the cellular complex, connections modulo gauge, and radiative modes retain the domains distinguished in Section 16.6; the radiative realization explicitly uses its dispersion relation and polarizations.

Subsection 30.3 proves, for the three observables F_N, F_E, F_Z , the passage from three-dimensional periodic sums to radial integrals as $L \rightarrow \infty$. Its domain requires $a, b > 0$, retains the fixed background constant mode, and uniformly controls the origin and the spectral tail. The result yields the number, excitation-energy, and logarithmic partition-function densities of the transverse modes; it is distinct from the finite spectral-capacity block.

Section 15 has as its domain the finite graded carrier, the microscopic and aggregated trees, the reversible memory lift, the capacity clock, and the thermometric charts declared there. Its spectral, transport, and variational identities are proved within those domains; physical recognition enters afterwards. The typed dimensional chart preserves the provenance of the marked intensity, energy chart, and physical temperature.

Section 29 works on the finite realization $\ell^2((\mathbb{Z}/27\mathbb{Z})^2)$ and its unitary Fourier transform. Its entropies are those of the distributions in two conjugate bases; the dodecaphase readers have the distinct domain $\{-1, 0, +1\}^{12}$ and provide a complementary ordered-memory result. Section 4.16 establishes the exact modal identity for each finite X ; its historical witness reaches 10^8 , and the documented independent reproduction reaches 10^6 . Promotion to the critical bound requires estimates uniform in the modes.

B Complete modal table for the prime-vacancy observer

The following table preserves the thirty evaluations recorded by the finite F051 witness at $X = 10^8$. Its columns are obtained from $S_k(X) = \sum_{p \leq X} (\log p) e^{2\pi i k p \delta}$, with $1 \leq k \leq 30$. They constitute numerical evidence for that execution within the finite window; modal selection, asymptotic promotion, and the internal construction of irreducibles remain defined by their respective mechanisms. The unrounded data are delivered in `RECUPERACION/F051/S_k_1_30_X1e8.csv`.

Numerical provenance. The historical package preserves `R_vac_powers.csv` through 10^8 and `S_k_1_30_X1e8.csv`, but it does not contain the program that produced the sweep. The documented independent reproduction reaches 10^6 ; the later cutoffs and the thirty modes retain the status of finite historical witnesses.

Table 13: Prime-vacancy modes recorded by F051 at $X = 10^8$.

k	$\text{Re } S_k$	$\text{Im } S_k$	$ S_k $	$ S_k /\sqrt{X}$	$ S_k /(\sqrt{X} \log X)$
1	40391.948	13384.211	42551.693	4.255	0.231
2	-17862.964	-15339.544	23545.427	2.355	0.128
3	342.273	36143.793	36145.414	3.615	0.196
4	39914.237	-13618.740	42173.645	4.217	0.229
5	-33890.198	14281.983	36776.631	3.678	0.200
6	14847.310	-946.591	14877.454	1.488	0.081
7	21809.183	7762.130	23149.322	2.315	0.126
8	9977.978	11021.054	14866.865	1.487	0.081
9	-16007.184	-34045.358	37620.690	3.762	0.204
10	15207.543	35816.973	38911.759	3.891	0.211
11	-24865.550	-97.564	24865.741	2.487	0.135
12	10436.509	-19270.200	21914.865	2.191	0.119
13	20804.377	14471.223	25342.423	2.534	0.138
14	-24098.417	-12513.403	27153.618	2.715	0.147
15	-19768.196	8389.656	21474.821	2.147	0.117
16	-29615.859	-1125.619	29637.242	2.964	0.161
17	14774.780	-23351.605	27633.161	2.763	0.150
18	-14522.533	-14506.892	20526.906	2.053	0.111
19	30646.583	-39786.843	50221.568	5.022	0.273
20	-13017.655	-9396.890	16054.933	1.605	0.087
21	1214.758	-44824.224	44840.681	4.484	0.243
22	-4553.717	-5978.105	7514.924	0.751	0.041
23	-15378.010	-4781.314	16104.166	1.610	0.087
24	-1325.006	47631.087	47649.513	4.765	0.259
25	4921.472	7202.528	8723.376	0.872	0.047
26	-26879.475	33364.677	42845.161	4.285	0.233
27	-796.083	-9671.604	9704.312	0.970	0.053
28	-19509.751	9075.771	21517.435	2.152	0.117
29	24446.295	17231.528	29908.977	2.991	0.162
30	-10661.806	-25194.821	27357.871	2.736	0.149

Within this finite window, $k = 19$ maximizes $|S_k|$ only among the thirty tabulated modes. The statement does not extend to other modes or other cutoffs X without an additional proof.

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